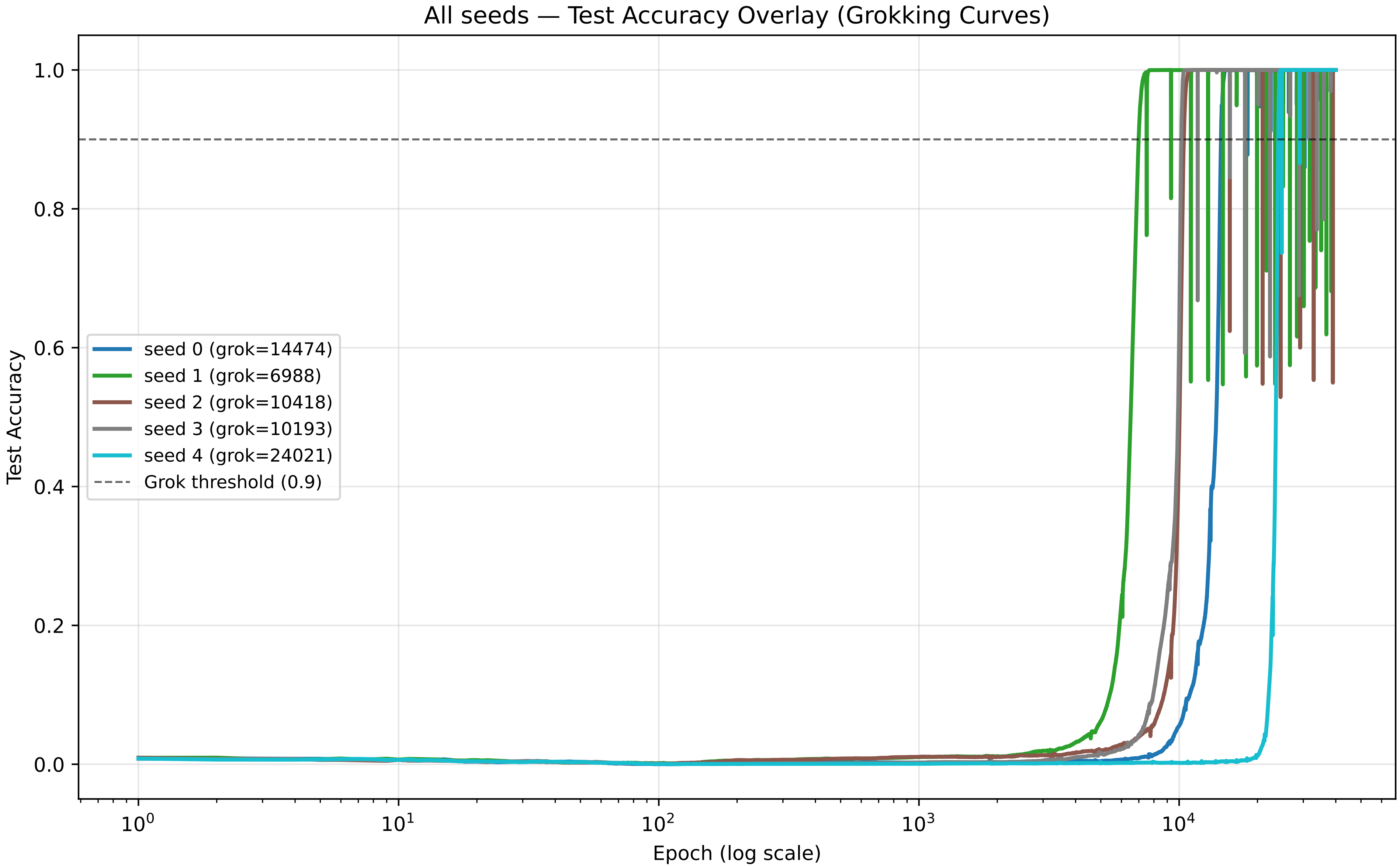
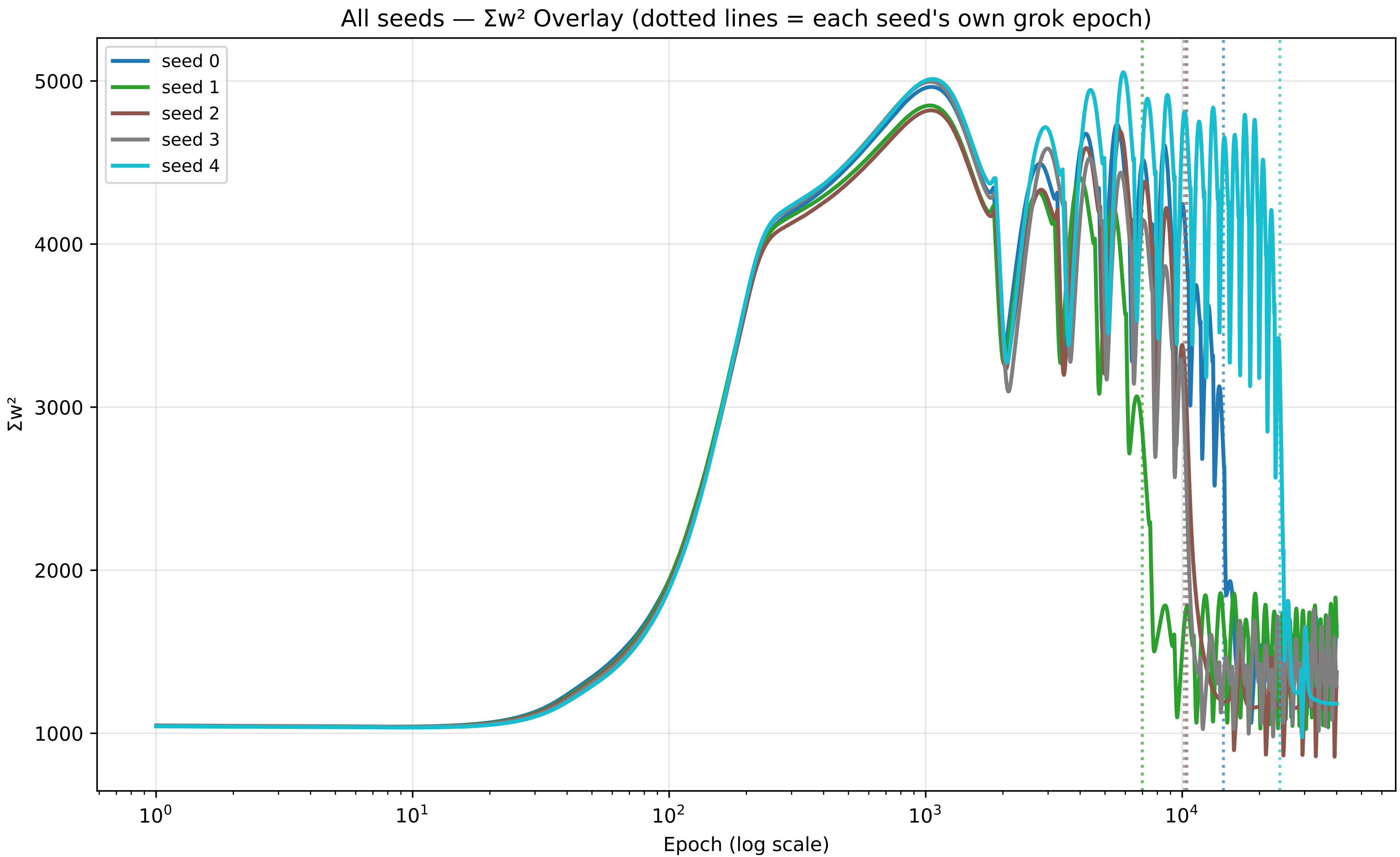
NANDA-UNIFIED BENCHMARK — RESULTS REPORT

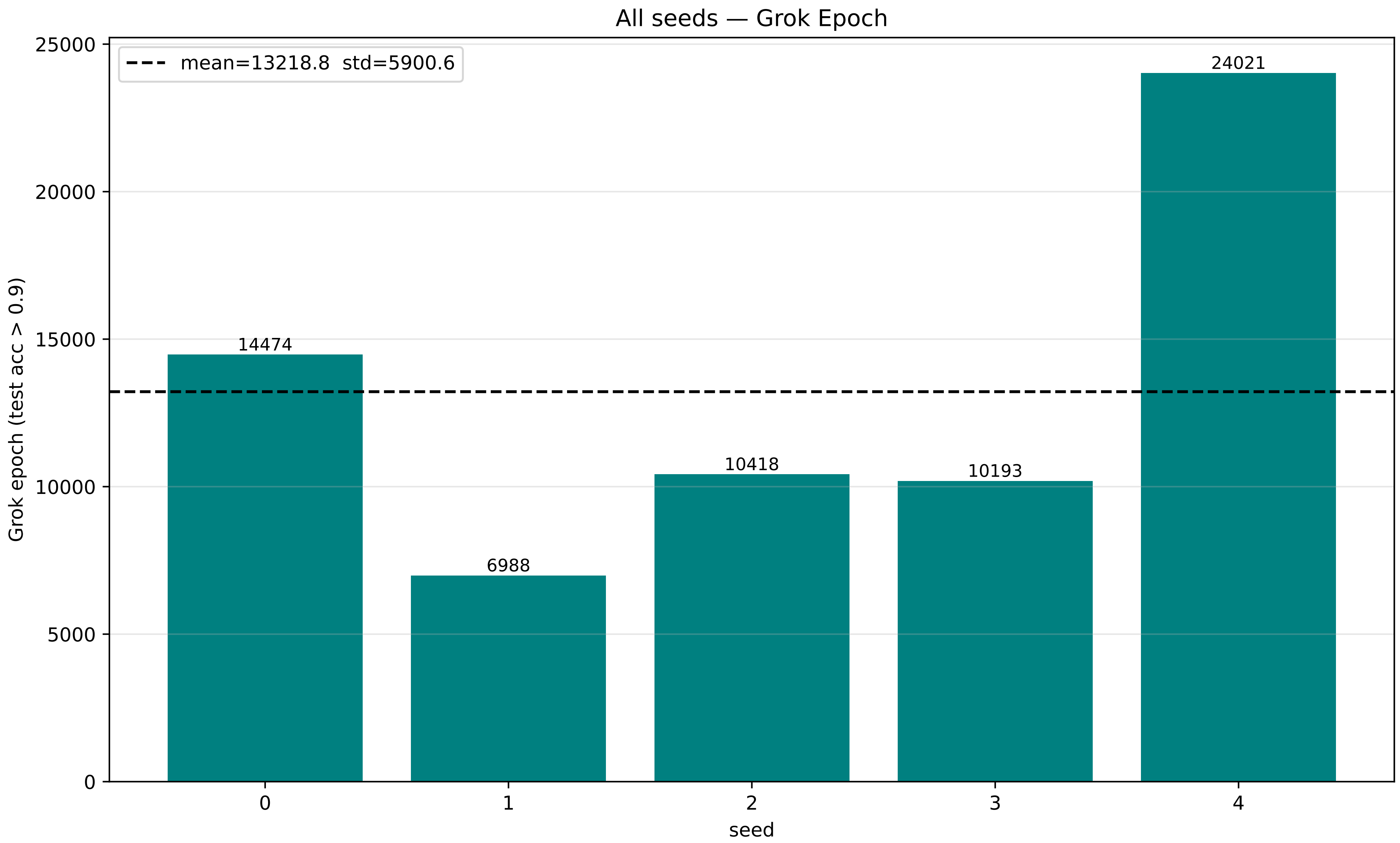
```
results_dir : /Users/jonathanjohn/Documents/grokking-benchmark/results/nanda_unified
seeds       : [0, 1, 2, 3, 4]

=== AGGREGATE SUMMARY ===
n_seeds          : 5
epochs           : 40000
grok_epoch_mean  : 13218.8
grok_epoch_std   : 5900.607880549257
grok_epochs      : [14474, 6988, 10418, 10193, 24021]
n_limit_cycle    : 0 / 5
```

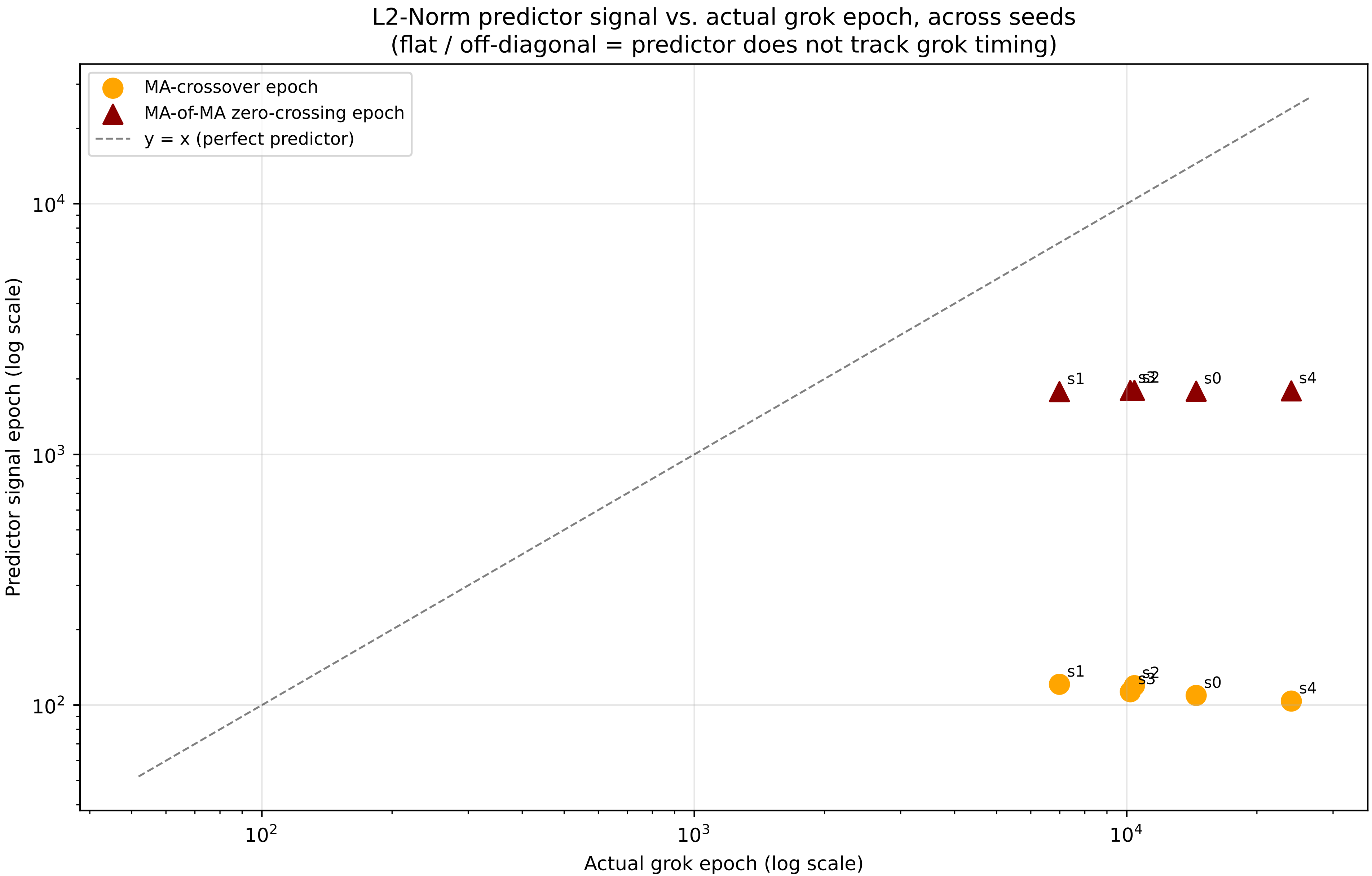


All seeds — Test Accuracy Overlay (Grokking Curves)
1.0
0.8
0.6
0.4
0.2
0.0
Test Accuracy
seed 0 (grok=14474)
seed 1 (grok=6988)
seed 2 (grok=10418)
seed 3 (grok=10193)
seed 4 (grok=24021)
Grok threshold (0.9)
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

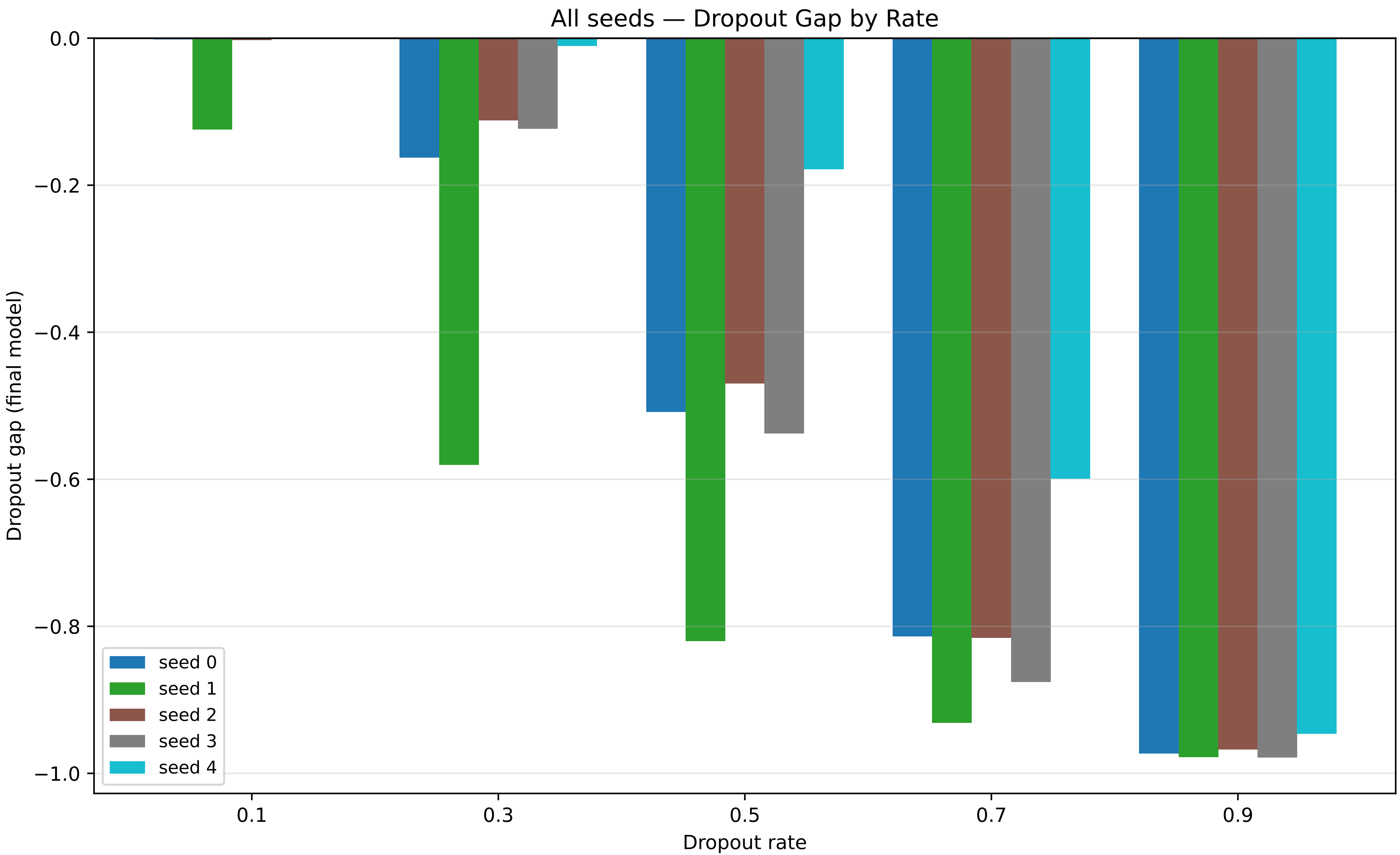


All seeds — Σw² Overlay (dotted lines = each seed's own grok epoch)
seed 0
seed 1
seed 2
seed 3
seed 4
5000
4000
3000
2000
1000
Σw²
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

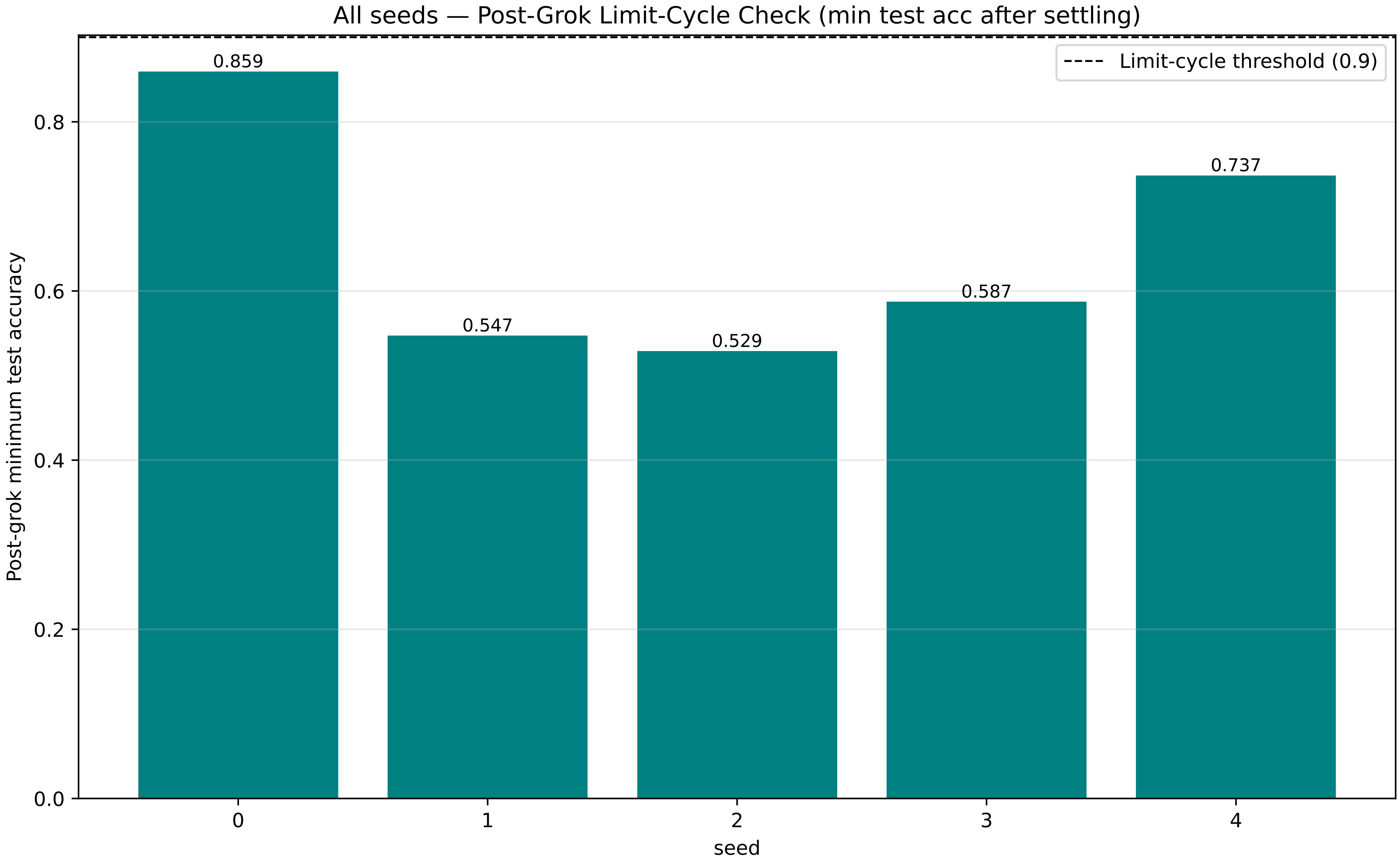

All seeds — Grok Epoch
mean=13218.8 std=5900.6
Grok epoch (test acc > 0.9)
seed
0
5000
10000
15000
20000
25000
0
1
2
3
4
14474
6988
10418
10193
24021



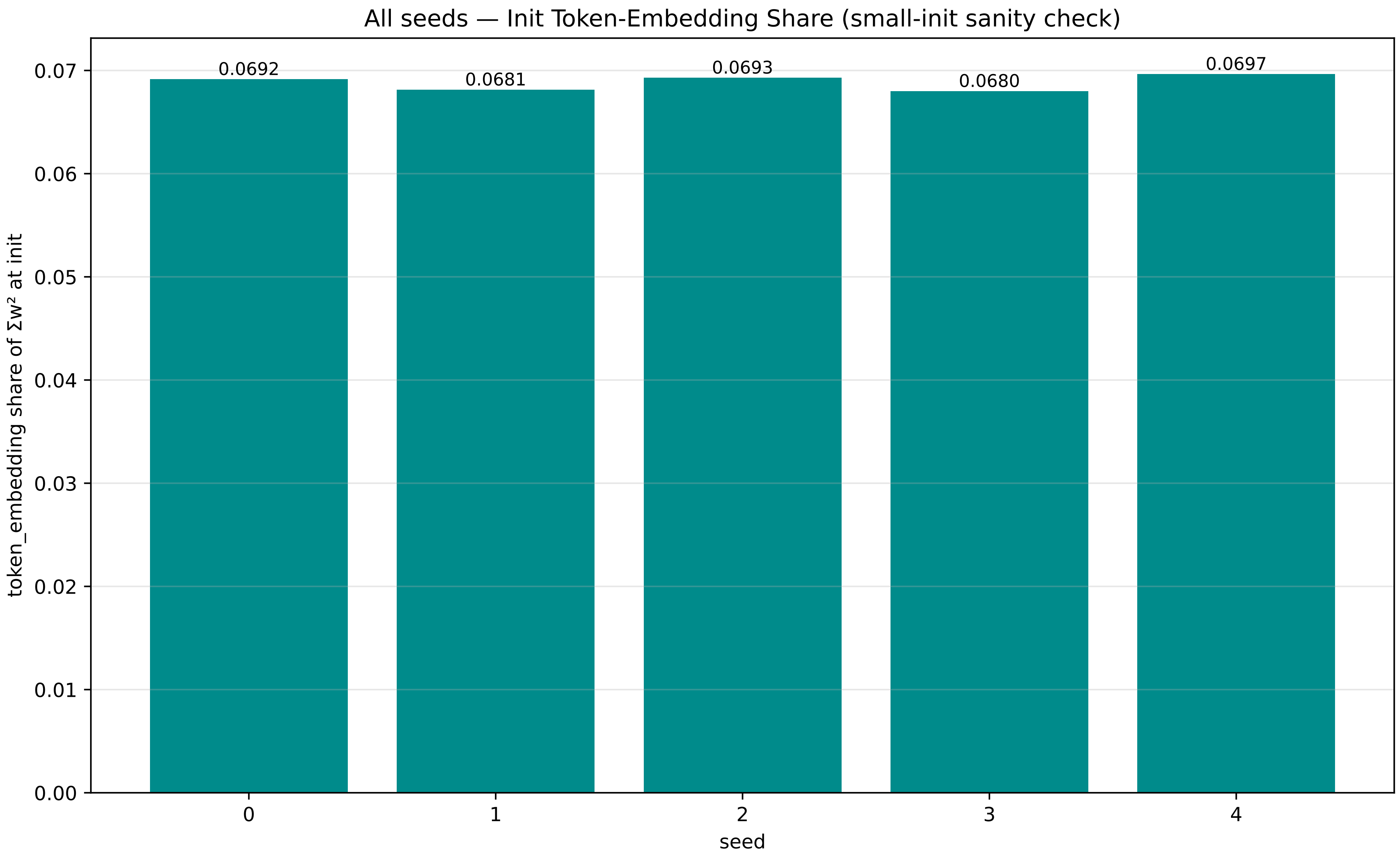
L2-Norm predictor signal vs. actual grok epoch, across seeds
(flat / off-diagonal = predictor does not track grok timing)
MA-crossover epoch
MA-of-MA zero-crossing epoch
y = x (perfect predictor)
Predictor signal epoch (log scale)
Actual grok epoch (log scale)
10^2
10^3
10^4
s0
s1
s2
s3
s4

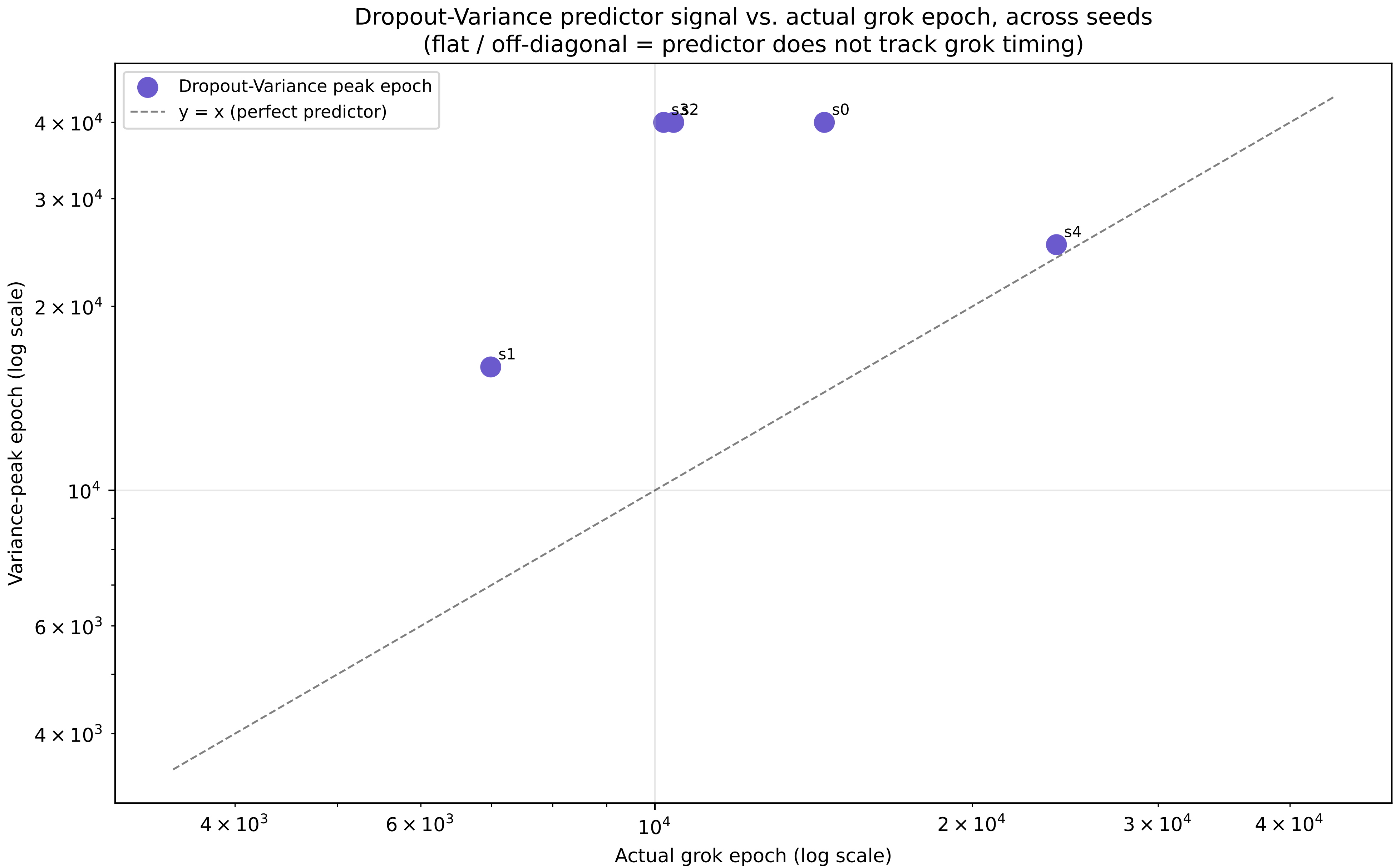


All seeds — Dropout Gap by Rate
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
Dropout gap (final model)
0.1
0.3
0.5
0.7
0.9
Dropout rate
seed 0
seed 1
seed 2
seed 3
seed 4



All seeds — Post-Grok Limit-Cycle Check (min test acc after settling)
Limit-cycle threshold (0.9)
0.859
0.547
0.529
0.587
0.737
Post-grok minimum test accuracy
0.0
0.2
0.4
0.6
0.8
0
1
2
3
4
seed

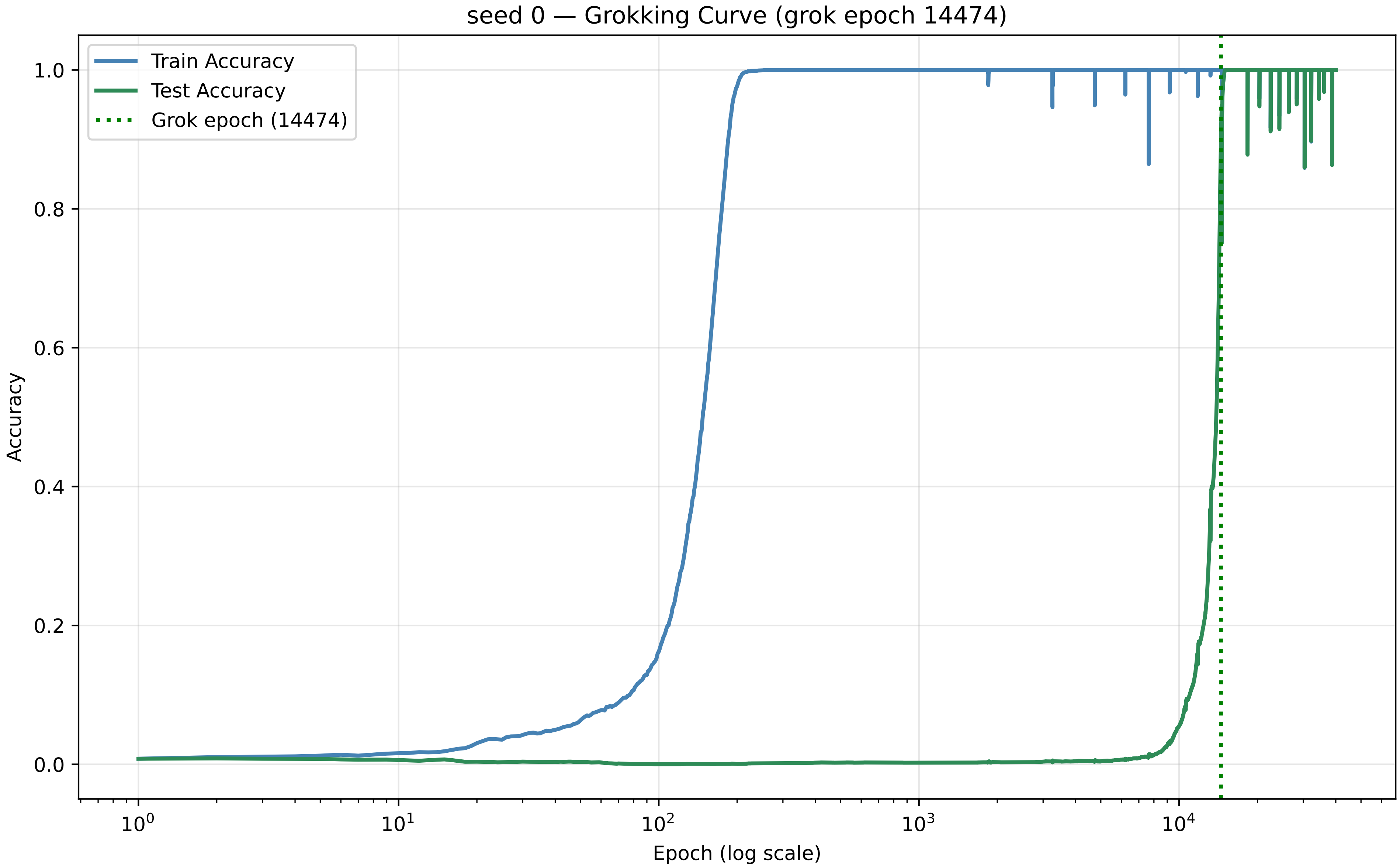

All seeds — Init Token-Embedding Share (small-init sanity check)
token_embedding share of Σw² at init
seed
0.0692
0.0681
0.0693
0.0680
0.0697
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0
1
2
3
4

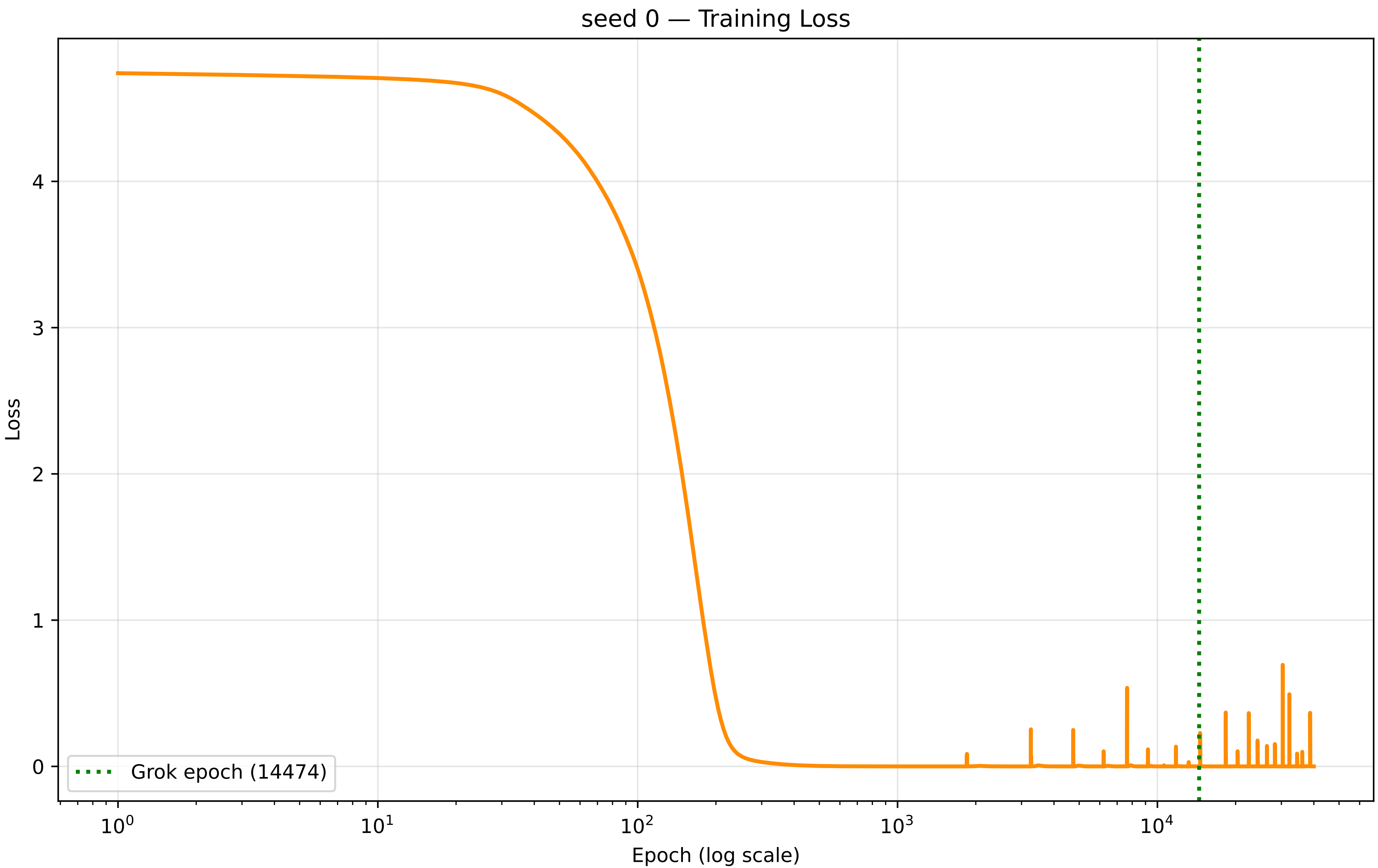


Dropout-Variance predictor signal vs. actual grok epoch, across seeds
(flat / off-diagonal = predictor does not track grok timing)
Dropout-Variance peak epoch
y = x (perfect predictor)
s3
s2
s0
s4
s1
Variance-peak epoch (log scale)
Actual grok epoch (log scale)
4×10^4
3×10^4
2×10^4
10^4
6×10^3
4×10^3

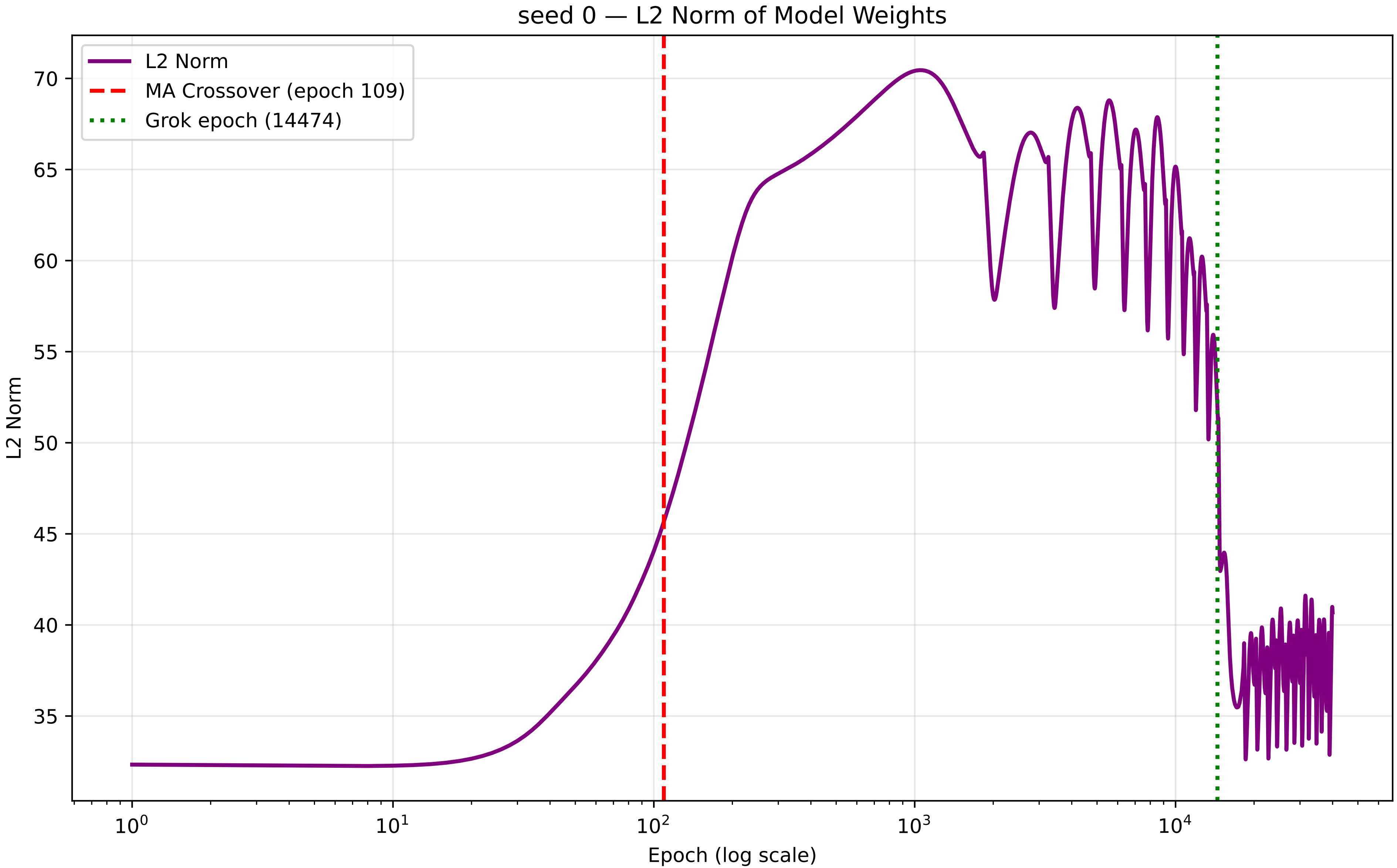
seed 0 — full numbers

```
seed 0  (modulus=113  epochs=40000  wall_time=4979.7s)

  grok_epoch                    : 14474
  final_train_acc               : 1.0000
  final_test_acc                : 1.0000
  l2_norm   init -> final       : 32.3335 -> 40.6897
  sum_w2    init -> final       : 1045.457 -> 1655.655
  token_embedding_share (init)  : 0.0692

  L2 predictor:
    MA-crossover epoch           : 109.25
    MA-of-MA zero-crossing epoch : 1787.48
    noise_floor                  : 0.000471
    windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

  Dropout gap by rate:
    p0.1=-0.0018  p0.3=-0.1625  p0.5=-0.5082  p0.7=-0.8135  p0.9=-0.9728

  Dropout-Variance predictor:
    variance_peak_epoch          : 39999
    peak_to_grok_ratio            : 2.7635
    rate=0.5  n_samples=30  num_checkpoints=24

  Limit-cycle check             : stable
    window_start=14974  post_grok_min=0.8594  post_grok_std=0.0029
    post_grok_final=1.0000  dips<0.9=6
```

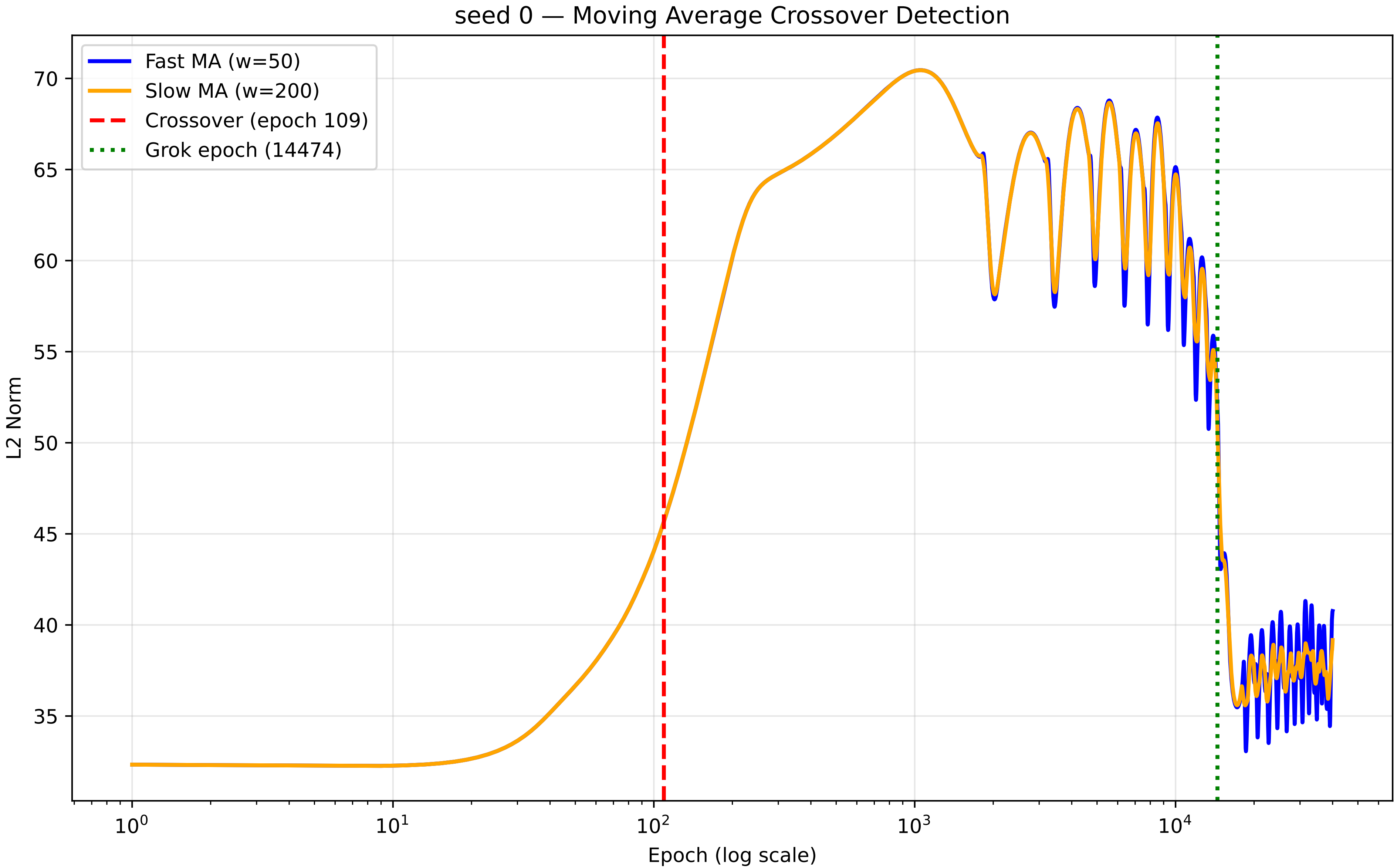

seed 0 — Grokking Curve (grok epoch 14474)
Train Accuracy
Test Accuracy
Grok epoch (14474)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

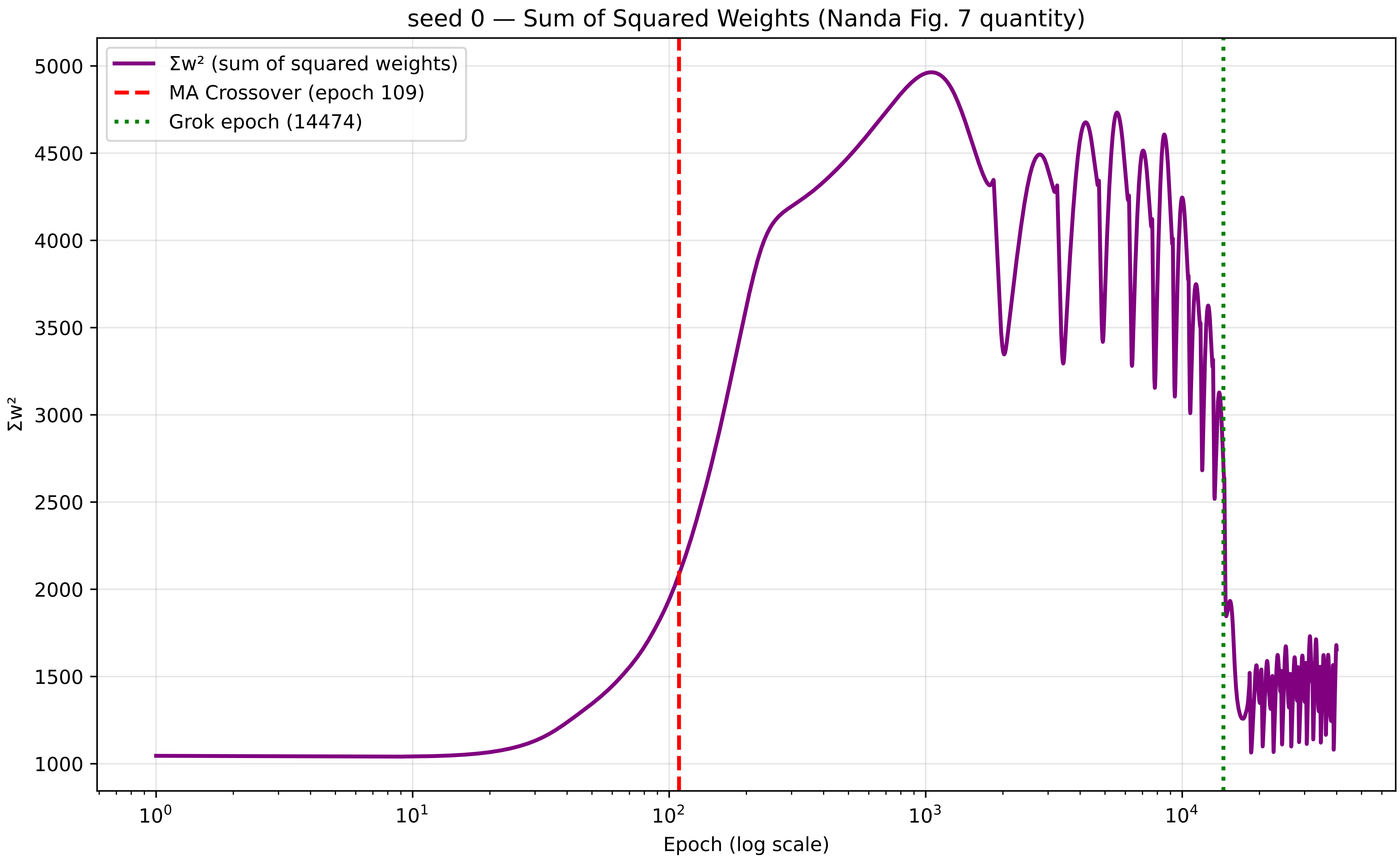


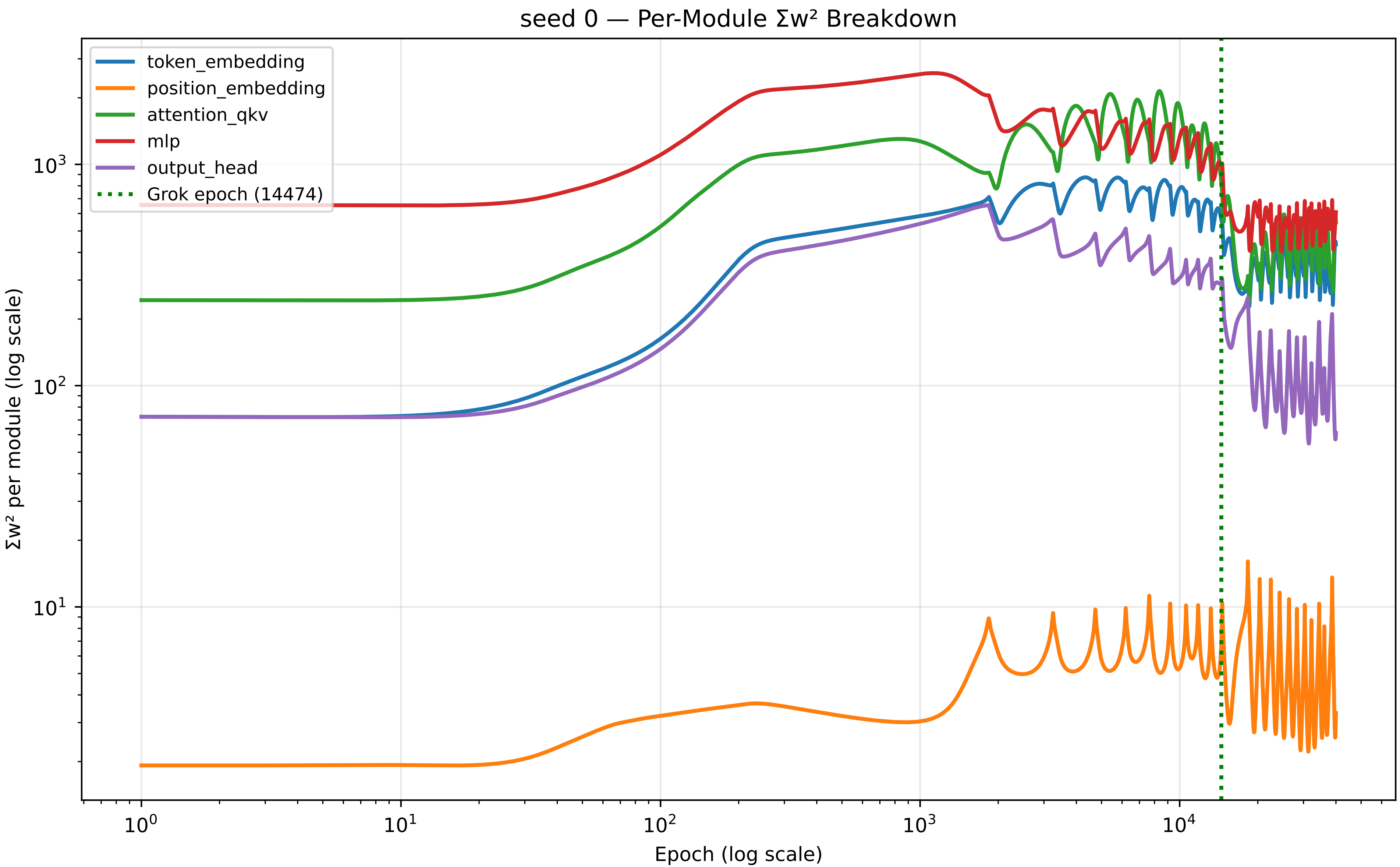
seed 0 — Training Loss
Loss
Epoch (log scale)
0
1
2
3
4
10^0
10^1
10^2
10^3
10^4
Grok epoch (14474)



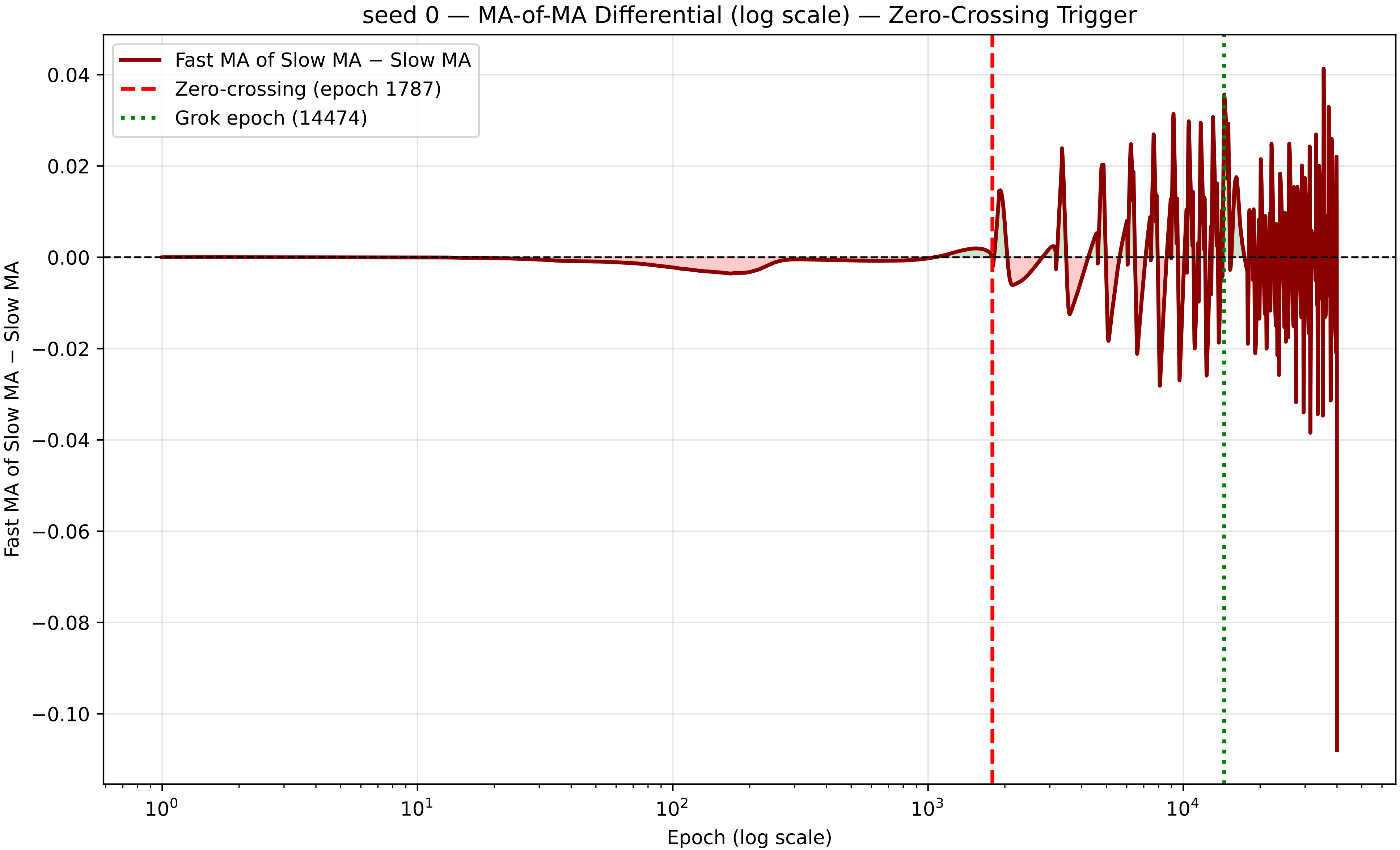
seed 0 — L2 Norm of Model Weights
L2 Norm
MA Crossover (epoch 109)
Grok epoch (14474)
70
65
60
55
50
45
40
35
L2 Norm
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)


seed 0 — Moving Average Crossover Detection
Fast MA (w=50)
Slow MA (w=200)
Crossover (epoch 109)
Grok epoch (14474)
L2 Norm
Epoch (log scale)
70
65
60
55
50
45
40
35
10^0
10^1
10^2
10^3
10^4

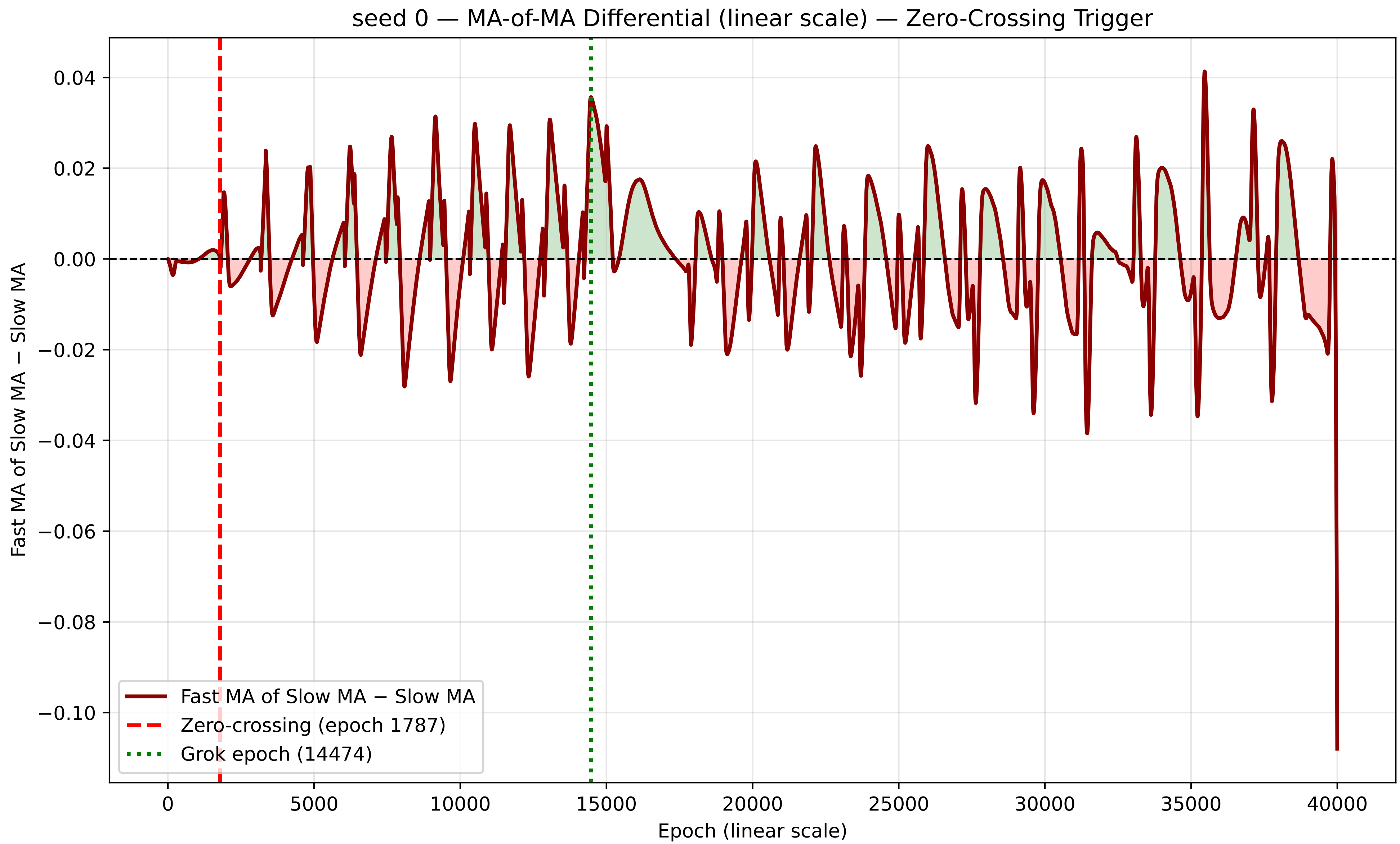

seed 0 — Sum of Squared Weights (Nanda Fig. 7 quantity)
Σw² (sum of squared weights)
MA Crossover (epoch 109)
Grok epoch (14474)
Σw²
Epoch (log scale)
5000
4500
4000
3500
3000
2500
2000
1500
1000
10^0
10^1
10^2
10^3
10^4

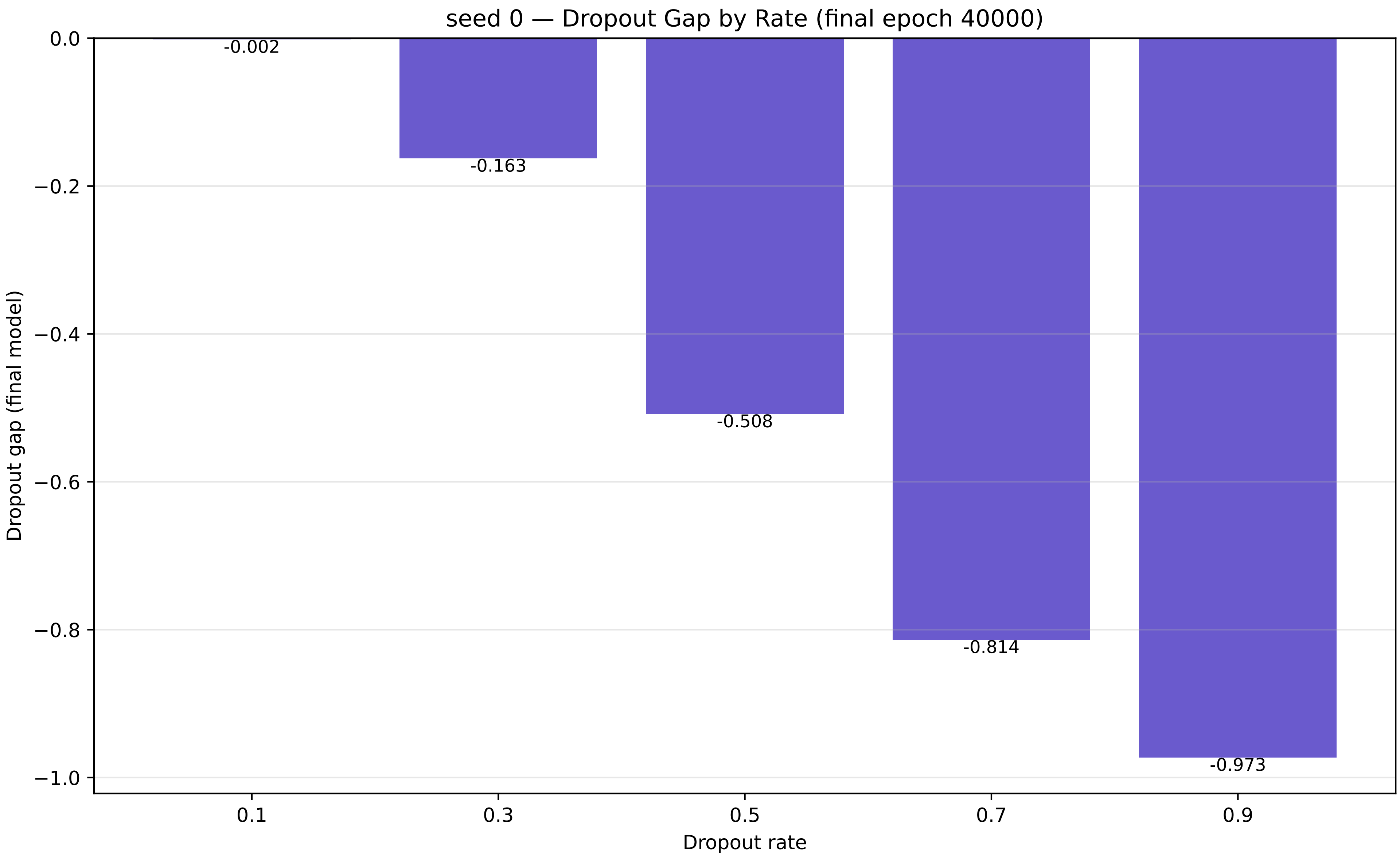


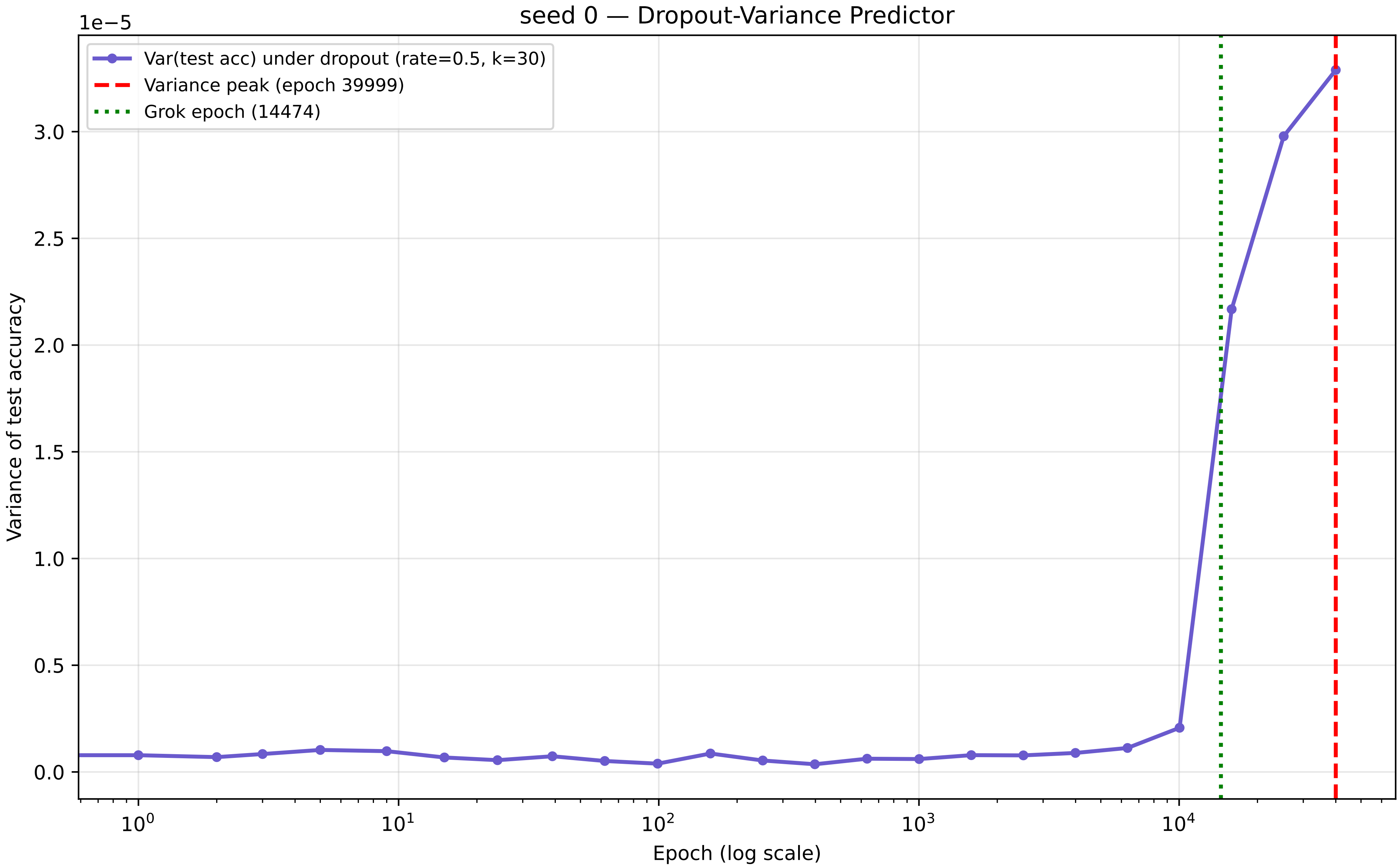
seed 0 — Per-Module Σw² Breakdown
token_embedding
position_embedding
attention_qkv
mlp
output_head
Grok epoch (14474)
Σw² per module (log scale)
Epoch (log scale)
10^3
10^2
10^1
10^0
10^1
10^2
10^3
10^4



seed 0 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1787)
Grok epoch (14474)
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
Fast MA of Slow MA − Slow MA
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)


seed 0 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Epoch (linear scale)
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
0
5000
10000
15000
20000
25000
30000
35000
40000
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1787)
Grok epoch (14474)

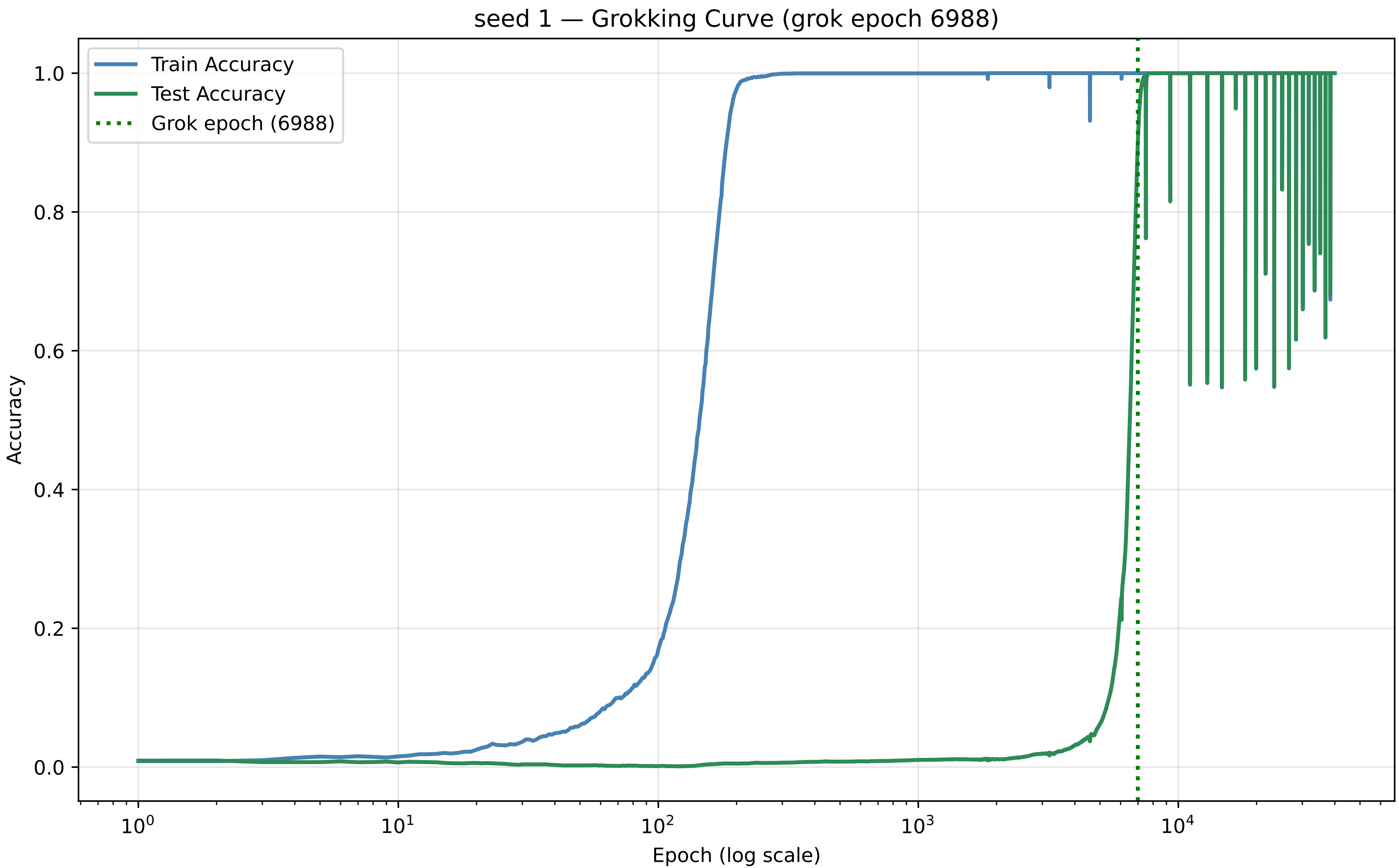

seed 0 — Dropout Gap by Rate (final epoch 40000)
Dropout gap (final model)
Dropout rate
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.1
0.3
0.5
0.7
0.9
-0.002
-0.163
-0.508
-0.814
-0.973

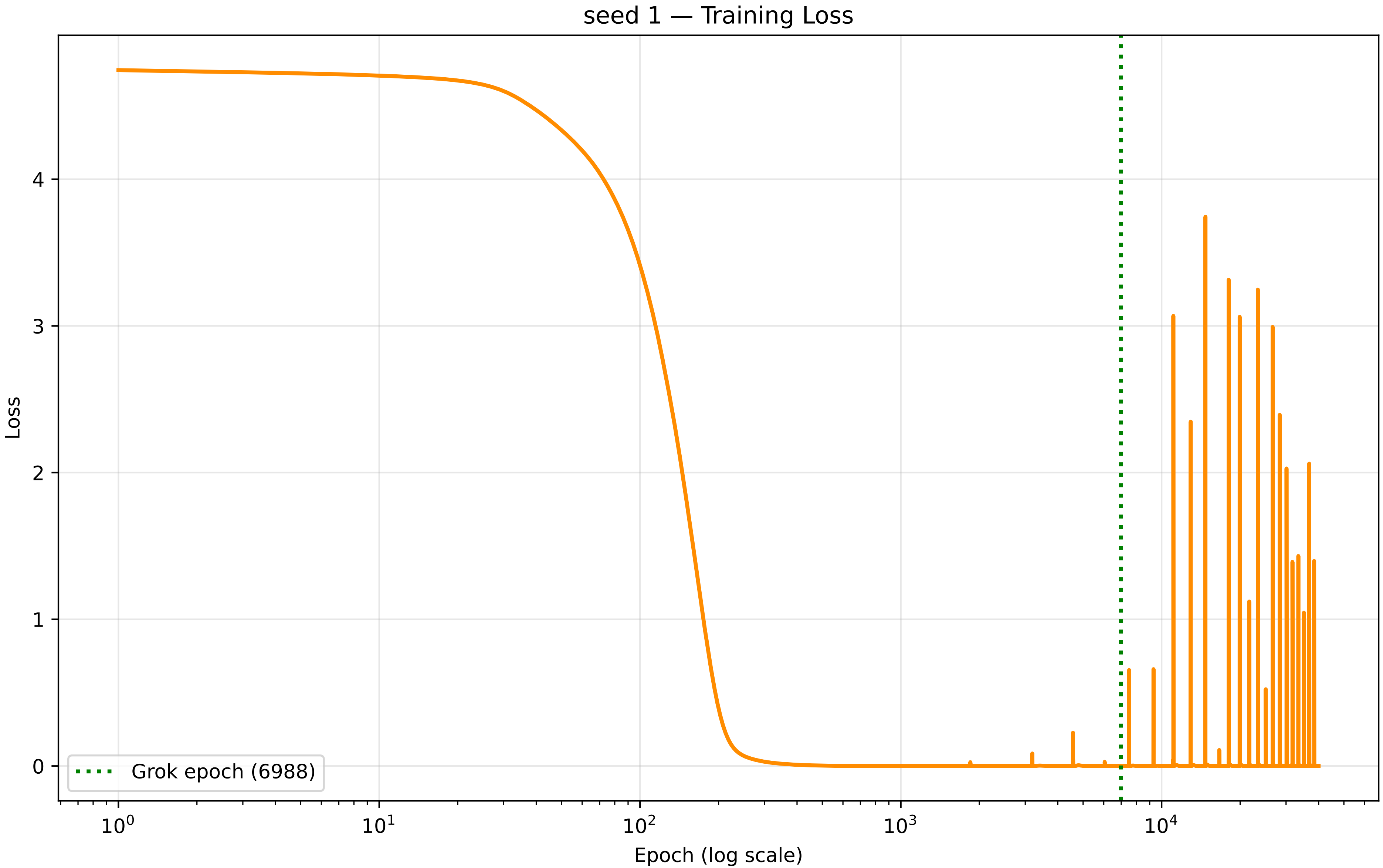


seed 0 — Dropout-Variance Predictor
1e−5
Var(test acc) under dropout (rate=0.5, k=30)
Variance peak (epoch 39999)
Grok epoch (14474)
Variance of test accuracy
0.0
0.5
1.0
1.5
2.0
2.5
3.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

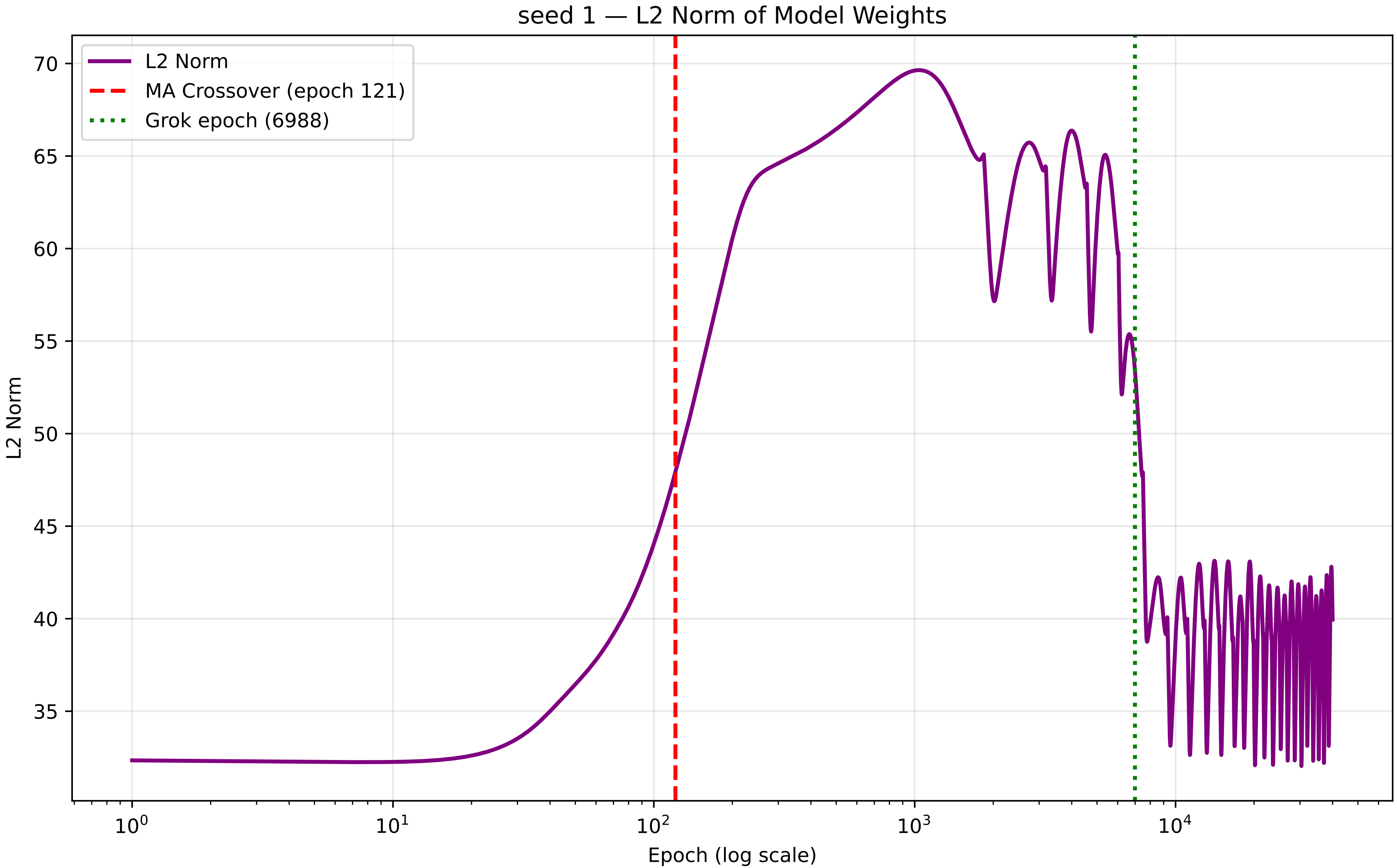
seed 1 — full numbers

```
seed 1  (modulus=113  epochs=40000  wall_time=4950.9s)

  grok_epoch                    : 6988
  final_train_acc               : 1.0000
  final_test_acc                : 1.0000
  l2_norm   init -> final       : 32.3437 -> 39.9515
  sum_w2    init -> final       : 1046.116 -> 1596.124
  token_embedding_share (init)  : 0.0681

  L2 predictor:
    MA-crossover epoch           : 121.01
    MA-of-MA zero-crossing epoch : 1779.92
    noise_floor                  : 0.000474
    windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

  Dropout gap by rate:
    p0.1=-0.1243  p0.3=-0.5806  p0.5=-0.8203  p0.7=-0.9314  p0.9=-0.9780

  Dropout-Variance predictor:
    variance_peak_epoch          : 15917
    peak_to_grok_ratio            : 2.2778
    rate=0.5  n_samples=30  num_checkpoints=24

  Limit-cycle check             : stable
    window_start=7488  post_grok_min=0.5472  post_grok_std=0.0134
    post_grok_final=1.0000  dips<0.9=80
```

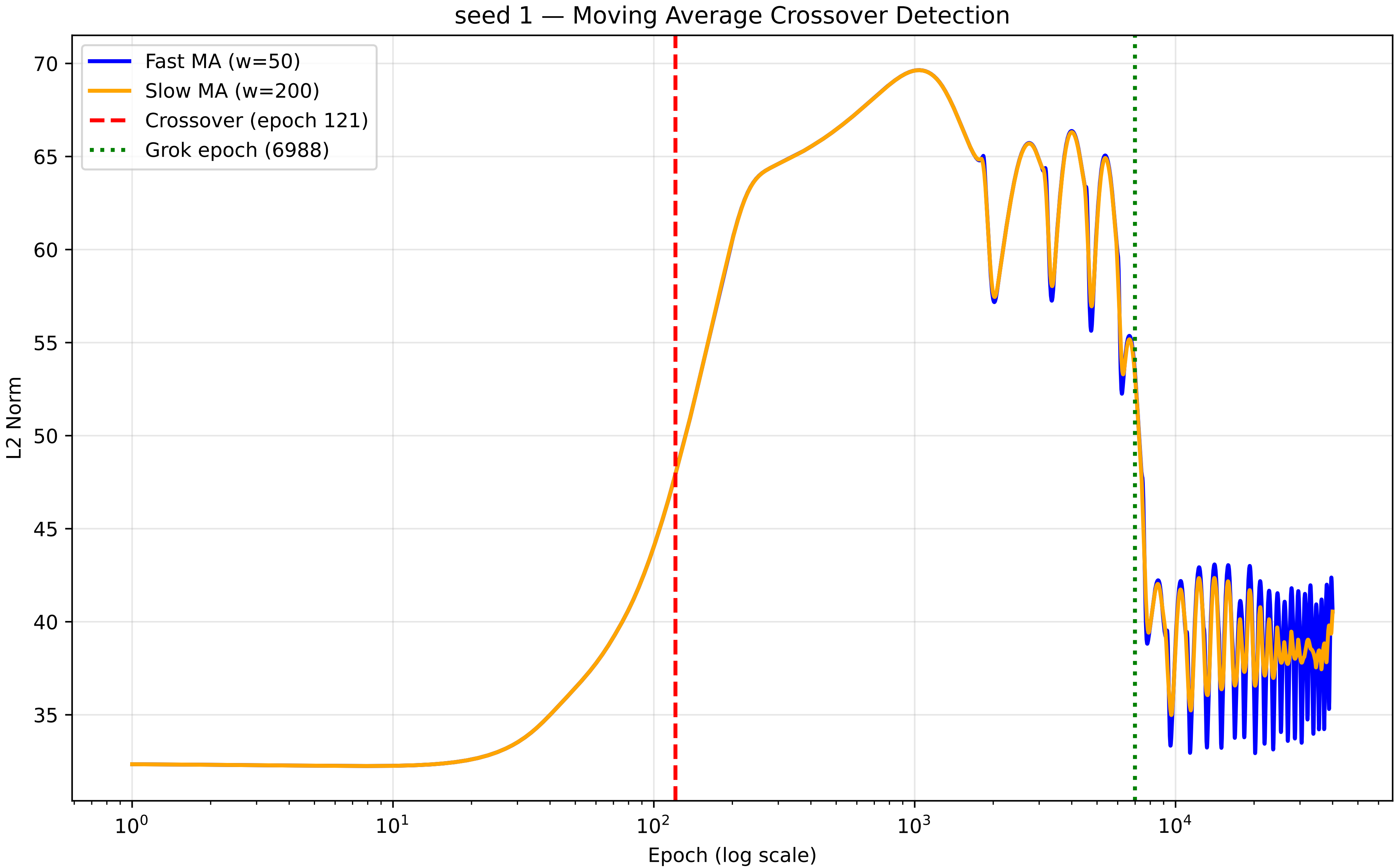

seed 1 — Grokking Curve (grok epoch 6988)
Train Accuracy
Test Accuracy
Grok epoch (6988)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

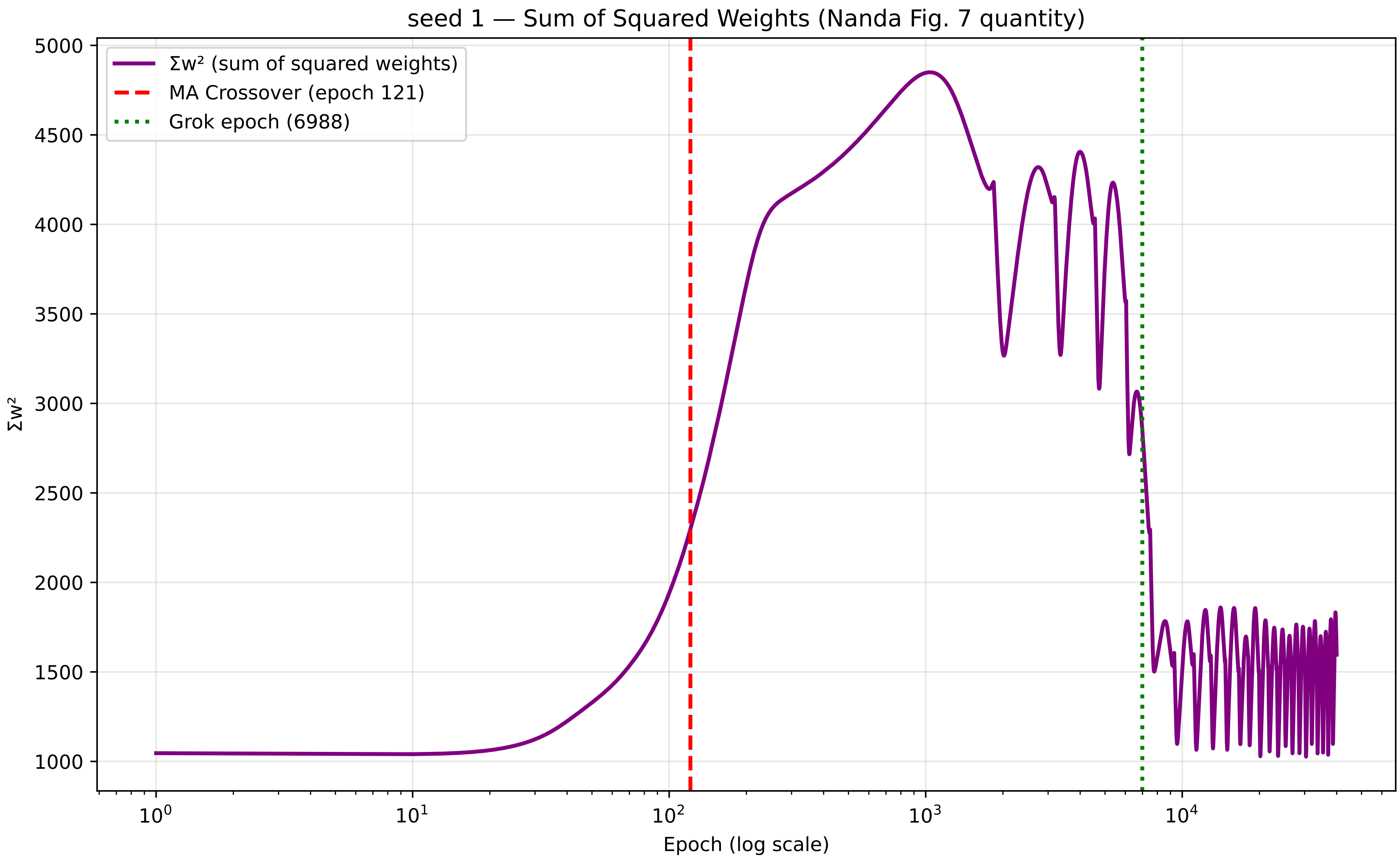


seed 1 — Training Loss
Loss
Epoch (log scale)
Grok epoch (6988)
4
3
2
1
0
10^0
10^1
10^2
10^3
10^4

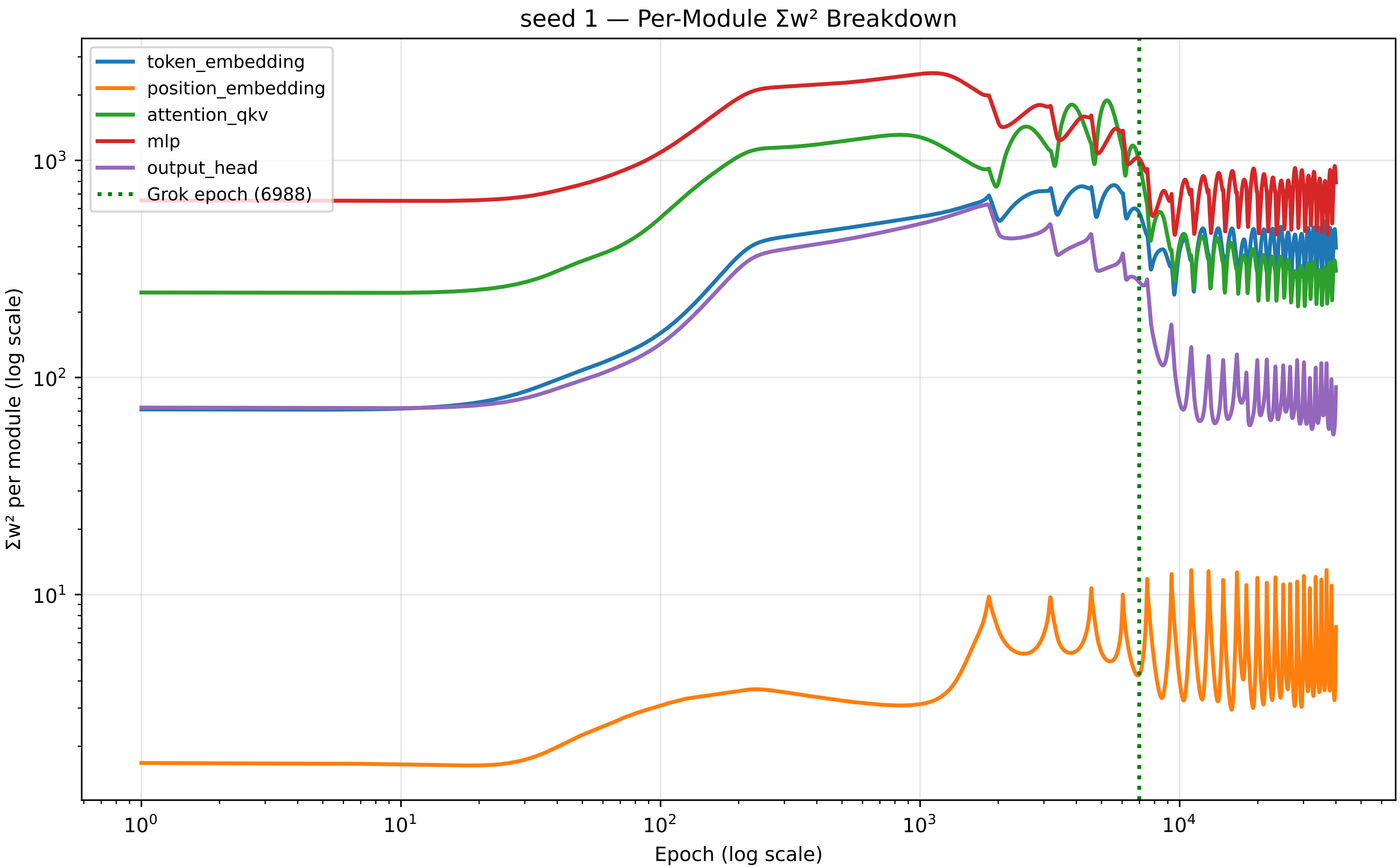


seed 1 — L2 Norm of Model Weights
L2 Norm
MA Crossover (epoch 121)
Grok epoch (6988)
70
65
60
55
50
45
40
35
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

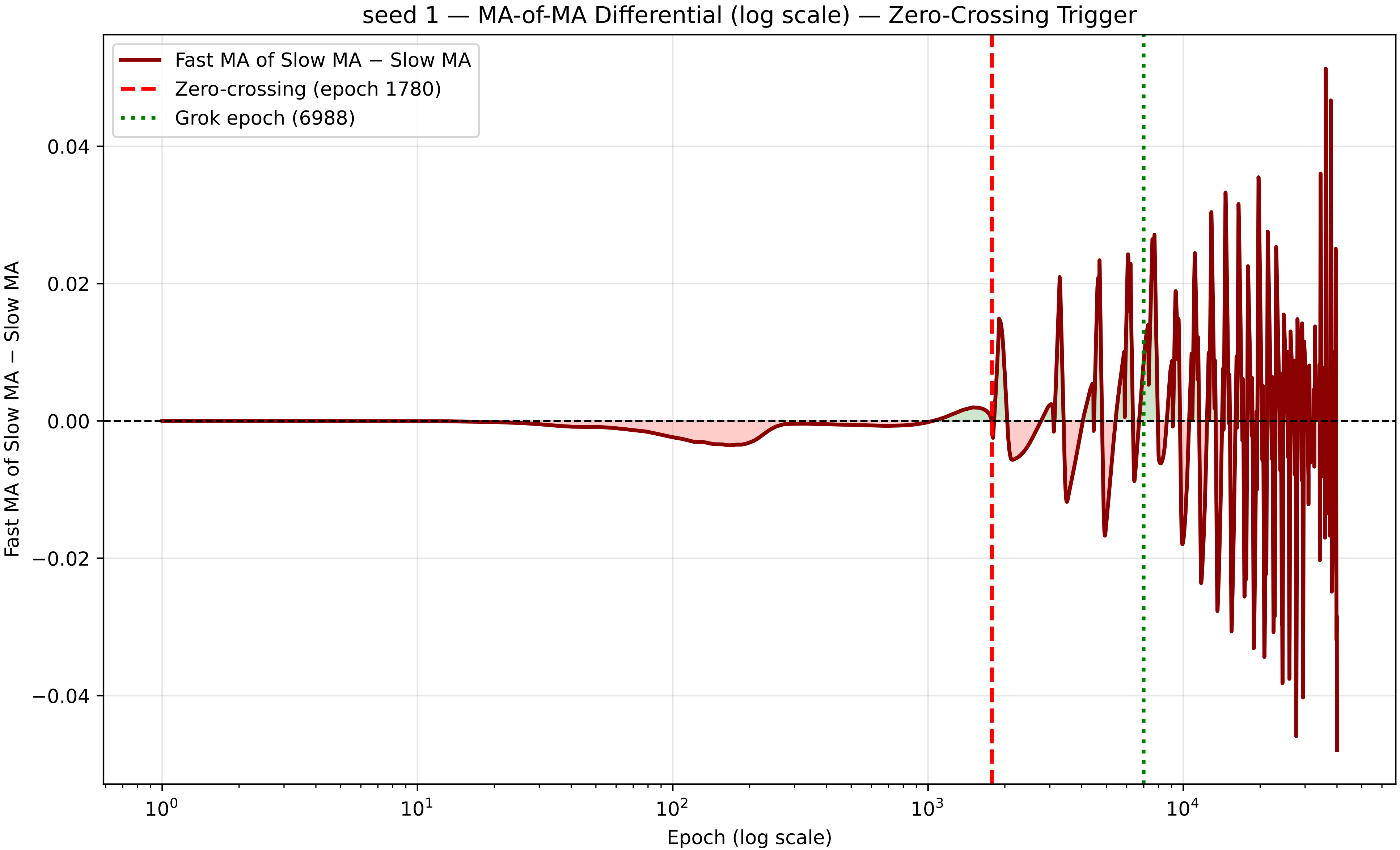

seed 1 — Moving Average Crossover Detection
Fast MA (w=50)
Slow MA (w=200)
Crossover (epoch 121)
Grok epoch (6988)
L2 Norm
Epoch (log scale)
70
65
60
55
50
45
40
35
10^0
10^1
10^2
10^3
10^4



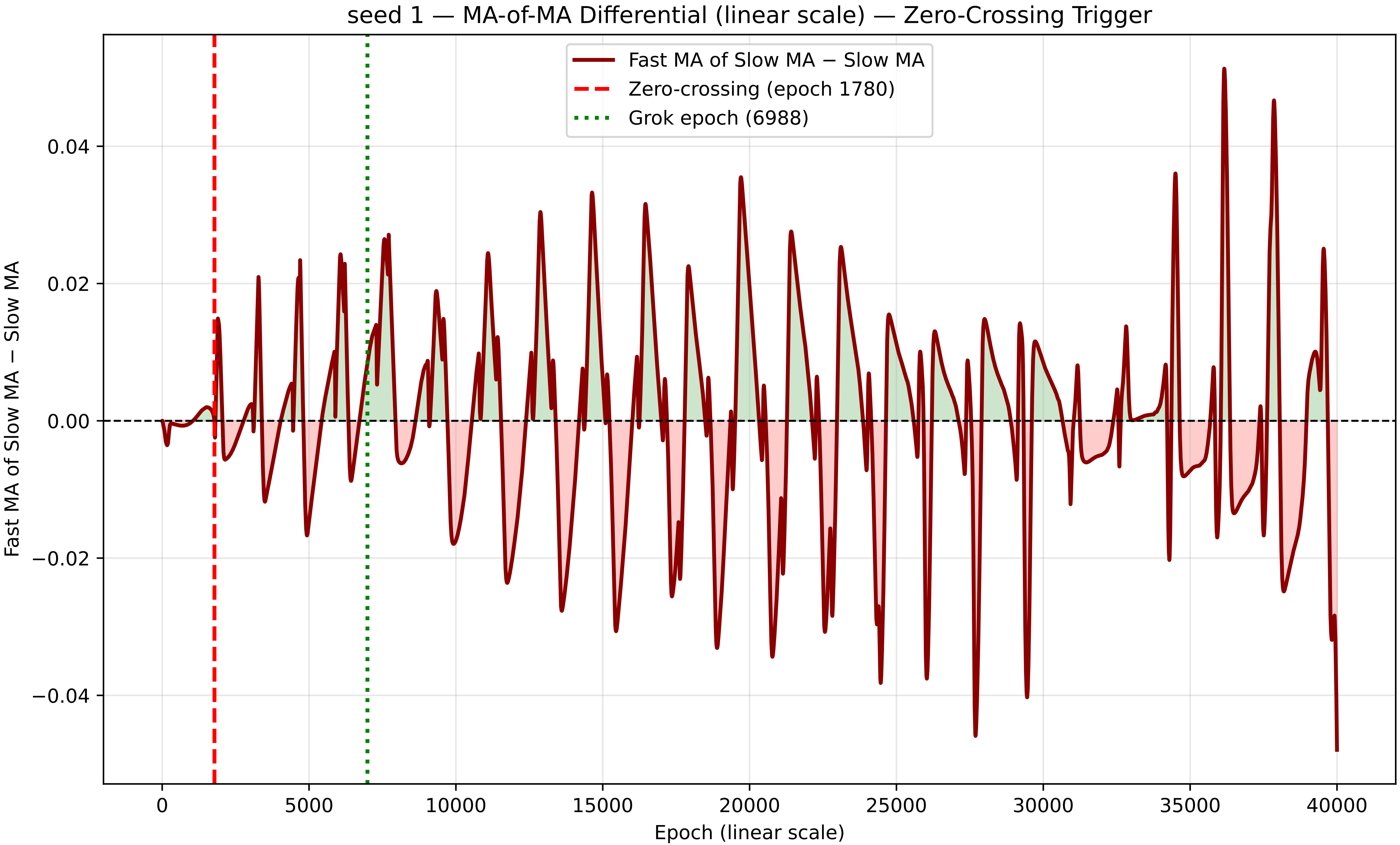
seed 1 — Sum of Squared Weights (Nanda Fig. 7 quantity)
Σw² (sum of squared weights)
MA Crossover (epoch 121)
Grok epoch (6988)
5000
4500
4000
3500
3000
2500
2000
1500
1000
Σw²
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



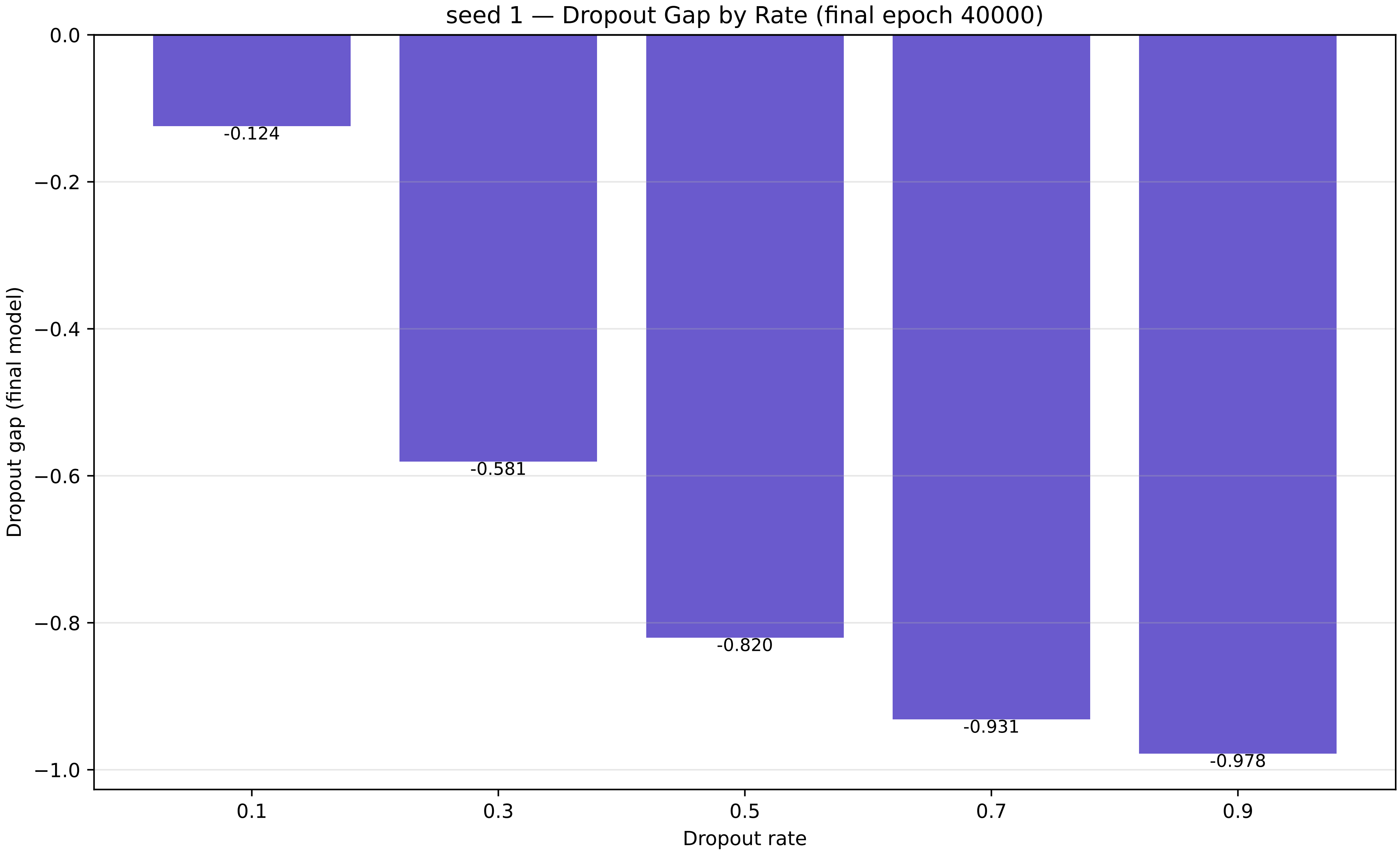
seed 1 — Per-Module Σw² Breakdown
token_embedding
position_embedding
attention_qkv
mlp
output_head
Grok epoch (6988)
Σw² per module (log scale)
Epoch (log scale)
10^3
10^2
10^1
10^0
10^1
10^2
10^3
10^4

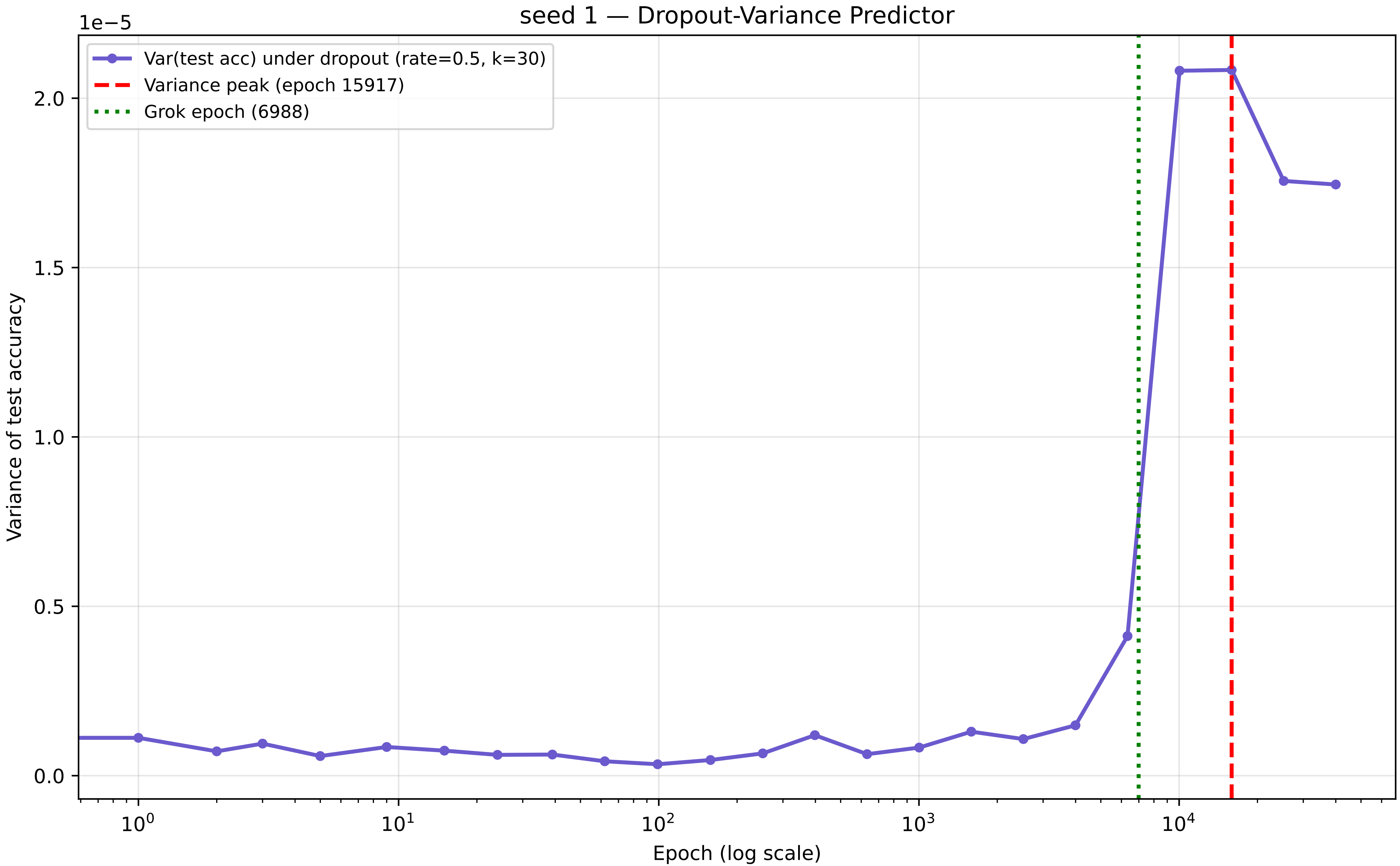


seed 1 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1780)
Grok epoch (6988)
0.04
0.02
0.00
−0.02
−0.04
Fast MA of Slow MA − Slow MA
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



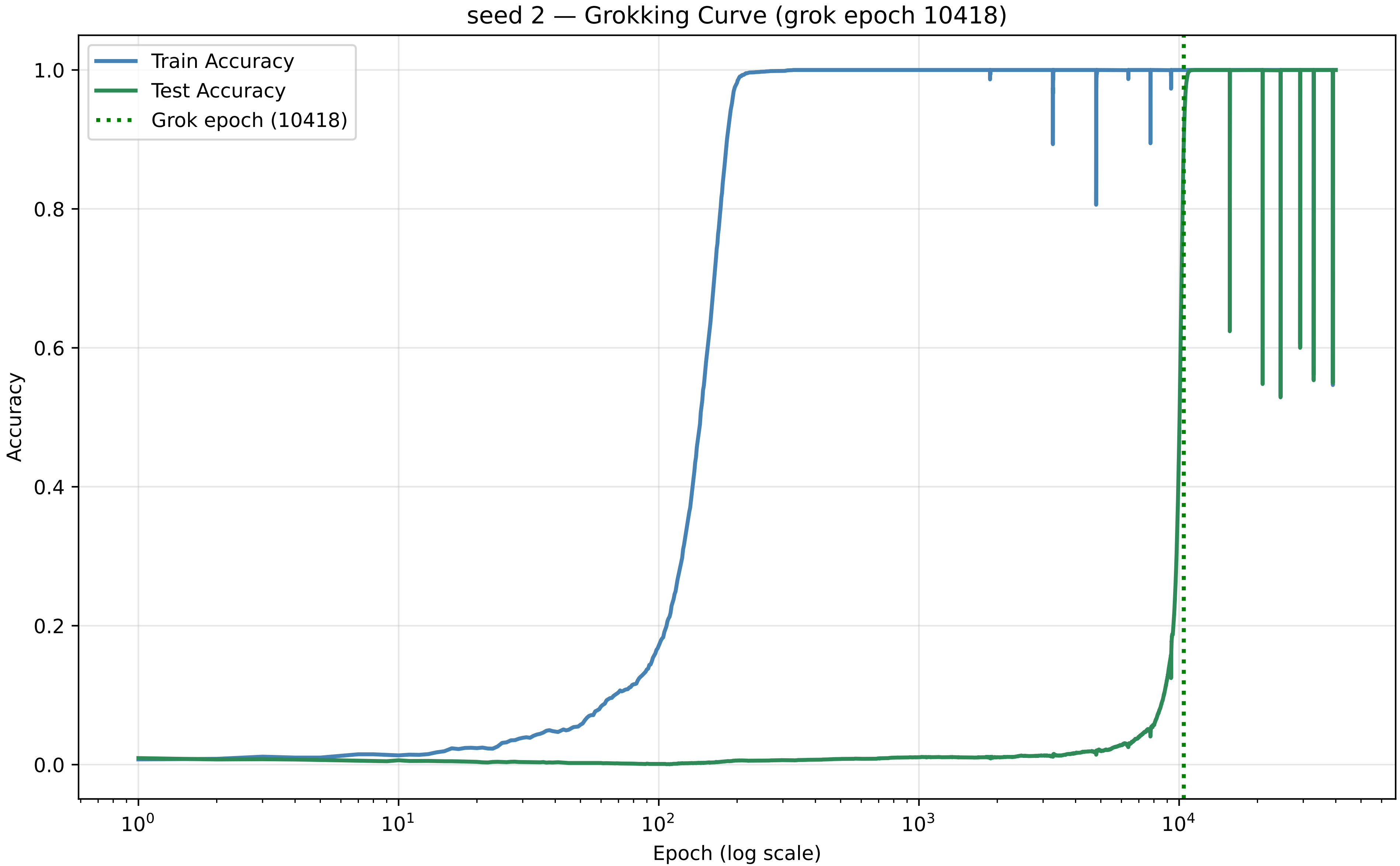
seed 1 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1780)
Grok epoch (6988)
Fast MA of Slow MA − Slow MA
0.04
0.02
0.00
−0.02
−0.04
0
5000
10000
15000
20000
25000
30000
35000
40000
Epoch (linear scale)

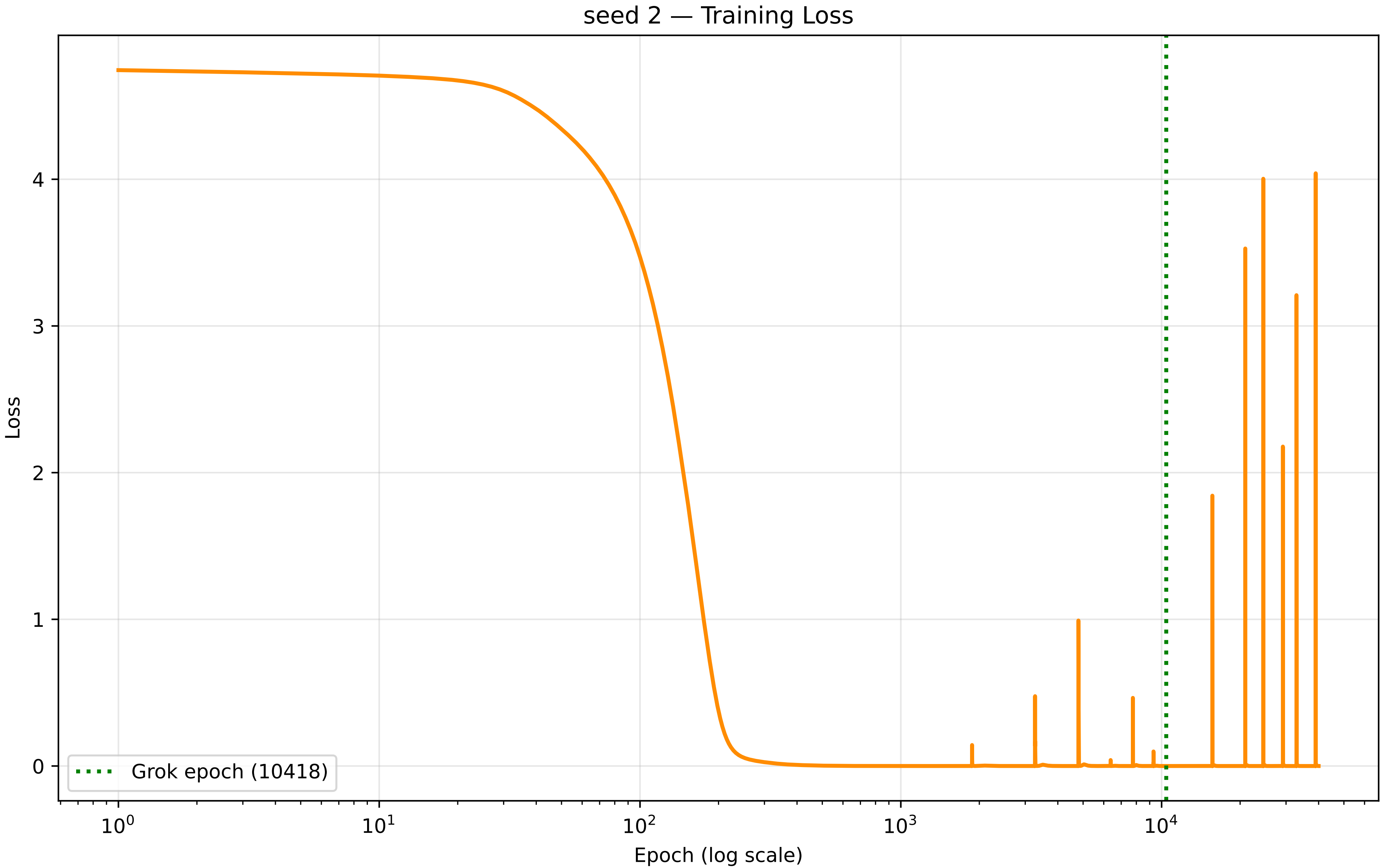

seed 1 — Dropout Gap by Rate (final epoch 40000)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
Dropout gap (final model)
-0.124
-0.581
-0.820
-0.931
-0.978
0.1
0.3
0.5
0.7
0.9
Dropout rate

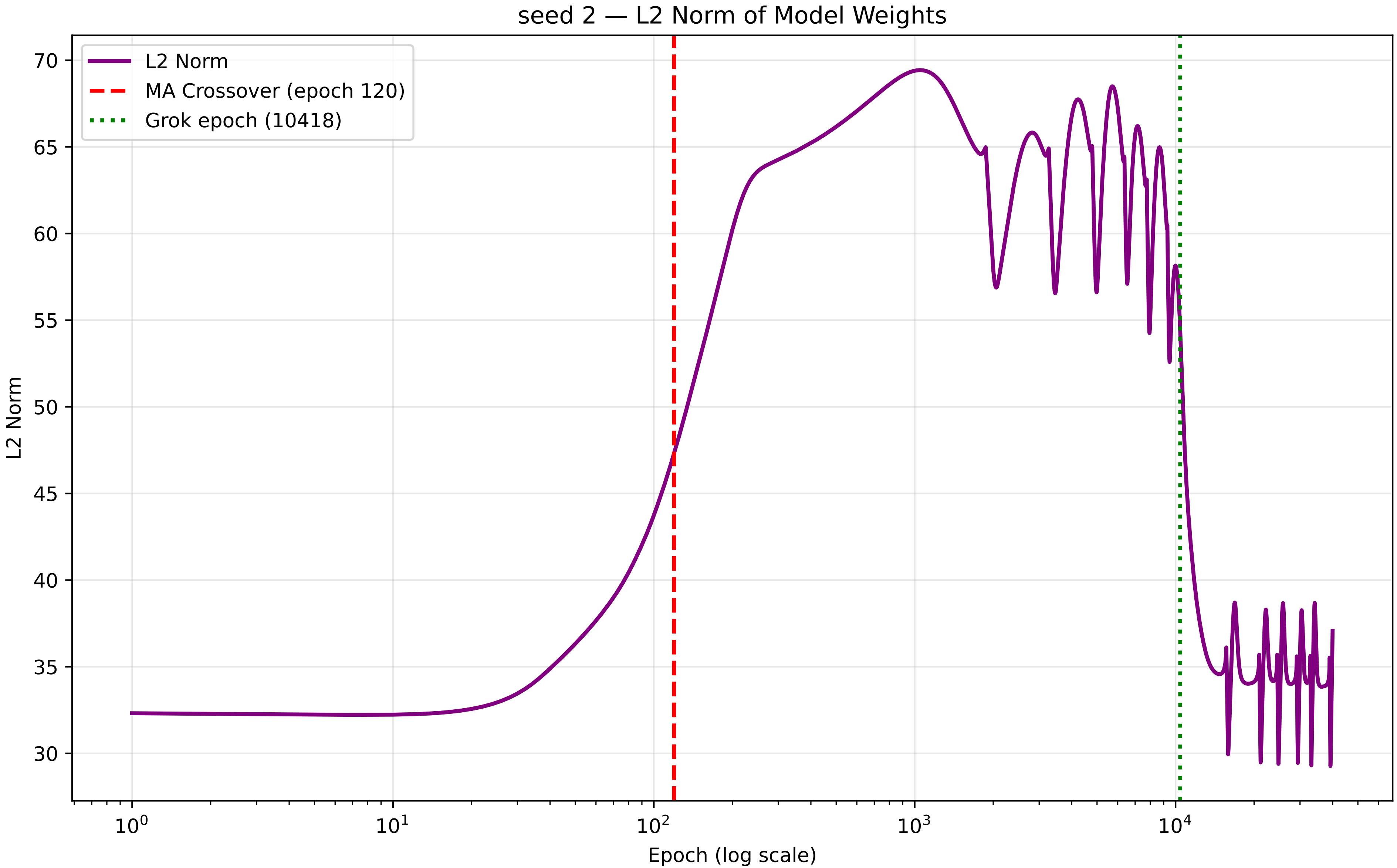

seed 1 — Dropout-Variance Predictor
1e−5
Var(test acc) under dropout (rate=0.5, k=30)
Variance peak (epoch 15917)
Grok epoch (6988)
Variance of test accuracy
0.0
0.5
1.0
1.5
2.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

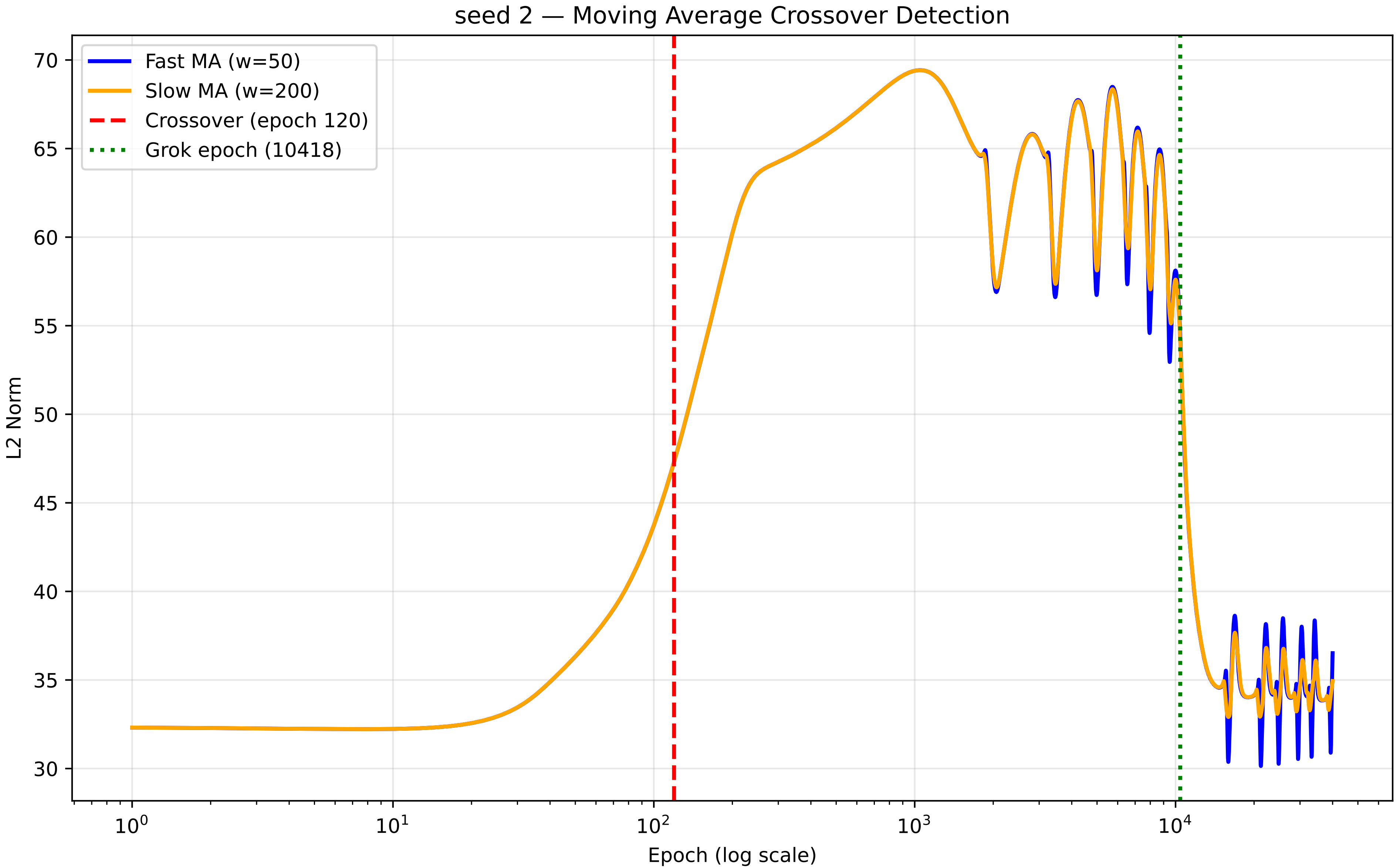
seed 2 — full numbers

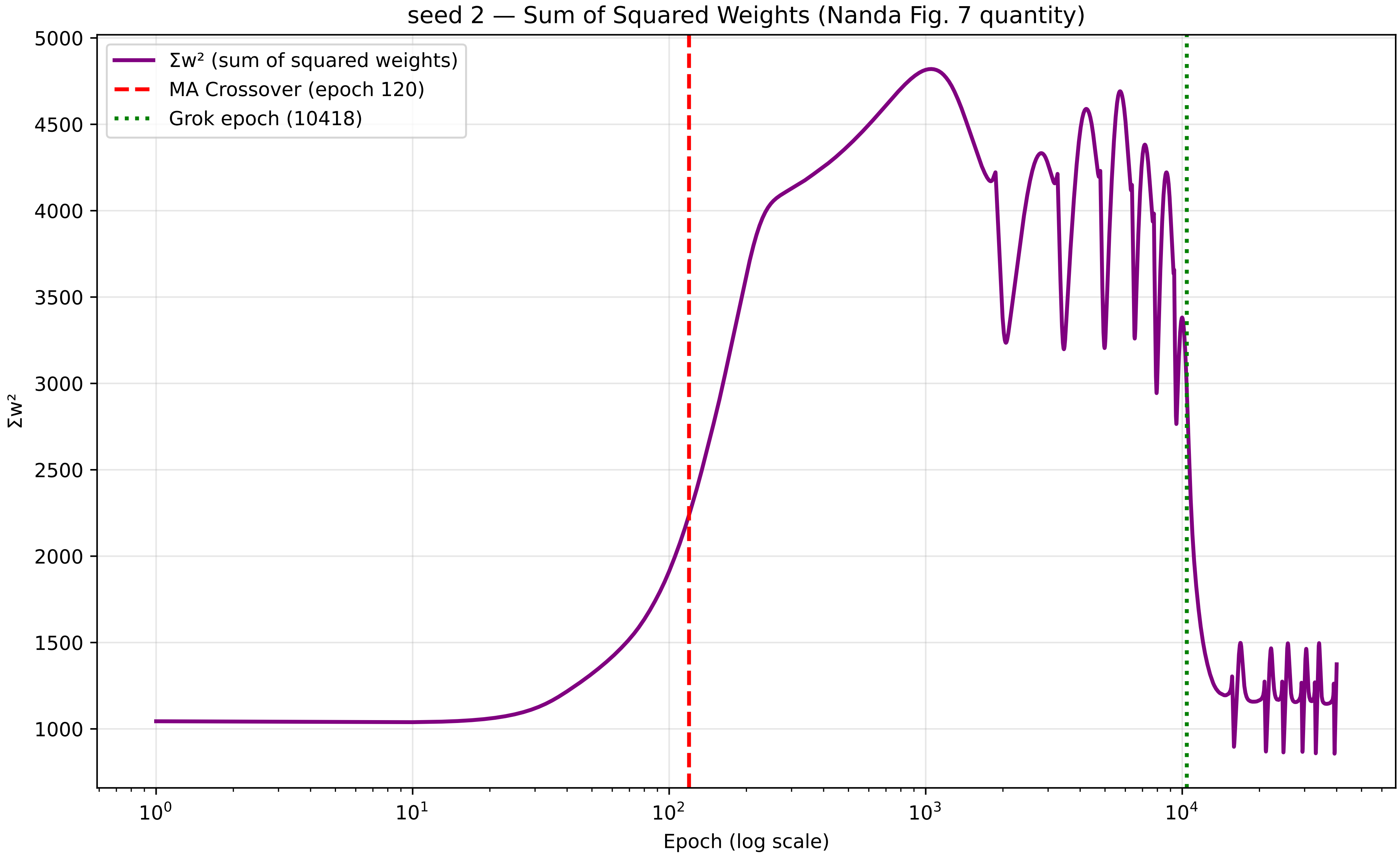
```
seed 2  (modulus=113  epochs=40000  wall_time=4952.6s)

  grok_epoch                    : 10418
  final_train_acc               : 1.0000
  final_test_acc                : 1.0000
  l2_norm   init -> final       : 32.3136 -> 37.0680
  sum_w2    init -> final       : 1044.168 -> 1374.034
  token_embedding_share (init)  : 0.0693

  L2 predictor:
    MA-crossover epoch           : 119.51
    MA-of-MA zero-crossing epoch : 1802.22
    noise_floor                  : 0.000464
    windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

  Dropout gap by rate:
    p0.1=-0.0029  p0.3=-0.1118  p0.5=-0.4696  p0.7=-0.8156  p0.9=-0.9678

  Dropout-Variance predictor:
    variance_peak_epoch          : 39999
    peak_to_grok_ratio            : 3.8394
    rate=0.5  n_samples=30  num_checkpoints=24

  Limit-cycle check             : stable
    window_start=10918  post_grok_min=0.5288  post_grok_std=0.0103
    post_grok_final=1.0000  dips<0.9=34
```

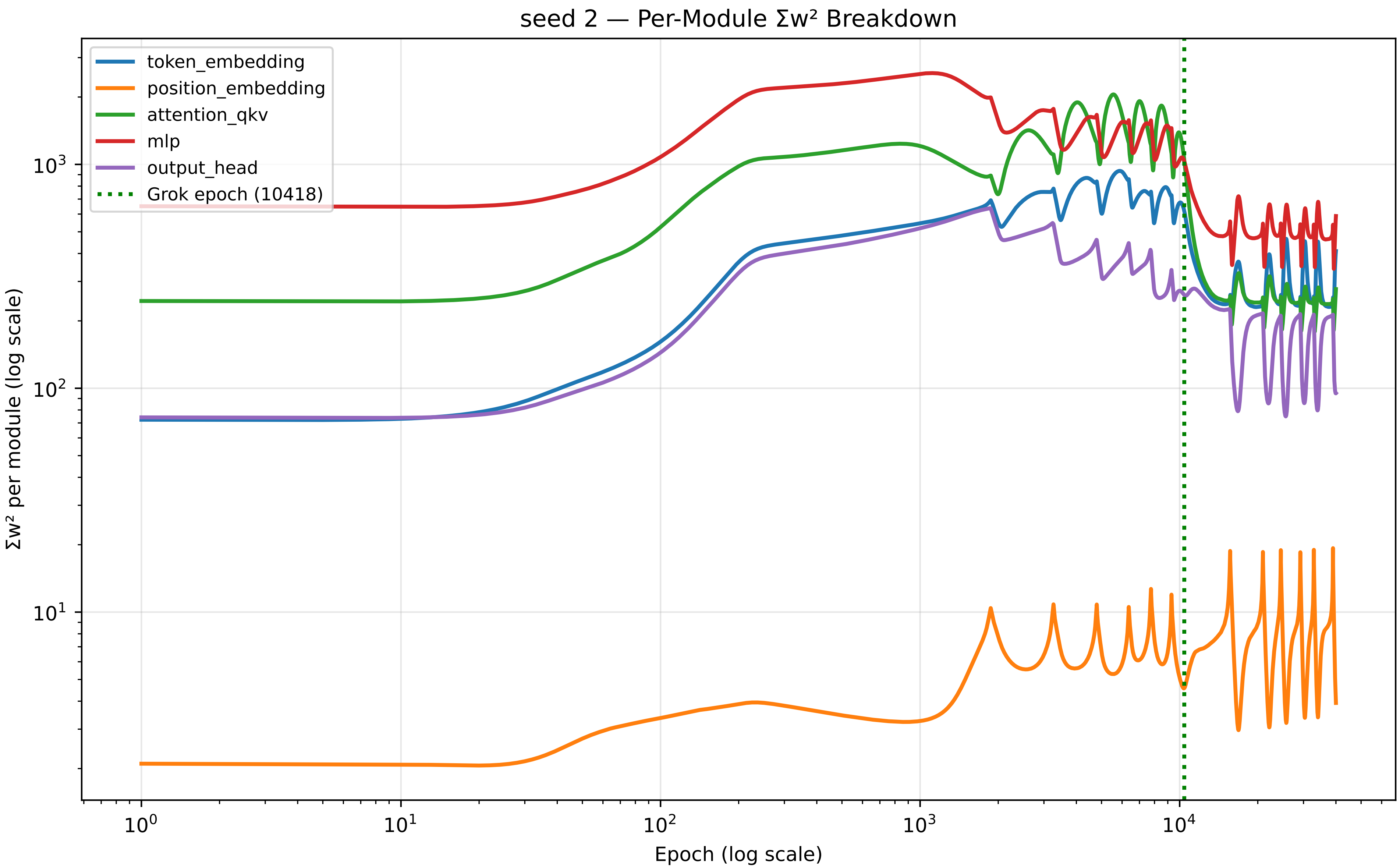

seed 2 — Grokking Curve (grok epoch 10418)
Train Accuracy
Test Accuracy
Grok epoch (10418)
Accuracy
0.0
0.2
0.4
0.6
0.8
1.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)


seed 2 — Training Loss
Loss
Epoch (log scale)
Grok epoch (10418)
0
1
2
3
4
10^0
10^1
10^2
10^3
10^4

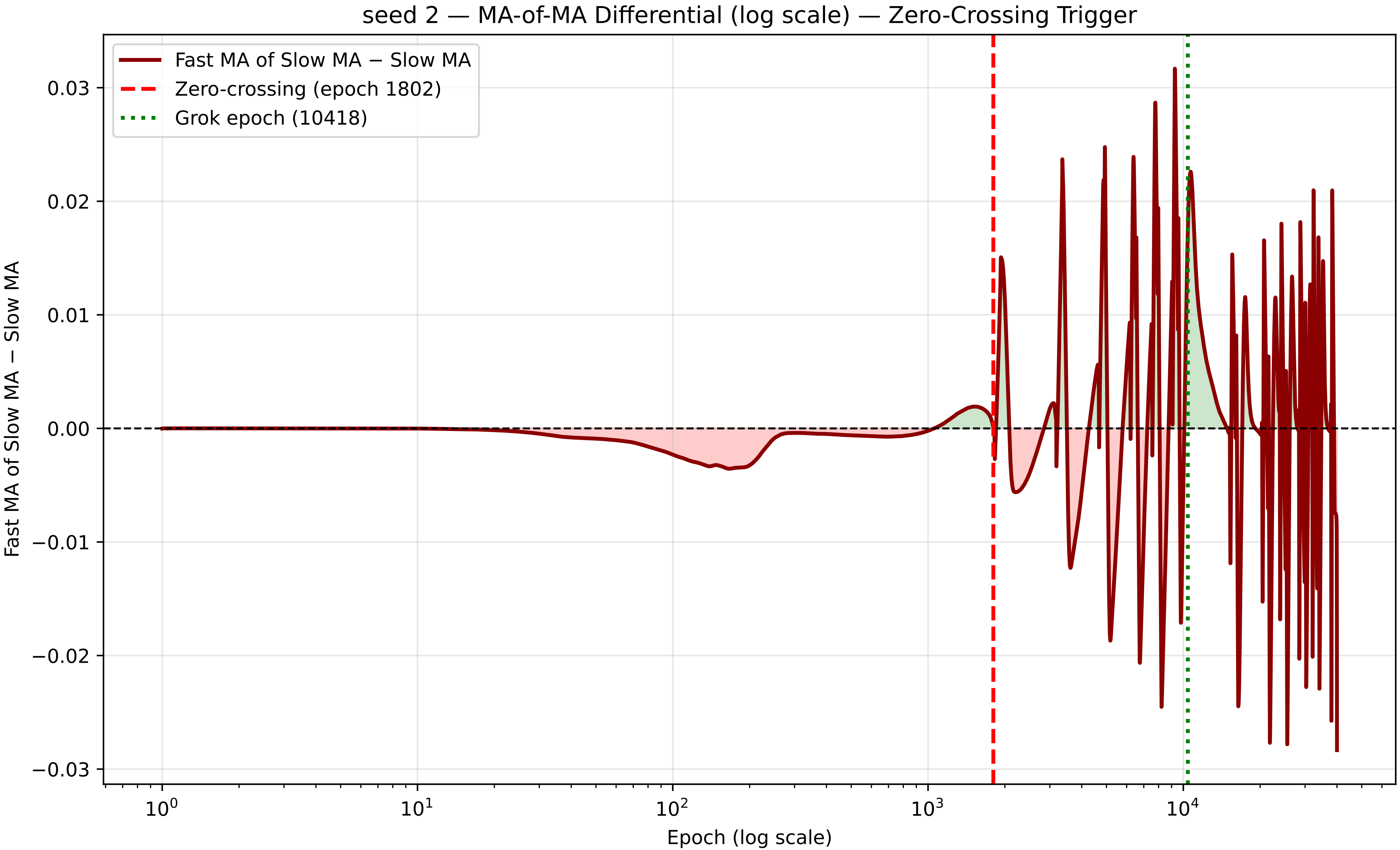

seed 2 — L2 Norm of Model Weights
L2 Norm
MA Crossover (epoch 120)
Grok epoch (10418)
70
65
60
55
50
45
40
35
30
L2 Norm
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)


seed 2 — Moving Average Crossover Detection
Fast MA (w=50)
Slow MA (w=200)
Crossover (epoch 120)
Grok epoch (10418)
L2 Norm
Epoch (log scale)
70
65
60
55
50
45
40
35
30
10^0
10^1
10^2
10^3
10^4

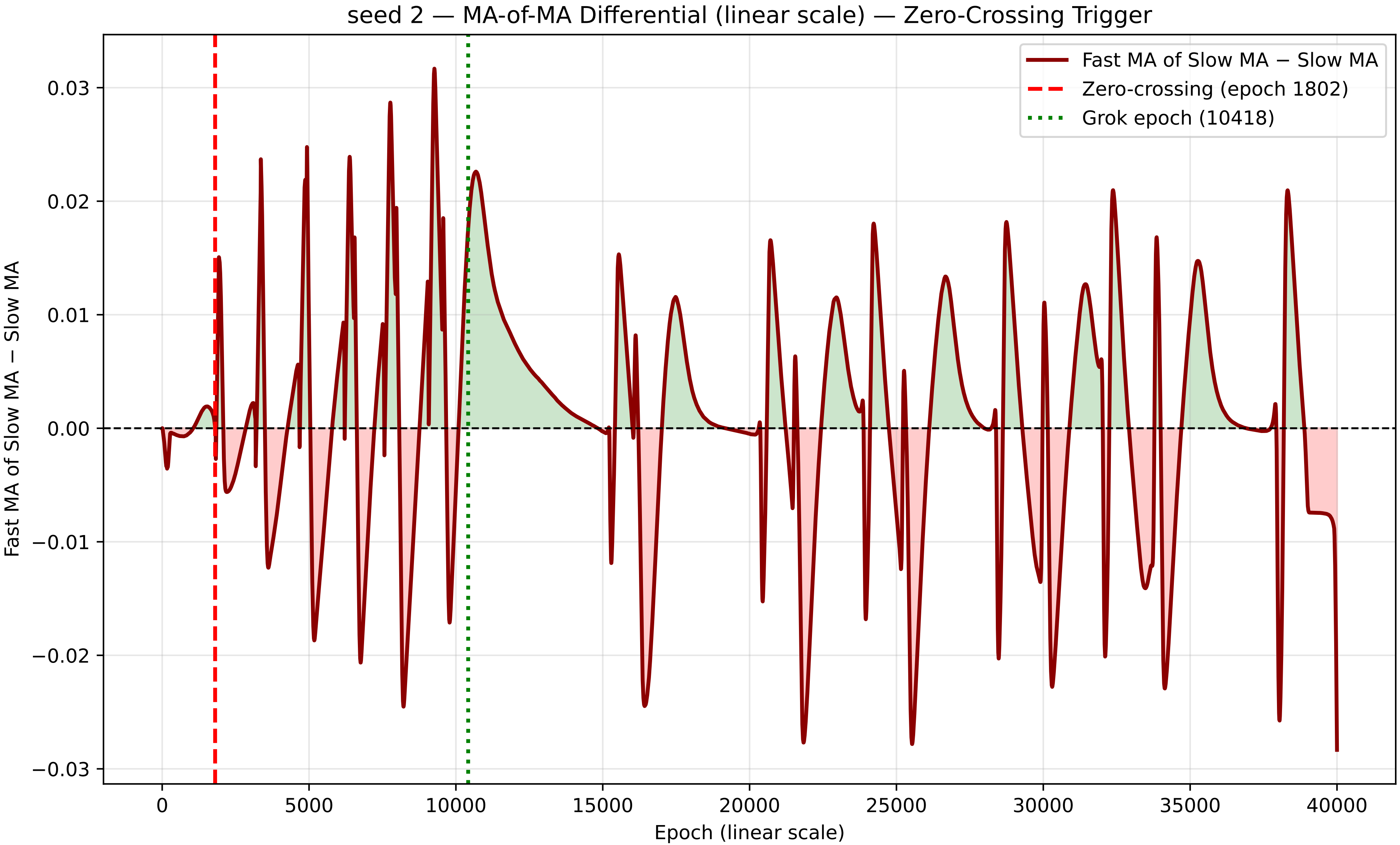

seed 2 — Sum of Squared Weights (Nanda Fig. 7 quantity)
Σw² (sum of squared weights)
MA Crossover (epoch 120)
Grok epoch (10418)
Σw²
5000
4500
4000
3500
3000
2500
2000
1500
1000
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



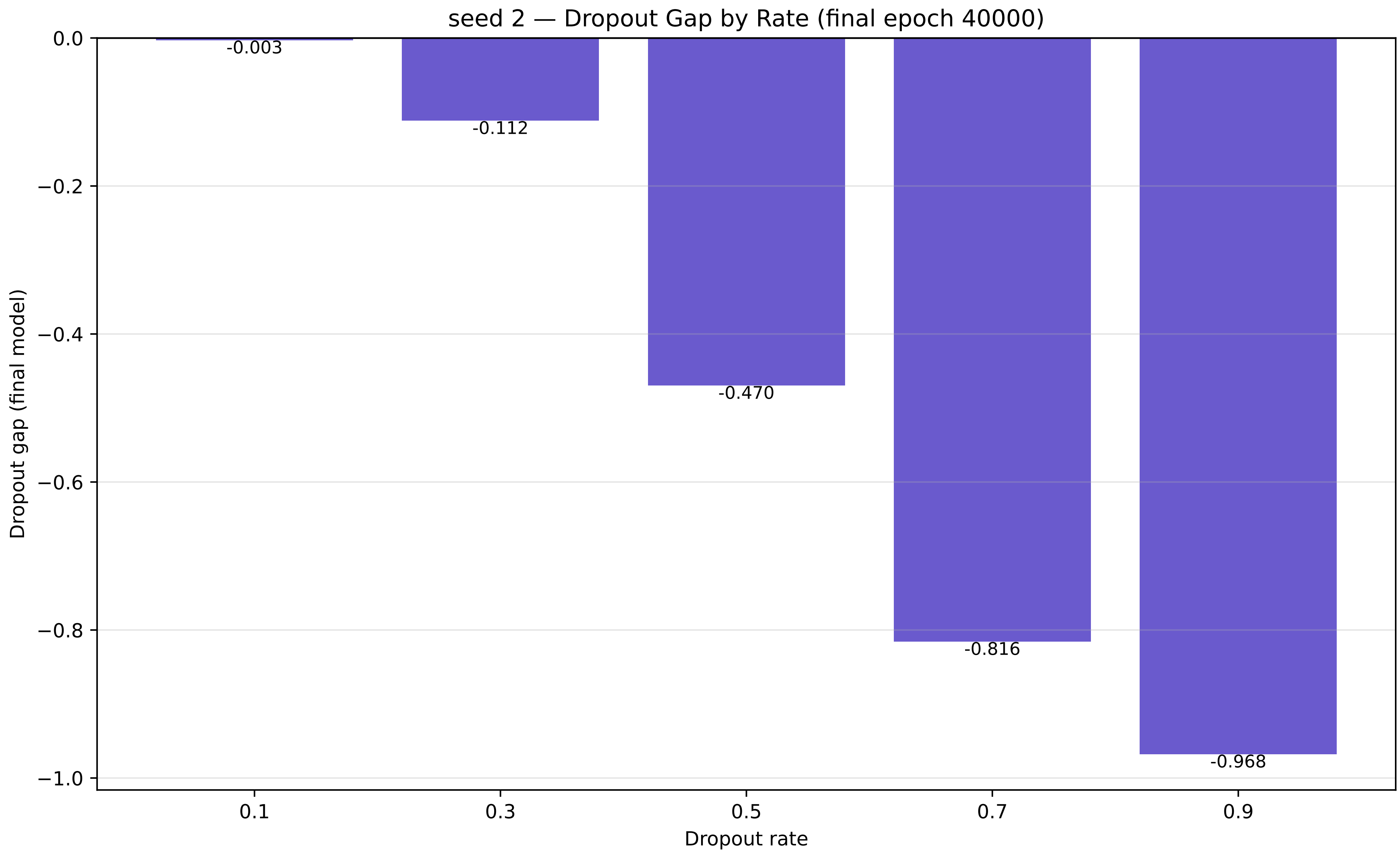
seed 2 — Per-Module Σw² Breakdown
token_embedding
position_embedding
attention_qkv
mlp
output_head
Grok epoch (10418)
Σw² per module (log scale)
Epoch (log scale)
10^3
10^2
10^1
10^0
10^1
10^2
10^3
10^4

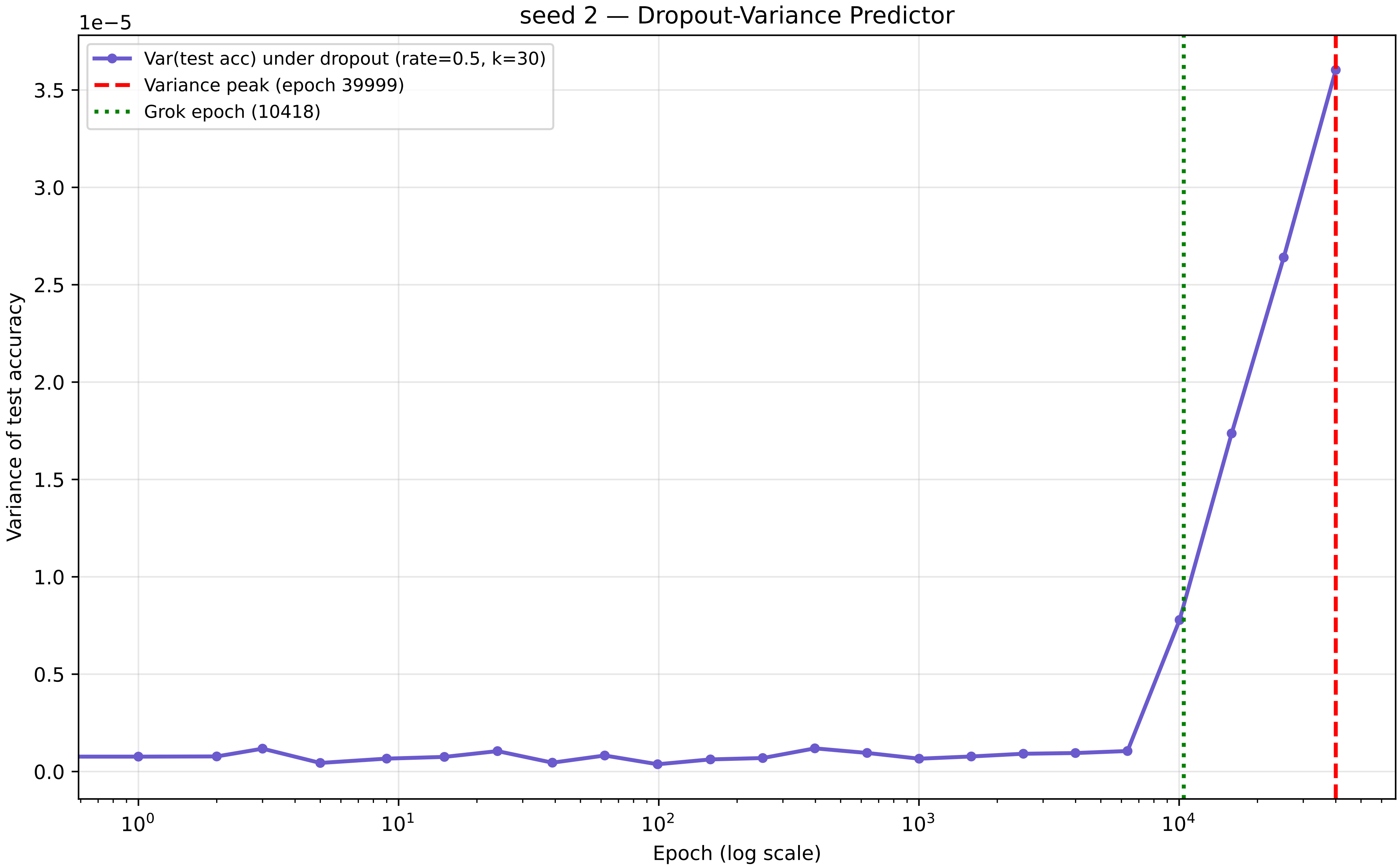


seed 2 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1802)
Grok epoch (10418)
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
Fast MA of Slow MA − Slow MA
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



seed 2 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1802)
Grok epoch (10418)
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
Fast MA of Slow MA − Slow MA
0
5000
10000
15000
20000
25000
30000
35000
40000
Epoch (linear scale)

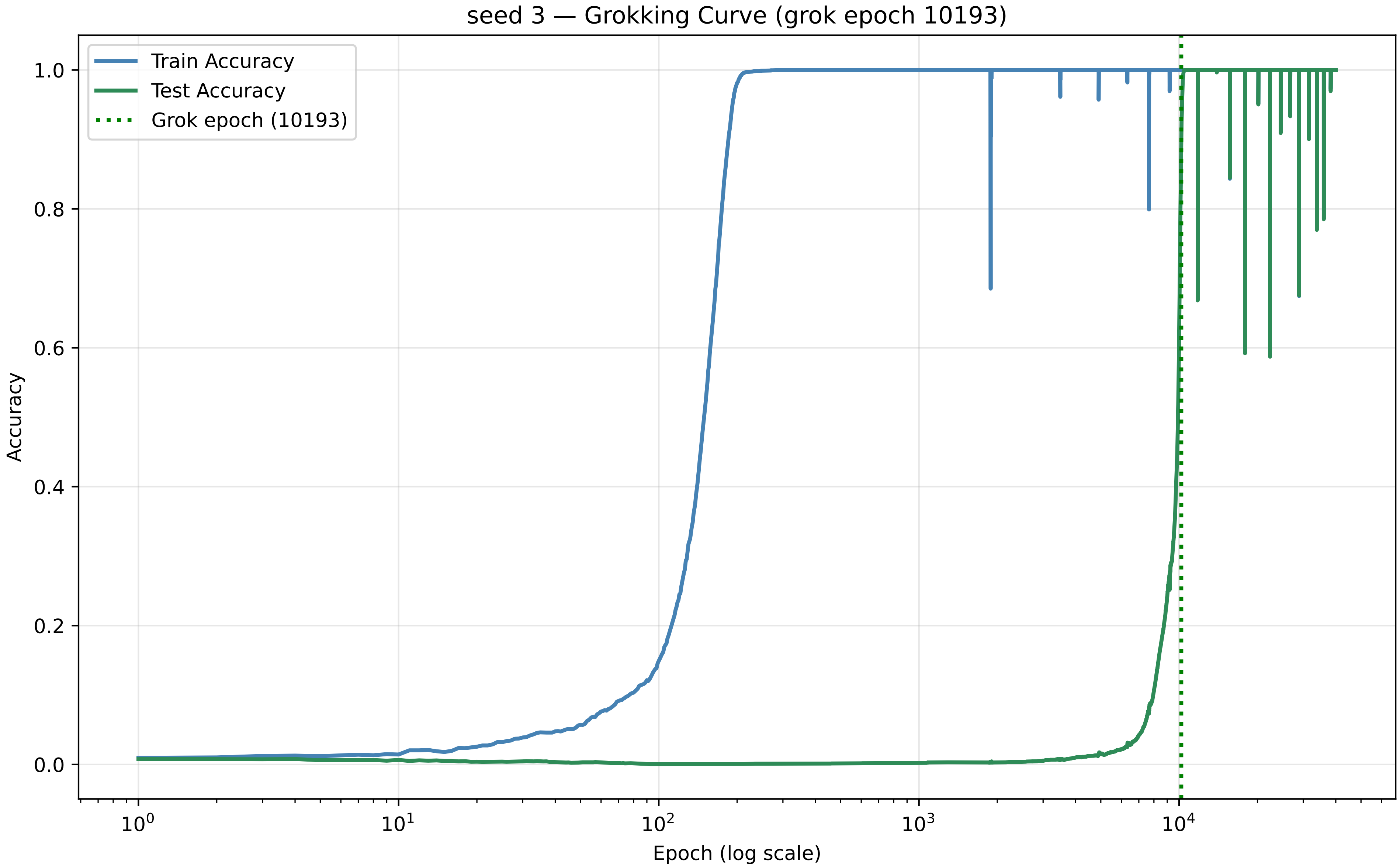

seed 2 — Dropout Gap by Rate (final epoch 40000)
Dropout gap (final model)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
-0.003
-0.112
-0.470
-0.816
-0.968
0.1
0.3
0.5
0.7
0.9
Dropout rate



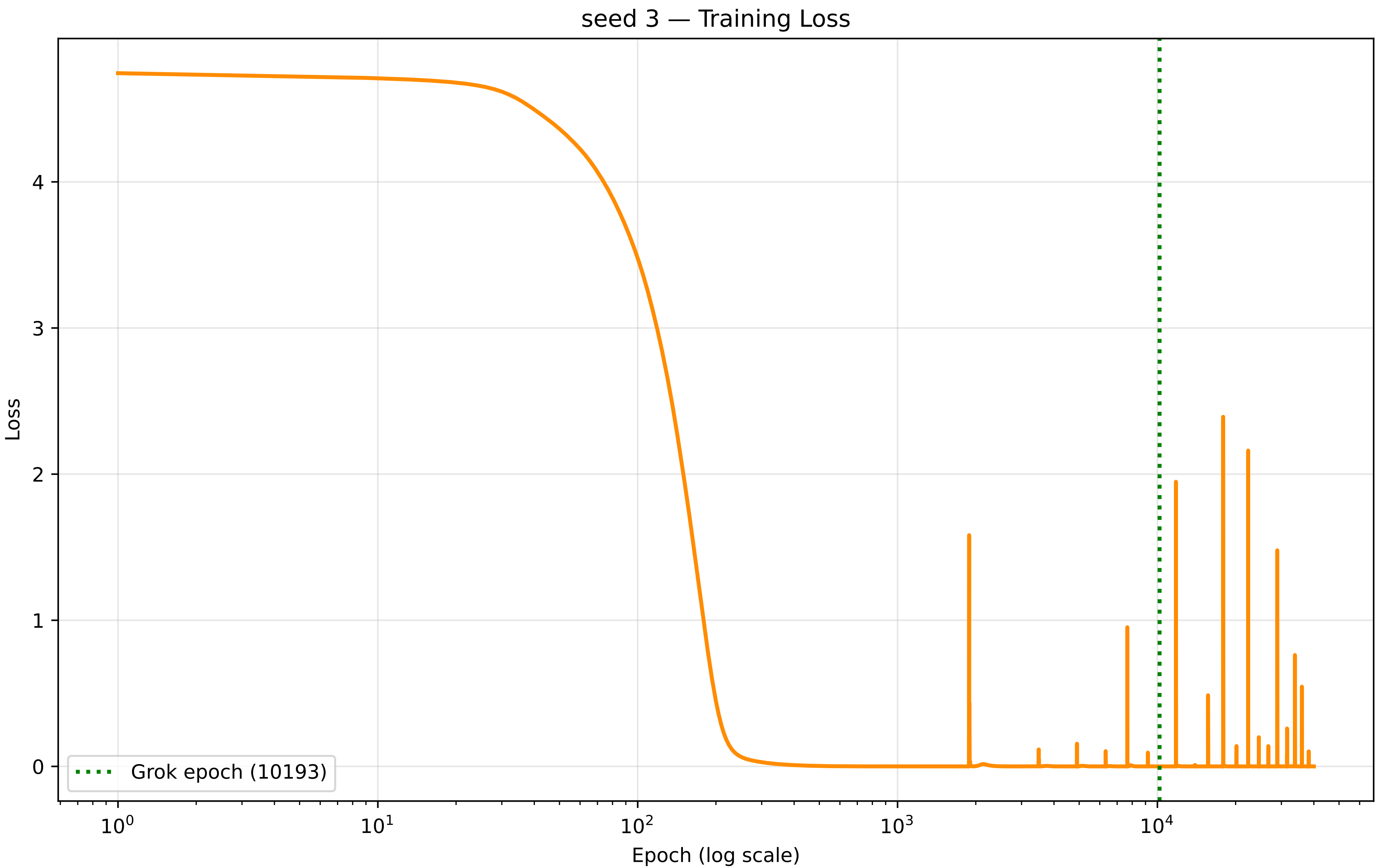
seed 2 — Dropout-Variance Predictor
1e−5
Var(test acc) under dropout (rate=0.5, k=30)
Variance peak (epoch 39999)
Grok epoch (10418)
Variance of test accuracy
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

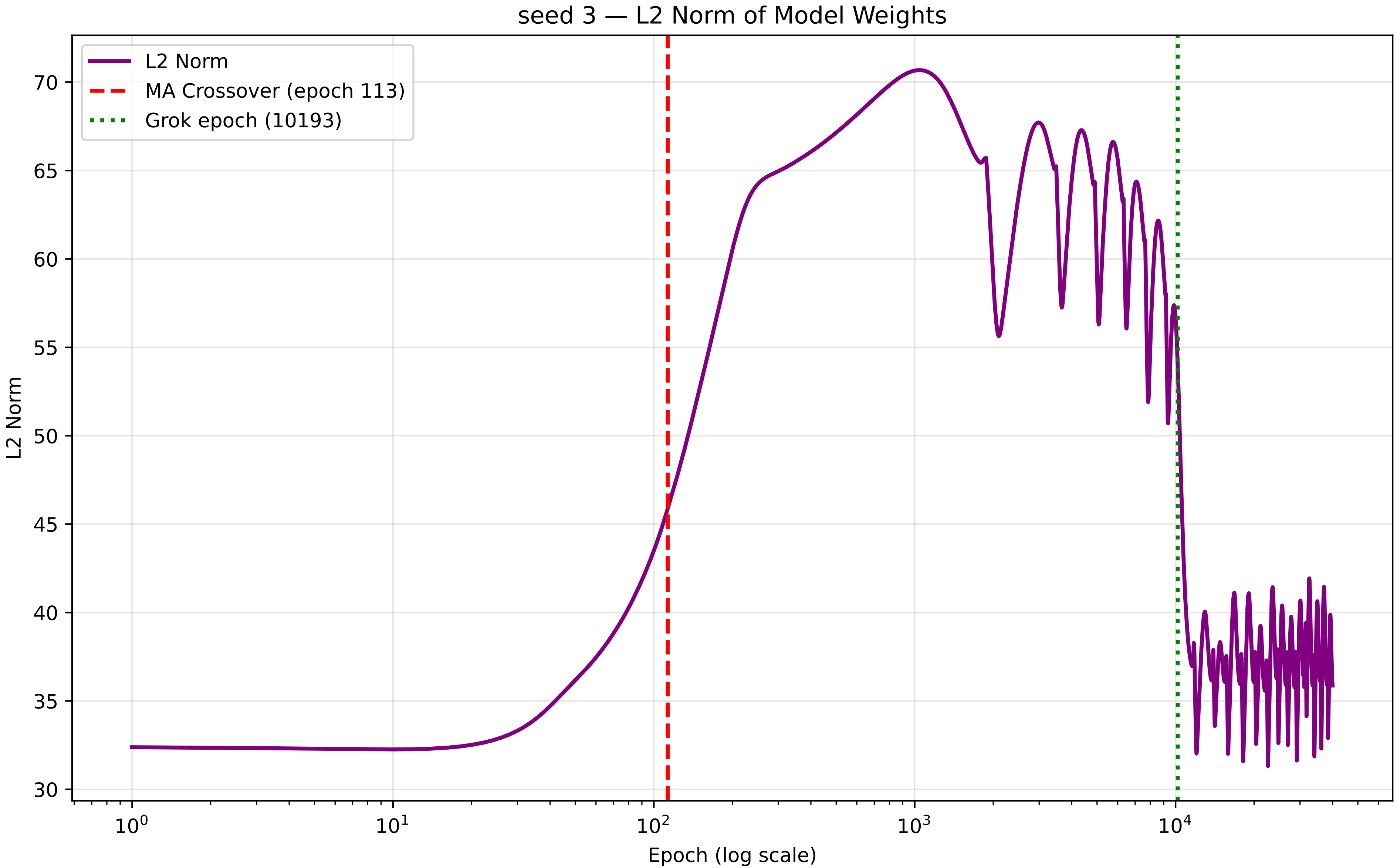
seed 3 — full numbers

```
seed 3  (modulus=113  epochs=40000  wall_time=4953.5s)

  grok_epoch                    : 10193
  final_train_acc               : 1.0000
  final_test_acc                : 1.0000
  l2_norm   init -> final       : 32.3835 -> 35.8760
  sum_w2    init -> final       : 1048.690 -> 1287.091
  token_embedding_share (init)  : 0.0680

  L2 predictor:
    MA-crossover epoch           : 112.99
    MA-of-MA zero-crossing epoch : 1800.32
    noise_floor                  : 0.000463
    windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

  Dropout gap by rate:
    p0.1=-0.0012  p0.3=-0.1231  p0.5=-0.5380  p0.7=-0.8758  p0.9=-0.9784

  Dropout-Variance predictor:
    variance_peak_epoch          : 39999
    peak_to_grok_ratio            : 3.9242
    rate=0.5  n_samples=30  num_checkpoints=24

  Limit-cycle check             : stable
    window_start=10693  post_grok_min=0.5872  post_grok_std=0.0070
    post_grok_final=1.0000  dips<0.9=20
```

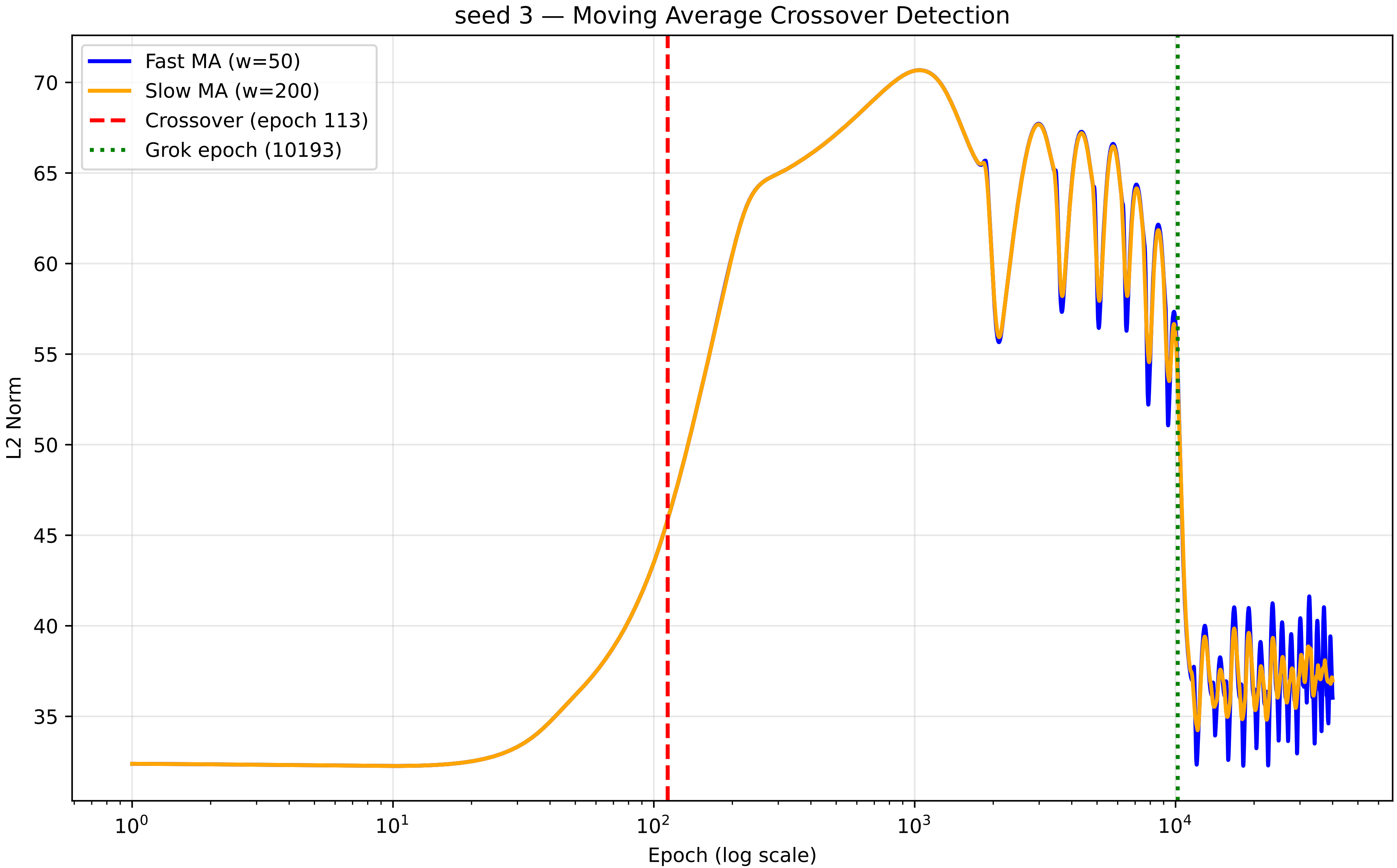


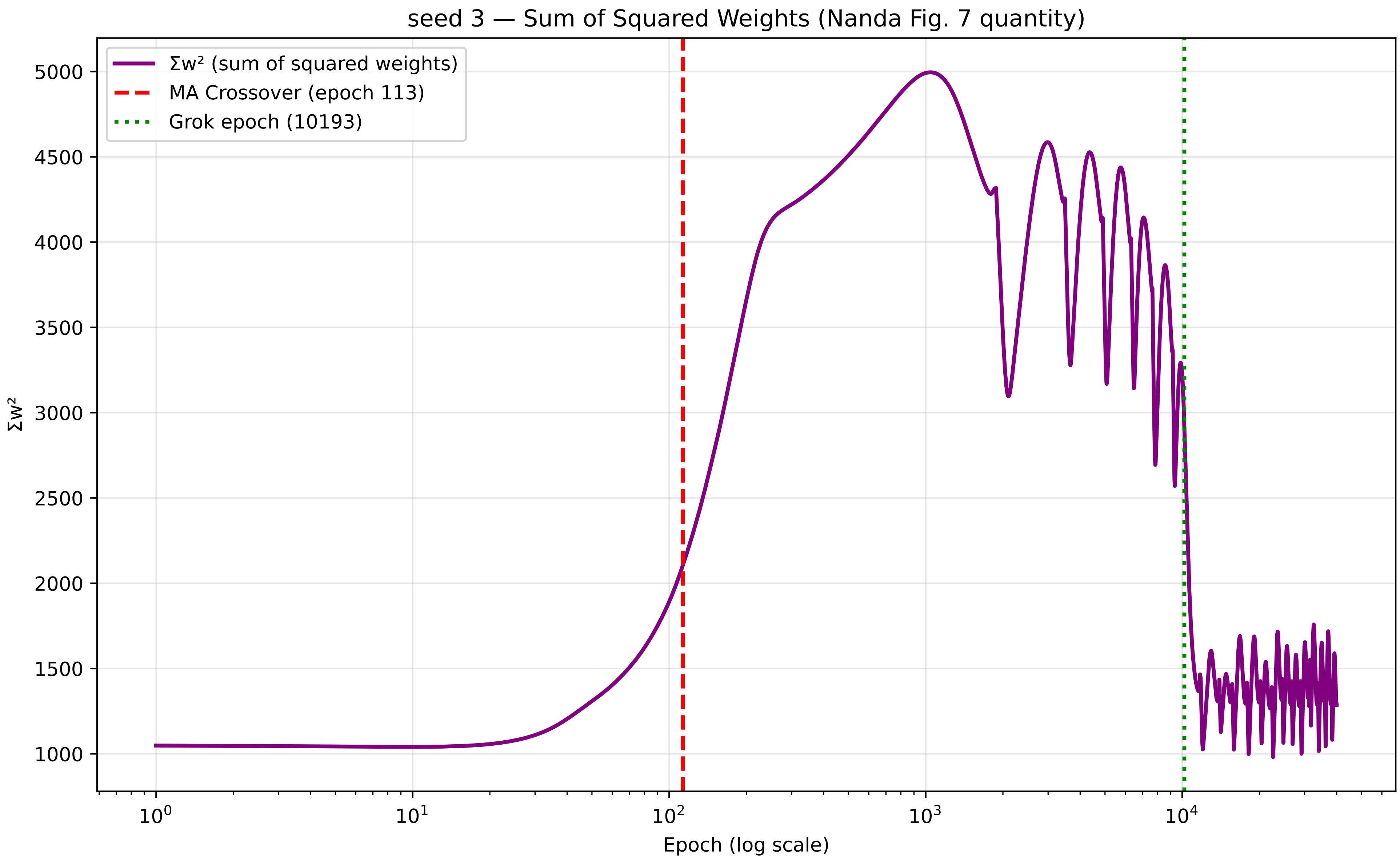
seed 3 — Grokking Curve (grok epoch 10193)
Train Accuracy
Test Accuracy
Grok epoch (10193)
Accuracy
Epoch (log scale)
0.0
0.2
0.4
0.6
0.8
1.0
10^0
10^1
10^2
10^3
10^4

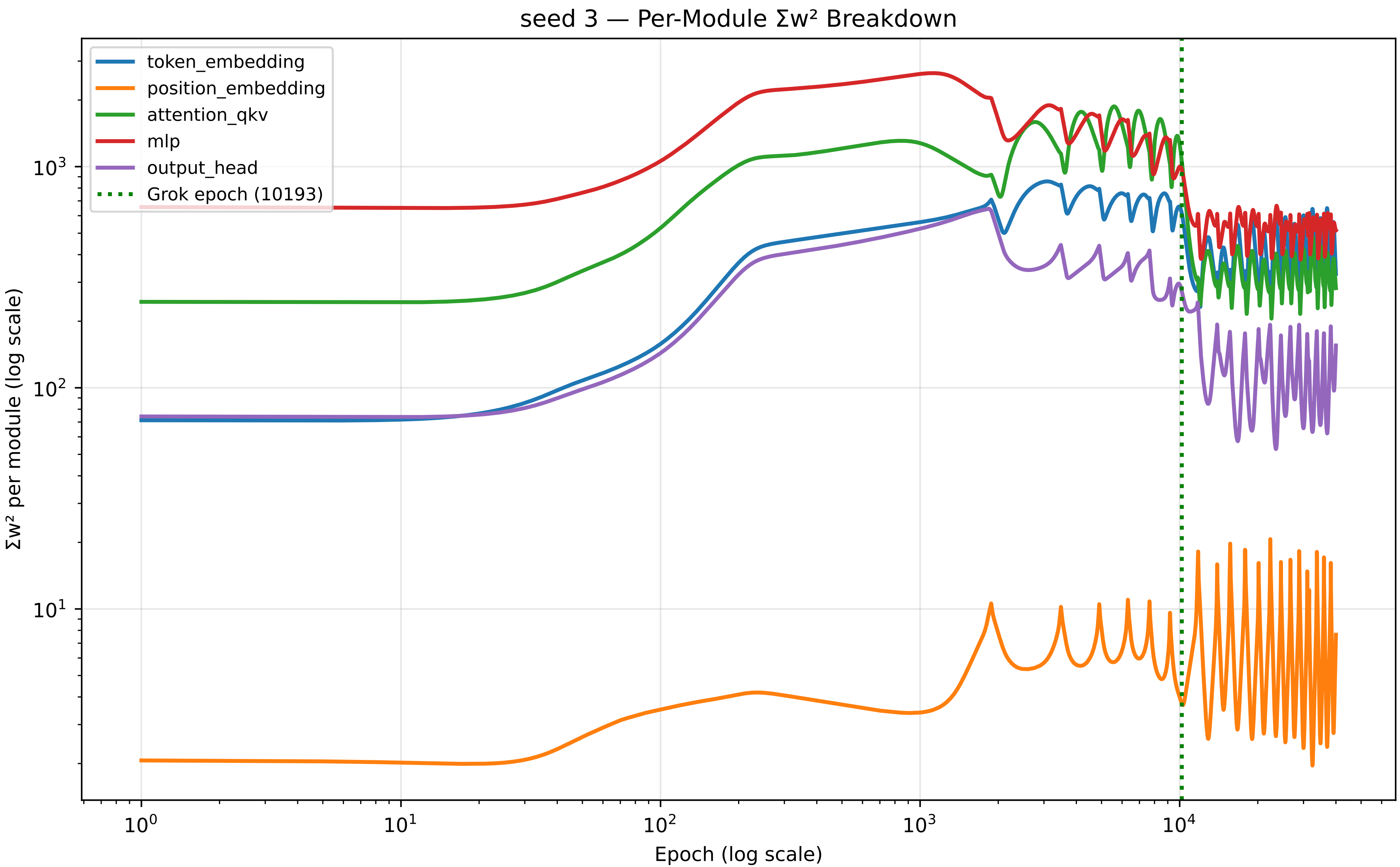

seed 3 — Training Loss
Loss
Epoch (log scale)
Grok epoch (10193)
0
1
2
3
4
10^0
10^1
10^2
10^3
10^4

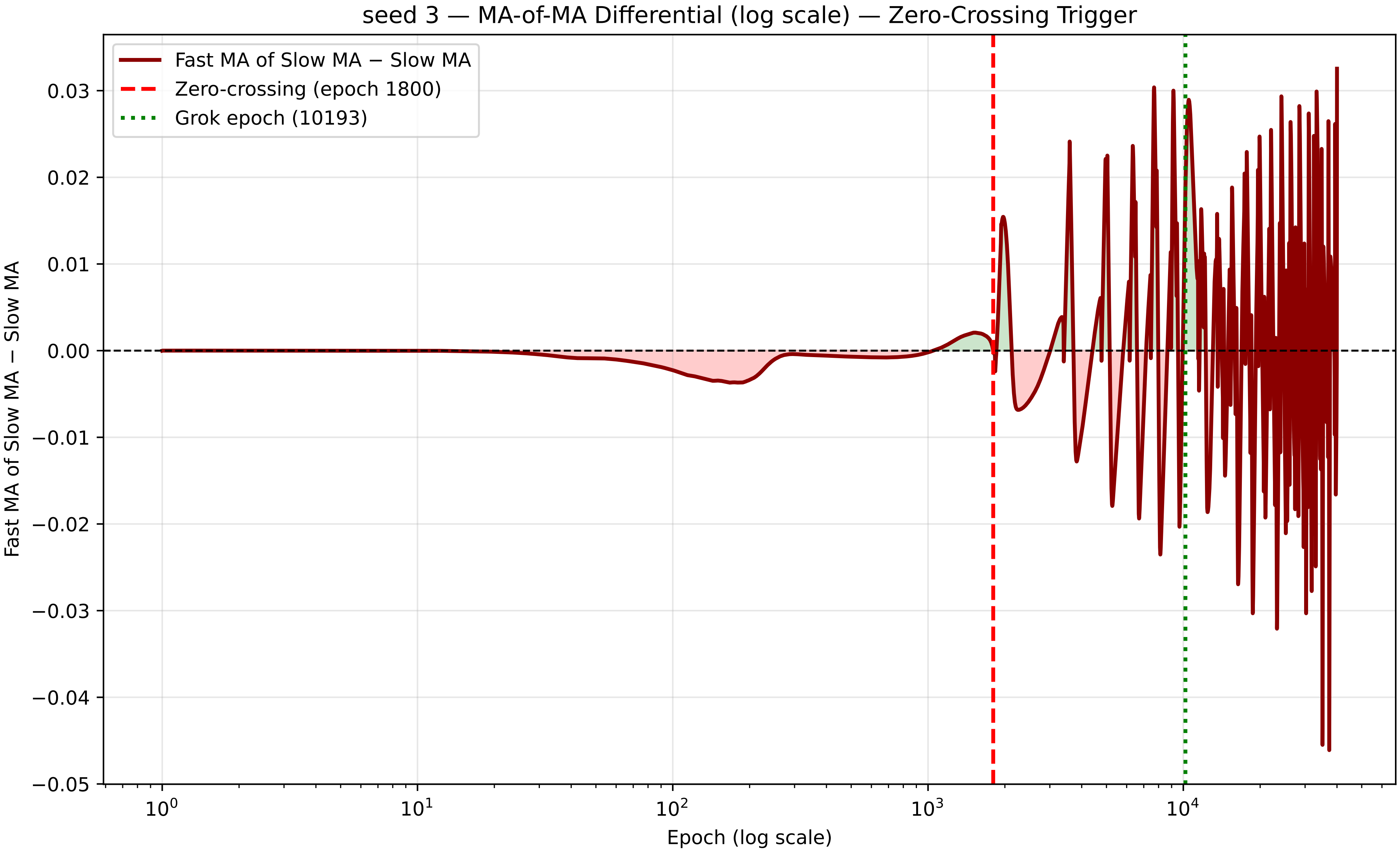


seed 3 — L2 Norm of Model Weights
L2 Norm
MA Crossover (epoch 113)
Grok epoch (10193)
70
65
60
55
50
45
40
35
30
L2 Norm
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)

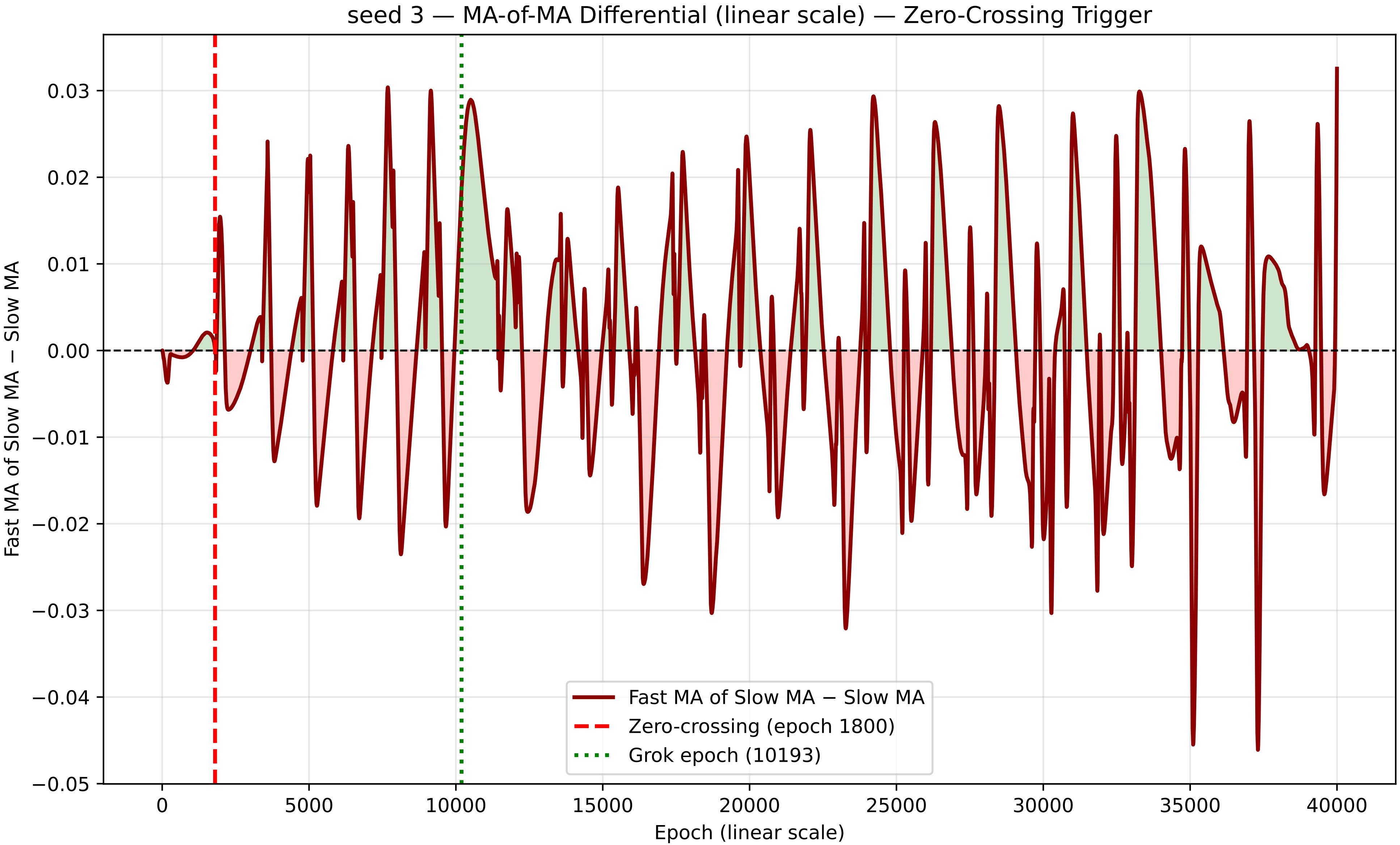

seed 3 — Moving Average Crossover Detection
Fast MA (w=50)
Slow MA (w=200)
Crossover (epoch 113)
Grok epoch (10193)
L2 Norm
Epoch (log scale)
70
65
60
55
50
45
40
35
10^0
10^1
10^2
10^3
10^4


seed 3 — Sum of Squared Weights (Nanda Fig. 7 quantity)
Σw² (sum of squared weights)
MA Crossover (epoch 113)
Grok epoch (10193)
Σw²
Epoch (log scale)
1000
1500
2000
2500
3000
3500
4000
4500
5000
10⁰
10¹
10²
10³
10⁴

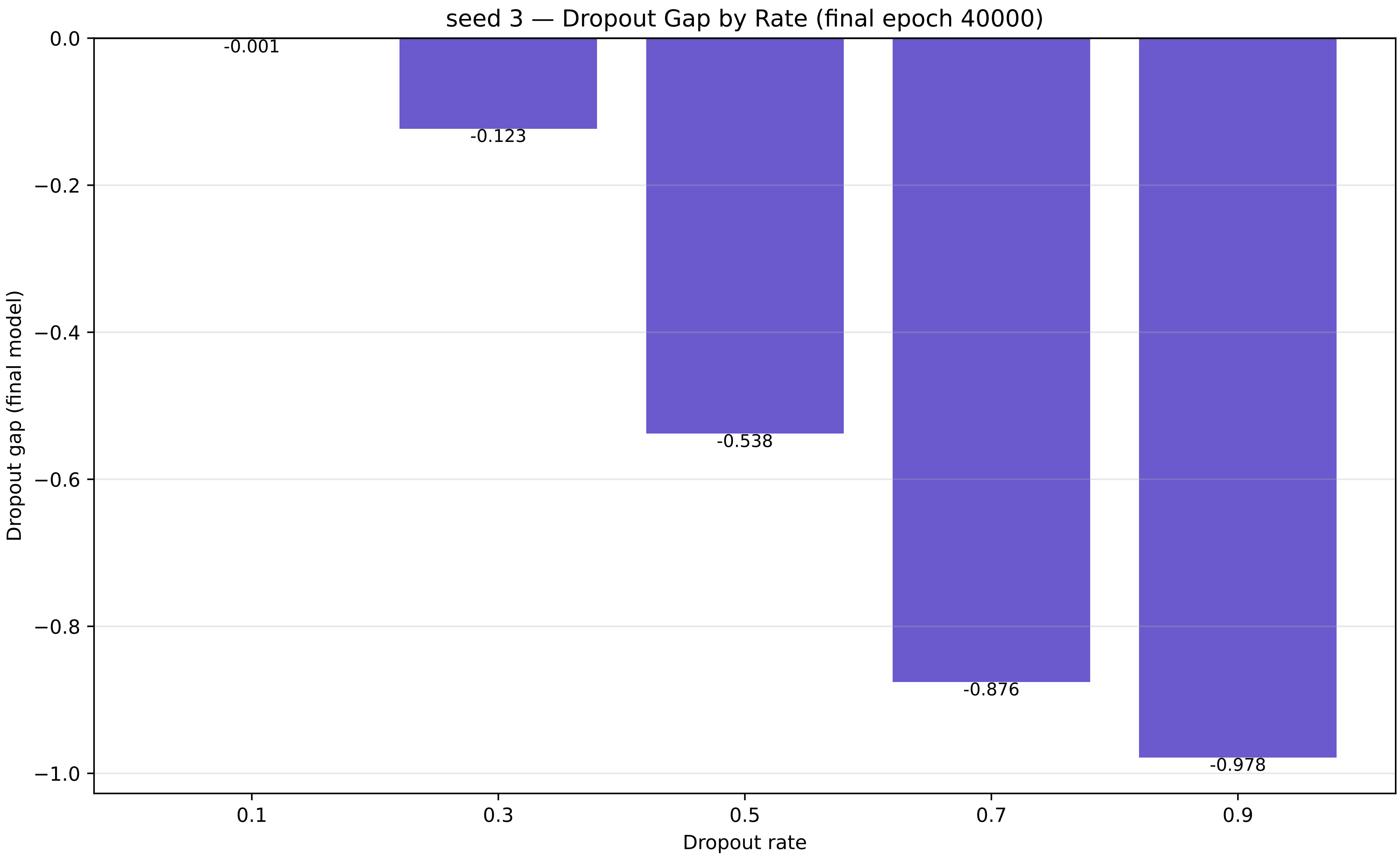

seed 3 — Per-Module Σw² Breakdown
token_embedding
position_embedding
attention_qkv
mlp
output_head
Grok epoch (10193)
Σw² per module (log scale)
Epoch (log scale)
10^0
10^1
10^2
10^3
10^4



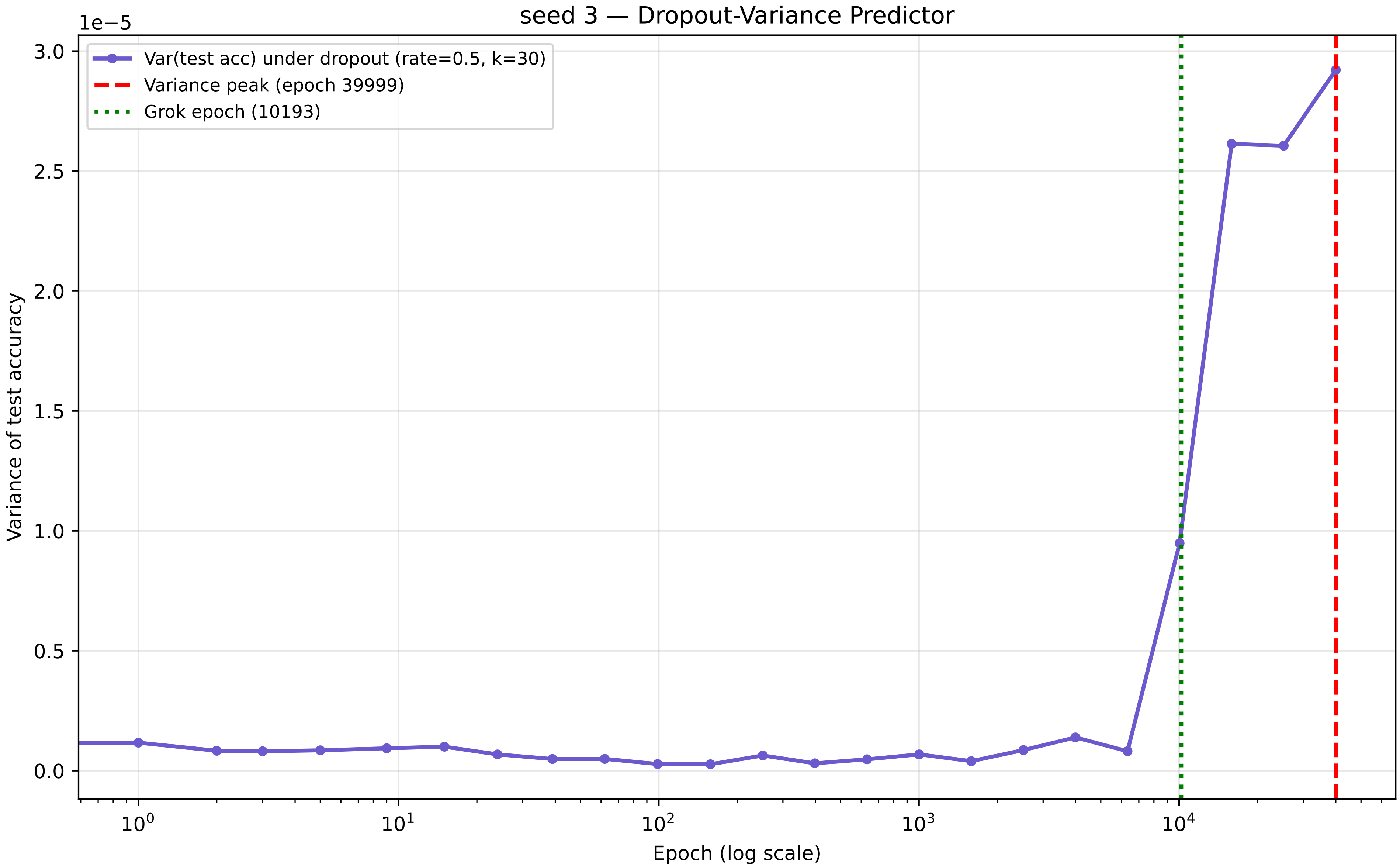
seed 3 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1800)
Grok epoch (10193)
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
Fast MA of Slow MA − Slow MA
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



seed 3 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Epoch (linear scale)
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
0
5000
10000
15000
20000
25000
30000
35000
40000
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1800)
Grok epoch (10193)

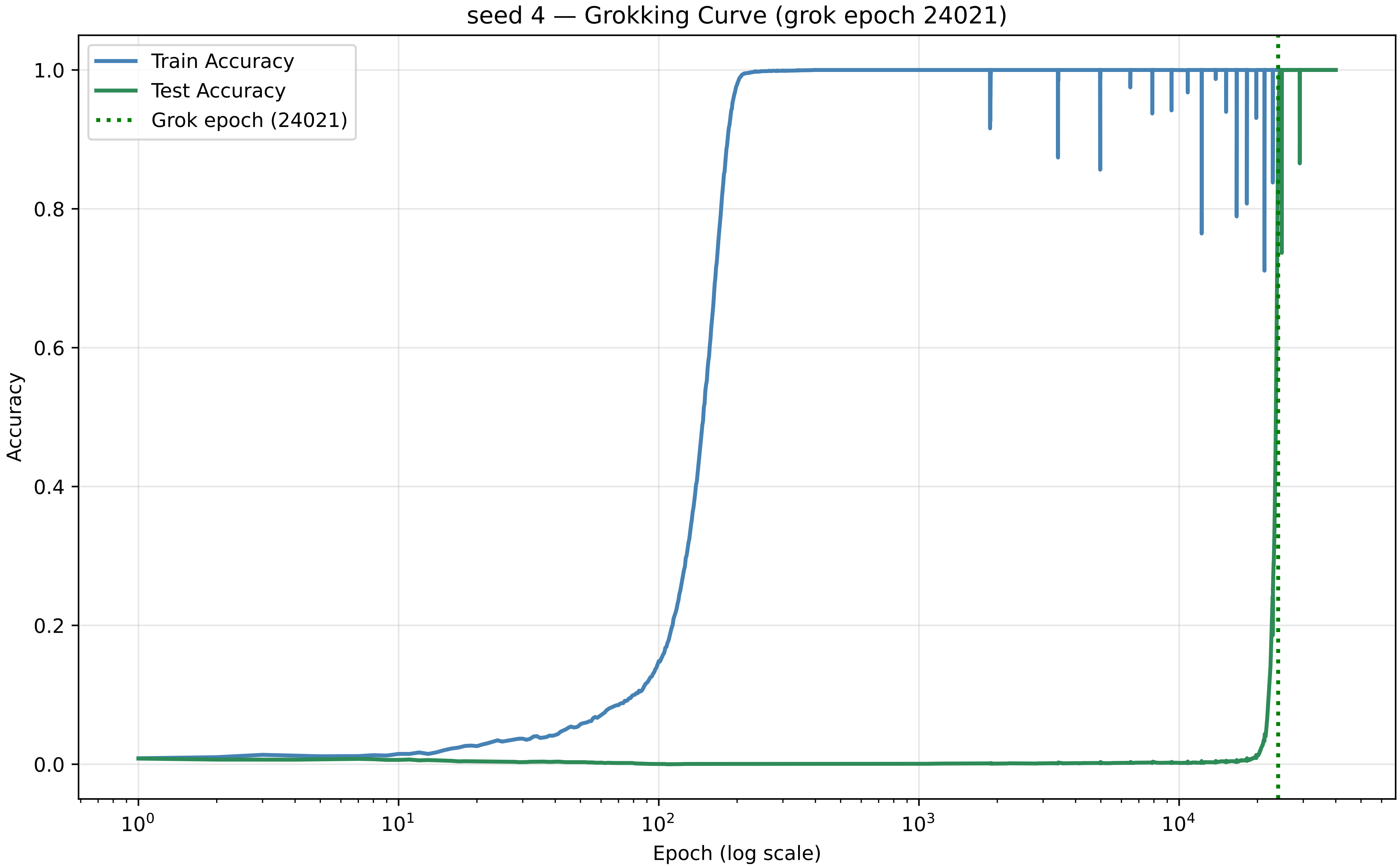


seed 3 — Dropout Gap by Rate (final epoch 40000)
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
Dropout gap (final model)
-0.001
-0.123
-0.538
-0.876
-0.978
0.1
0.3
0.5
0.7
0.9
Dropout rate

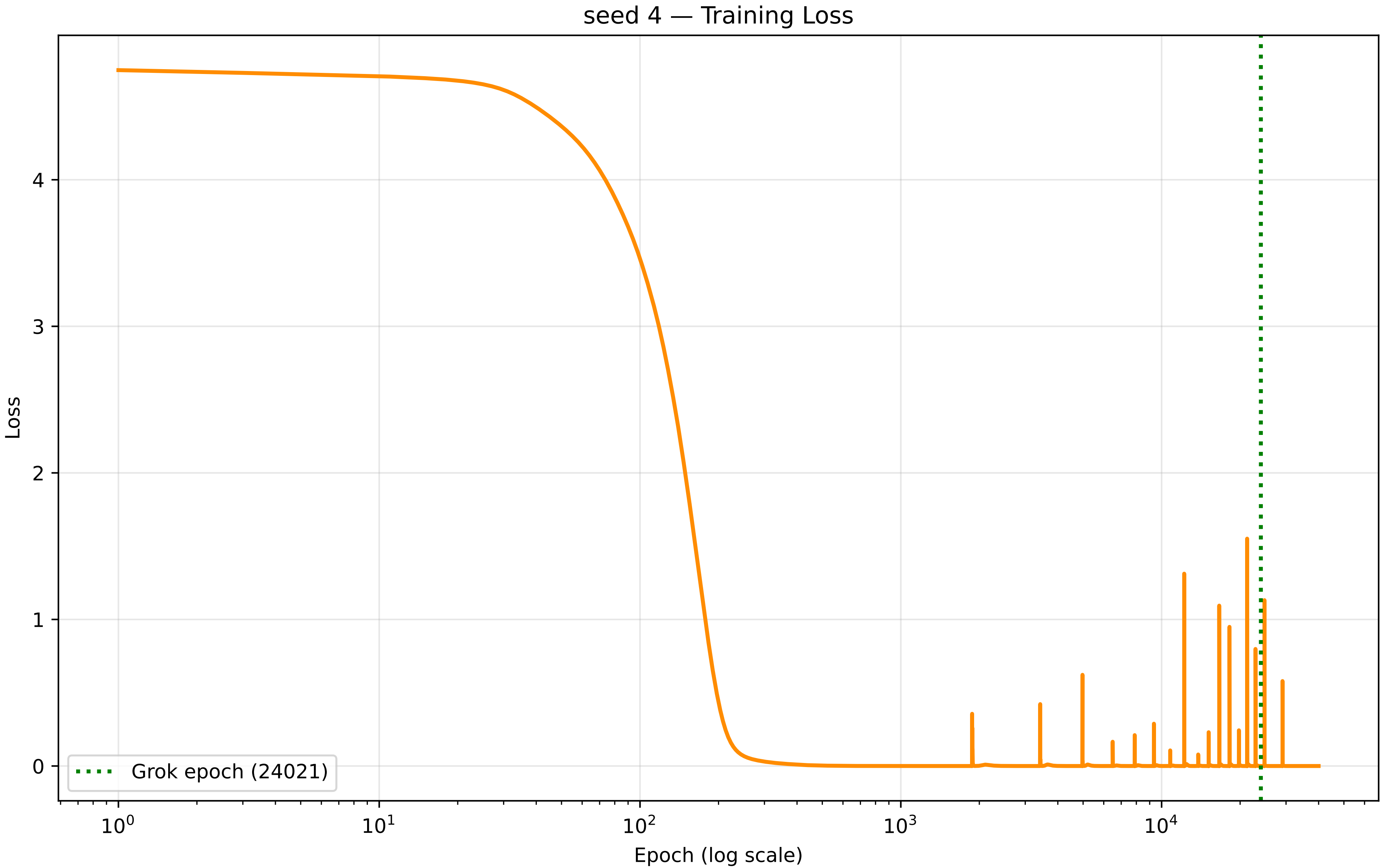

seed 3 — Dropout-Variance Predictor
1e−5
Var(test acc) under dropout (rate=0.5, k=30)
Variance peak (epoch 39999)
Grok epoch (10193)
Variance of test accuracy
Epoch (log scale)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
10^0
10^1
10^2
10^3
10^4

seed 4 — full numbers

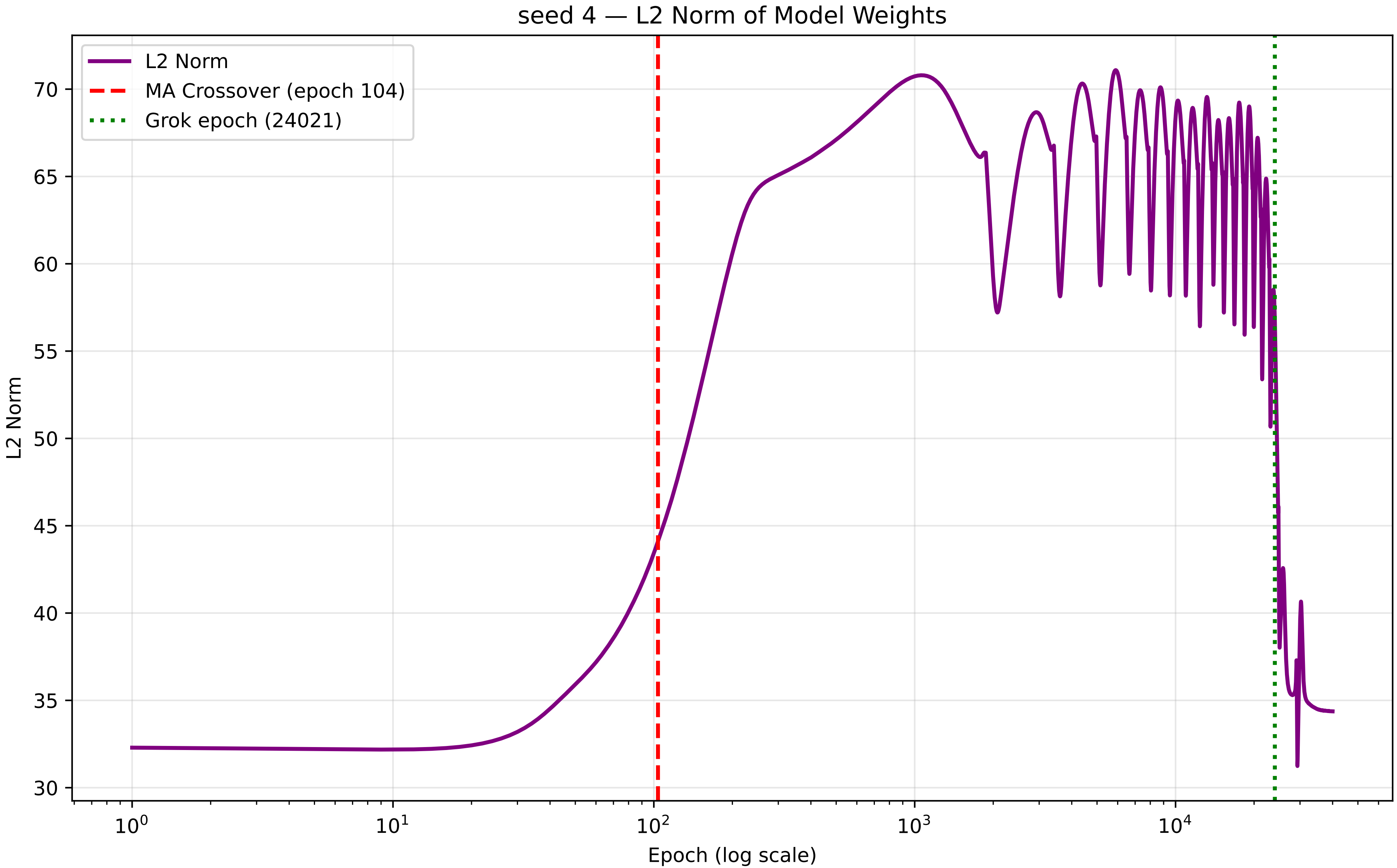
```
seed 4  (modulus=113  epochs=40000  wall_time=4948.9s)

  grok_epoch                    : 24021
  final_train_acc               : 1.0000
  final_test_acc                : 1.0000
  l2_norm   init -> final       : 32.2933 -> 34.3719
  sum_w2    init -> final       : 1042.858 -> 1181.431
  token_embedding_share (init)  : 0.0697

  L2 predictor:
    MA-crossover epoch           : 103.72
    MA-of-MA zero-crossing epoch : 1791.75
    noise_floor                  : 0.000469
    windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

  Dropout gap by rate:
    p0.1=+0.0000  p0.3=-0.0105  p0.5=-0.1783  p0.7=-0.5991  p0.9=-0.9464

  Dropout-Variance predictor:
    variance_peak_epoch          : 25232
    peak_to_grok_ratio            : 1.0504
    rate=0.5  n_samples=30  num_checkpoints=24

  Limit-cycle check             : stable
    window_start=24521  post_grok_min=0.7365  post_grok_std=0.0035
    post_grok_final=1.0000  dips<0.9=6
```



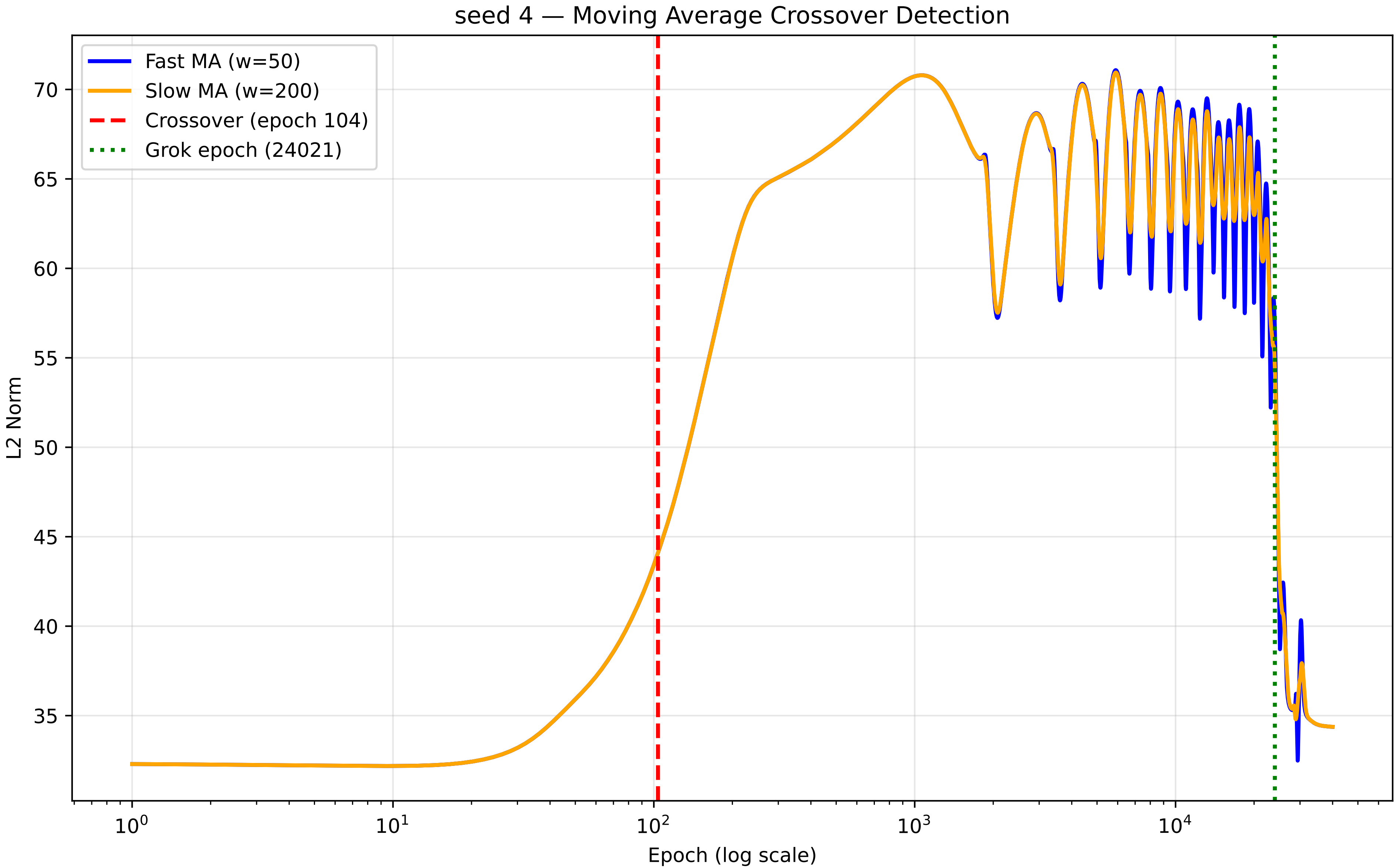
seed 4 — Grokking Curve (grok epoch 24021)
Train Accuracy
Test Accuracy
Grok epoch (24021)
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



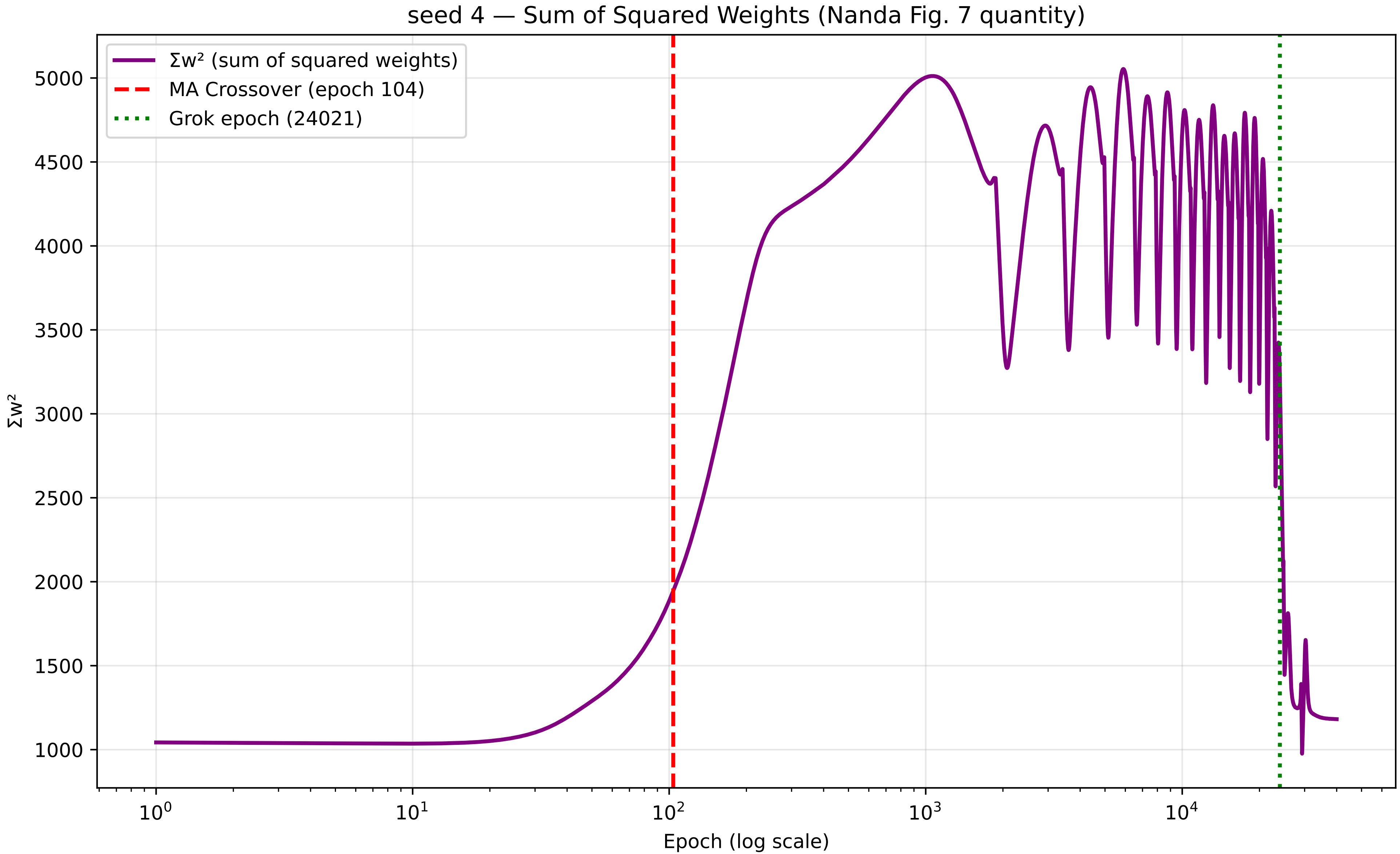
seed 4 — Training Loss
Loss
Epoch (log scale)
Grok epoch (24021)
0
1
2
3
4
10^0
10^1
10^2
10^3
10^4



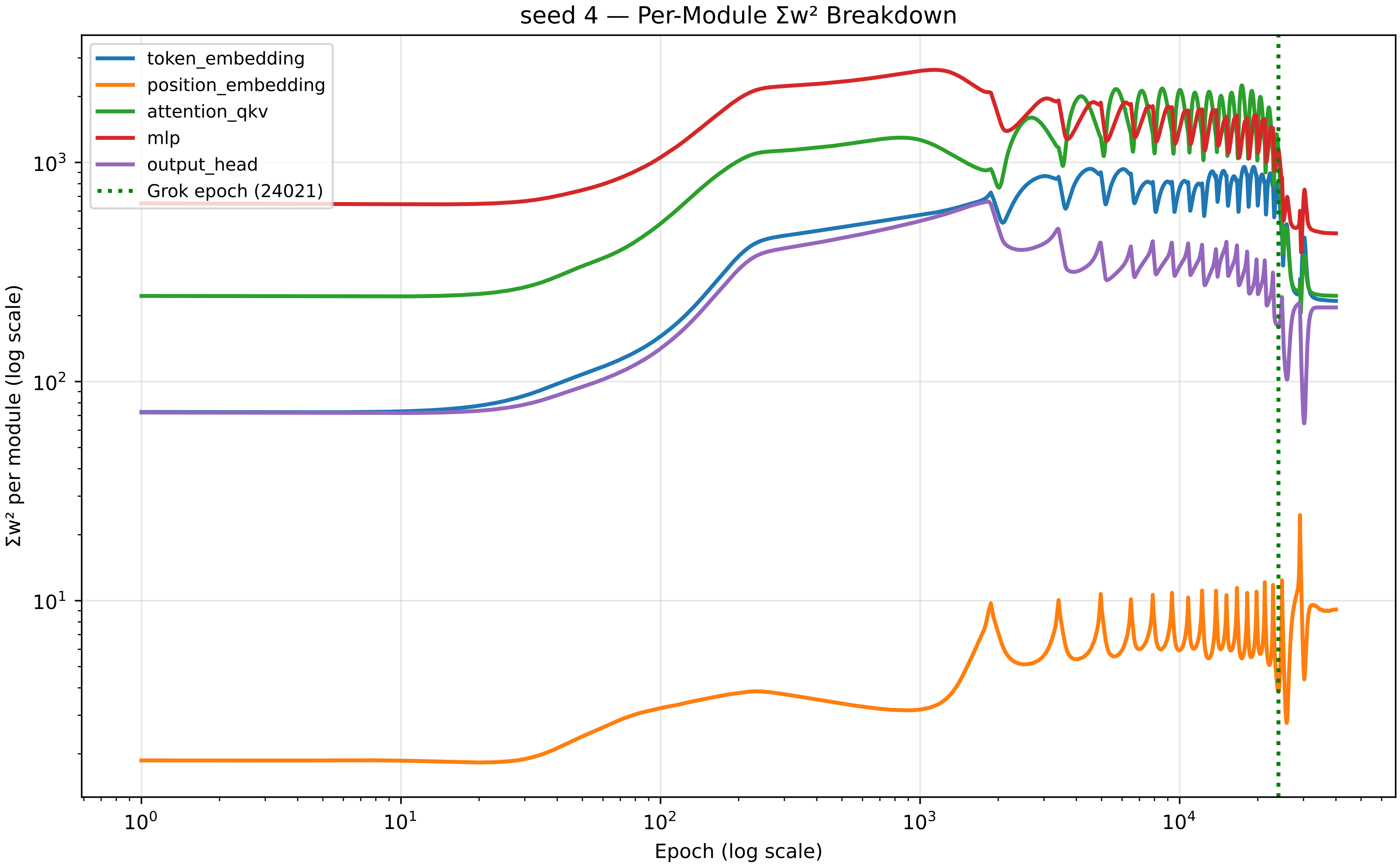
seed 4 — L2 Norm of Model Weights
L2 Norm
MA Crossover (epoch 104)
Grok epoch (24021)
L2 Norm
Epoch (log scale)
70
65
60
55
50
45
40
35
30
10^0
10^1
10^2
10^3
10^4



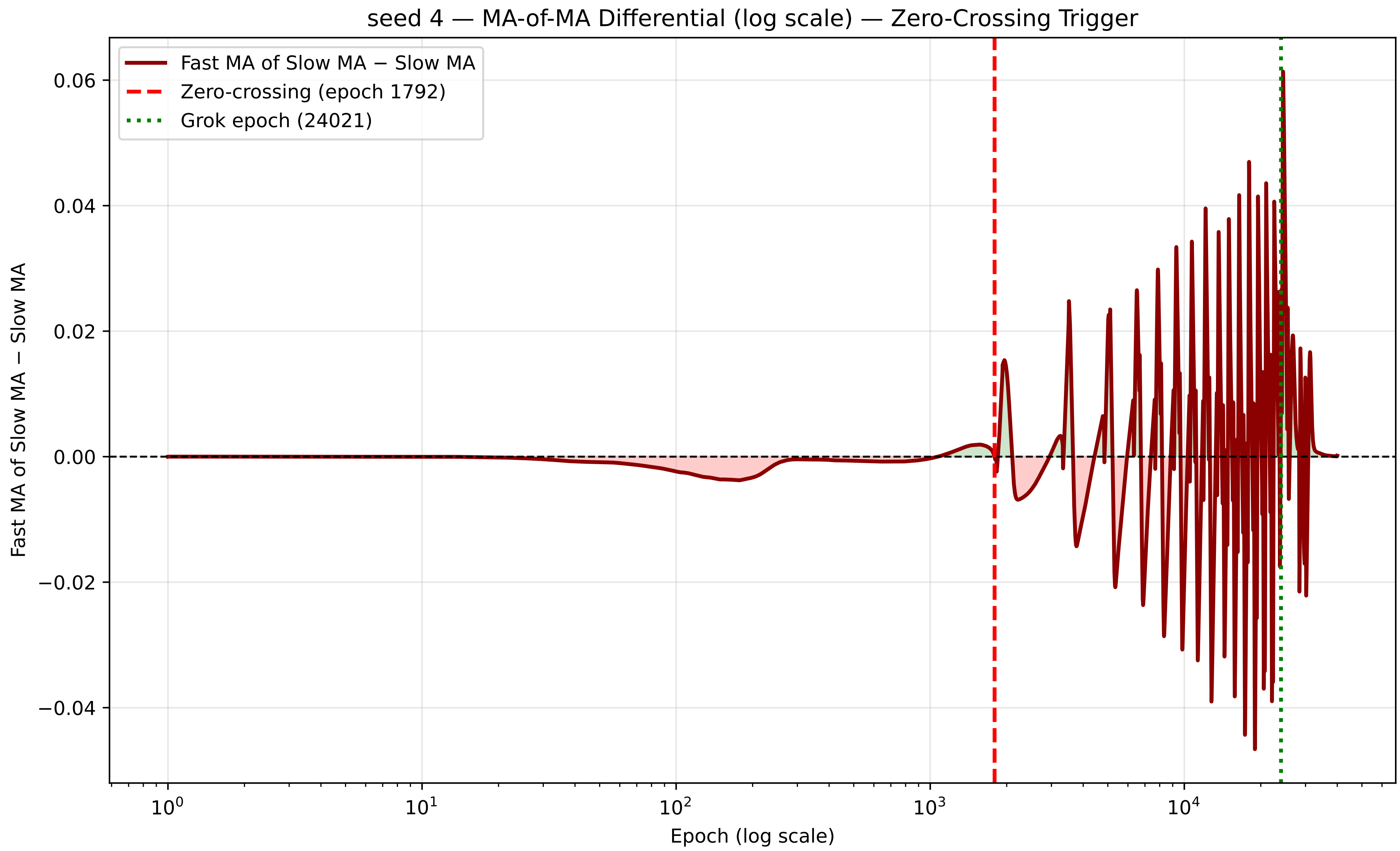
seed 4 — Moving Average Crossover Detection
Fast MA (w=50)
Slow MA (w=200)
Crossover (epoch 104)
Grok epoch (24021)
L2 Norm
Epoch (log scale)
70
65
60
55
50
45
40
35
10^0
10^1
10^2
10^3
10^4



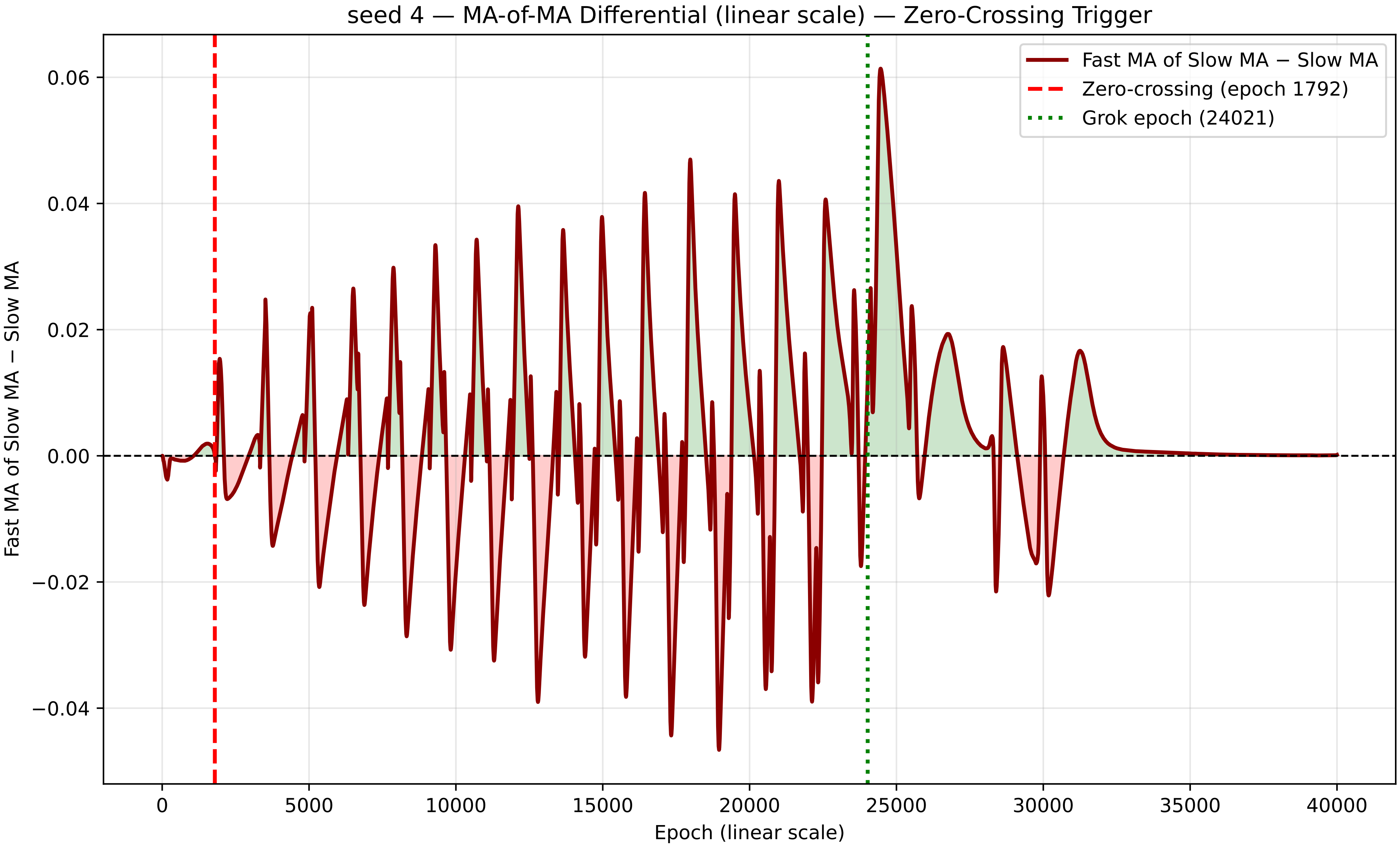
seed 4 — Sum of Squared Weights (Nanda Fig. 7 quantity)
Σw² (sum of squared weights)
MA Crossover (epoch 104)
Grok epoch (24021)
5000
4500
4000
3500
3000
2500
2000
1500
1000
Σw²
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



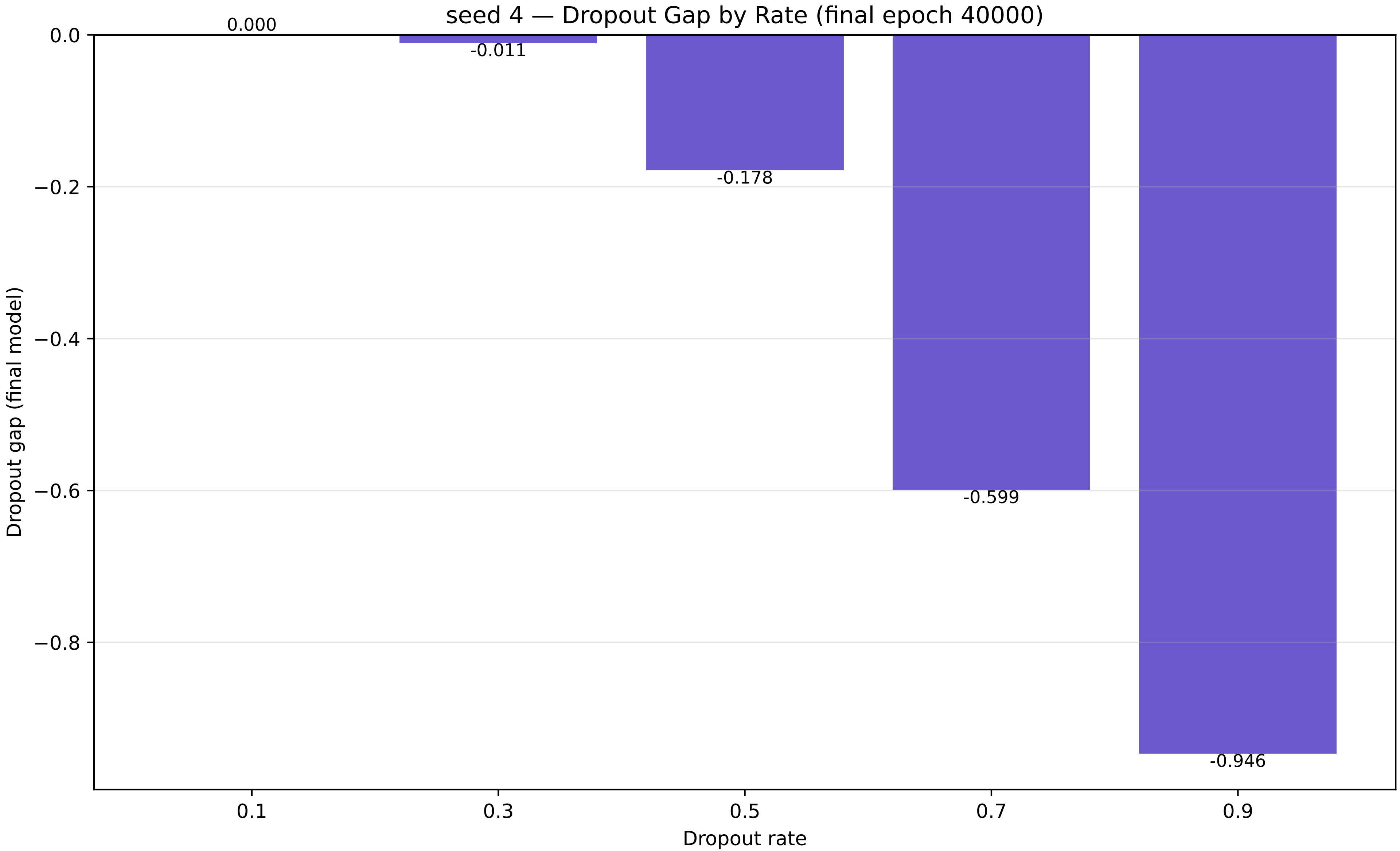
seed 4 — Per-Module Σw² Breakdown
token_embedding
position_embedding
attention_qkv
mlp
output_head
Grok epoch (24021)
Σw² per module (log scale)
Epoch (log scale)
10^0
10^1
10^2
10^3
10^4

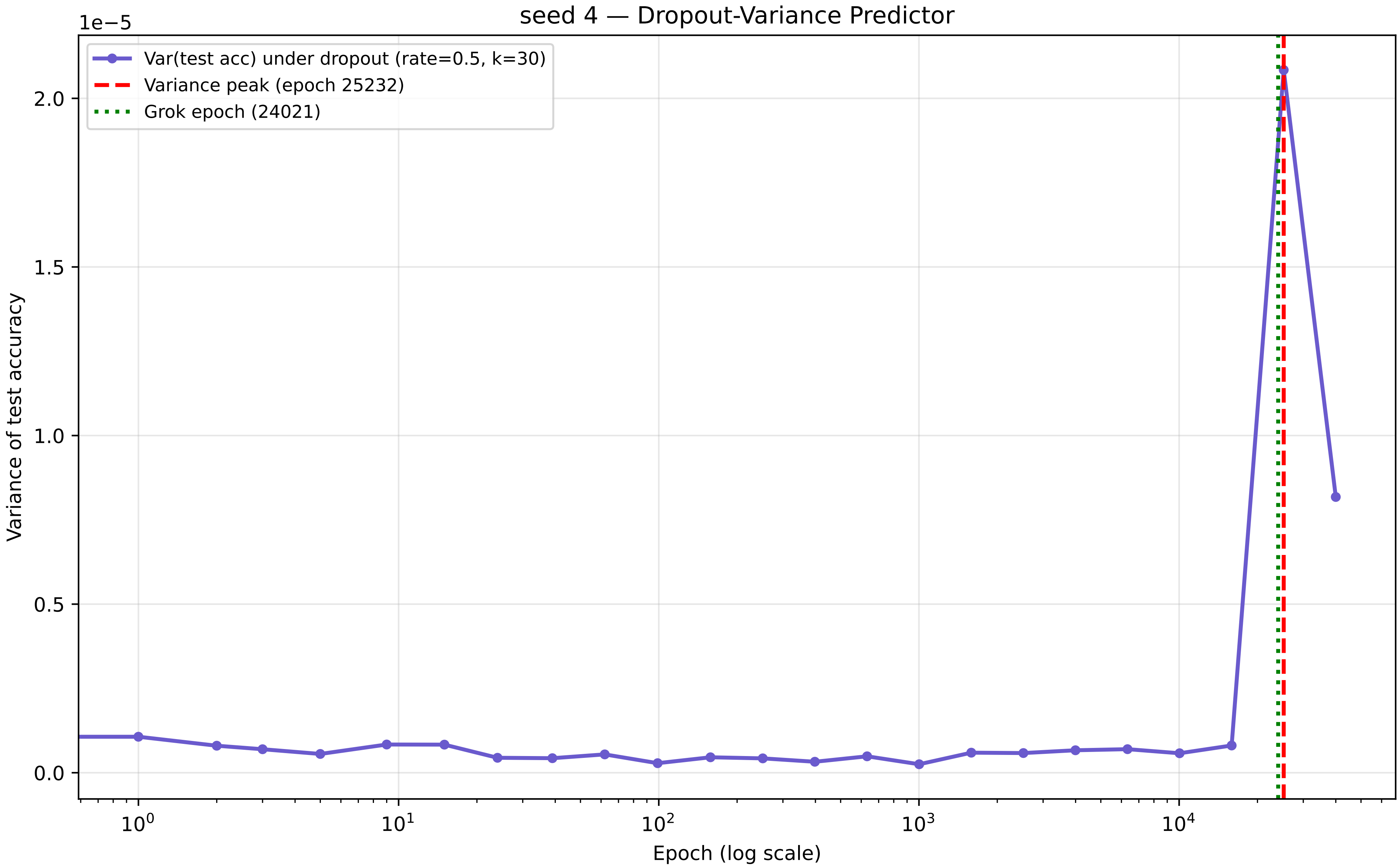


seed 4 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1792)
Grok epoch (24021)
0.06
0.04
0.02
0.00
−0.02
−0.04
Fast MA of Slow MA − Slow MA
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)



seed 4 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger
Fast MA of Slow MA − Slow MA
Zero-crossing (epoch 1792)
Grok epoch (24021)
0.06
0.04
0.02
0.00
−0.02
−0.04
0
5000
10000
15000
20000
25000
30000
35000
40000
Epoch (linear scale)
Fast MA of Slow MA − Slow MA


seed 4 — Dropout Gap by Rate (final epoch 40000)
0.000
-0.011
-0.178
-0.599
-0.946
0.0
−0.2
−0.4
−0.6
−0.8
Dropout gap (final model)
0.1
0.3
0.5
0.7
0.9
Dropout rate


seed 4 — Dropout-Variance Predictor
1e−5
Var(test acc) under dropout (rate=0.5, k=30)
Variance peak (epoch 25232)
Grok epoch (24021)
0.0
0.5
1.0
1.5
2.0
Variance of test accuracy
10^0
10^1
10^2
10^3
10^4
Epoch (log scale)